



UNIVERSITAS INDONESIA

**INTELLIGENCE ANALYSIS OF CYBERATTACKS ON THE SECURITY OF
BANKING INSTITUTIONS IN INDONESIA**

THESIS

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**SCHOOL OF STRATEGIC AND GLOBAL STUDIES
NATIONAL RESILIENCE STUDIES PROGRAM
INTELLIGENCE STRATEGIC STUDIES CONCENTRATION
JAKARTA, JUNE 2025**



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**This thesis is submitted as one of the requirements for obtaining the degree of
Master of Science (M.Si)**

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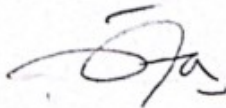
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Telah dibaca, dikoreksi dan diterima sebagai persyaratan yang diperlukan untuk memperoleh gelar Magister Sains pada Program Studi Kajian Ketahanan Nasional, Sekolah Kajian Strategik dan Global Universitas Indonesia

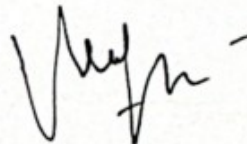
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PREFACE

Praise be to God Almighty for His abundant grace and strength, which enabled the completion of the thesis titled “Intelligence Analysis of Cyberattacks on the Security of Banking Institutions in Indonesia” as part of the requirements for obtaining the degree of Master of Science in the National Resilience Studies Program, School of Strategic and Global Studies, Universitas Indonesia.

The completion of this thesis would not have been possible without the support and contribution of various parties. The author extends sincere gratitude to both parents for their unceasing prayers and support. Deep appreciation is also extended to Dr. Puspitasari, S.Sos., M.Si., as First Supervisor, and Dr. Aloysius Mado, B.Sc, S.Sos, M.Han., as Second Supervisor, for their invaluable guidance and direction throughout the research process.

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The author acknowledges that this work still has limitations. Criticism and suggestions from various parties are therefore greatly welcomed for further development. It is hoped that this thesis can make a positive contribution to the study of strategic intelligence and cyber resilience in Indonesia's banking sector.

**STATEMENT OF APPROVAL FOR THE PUBLICATION OF ACADEMIC
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This research aims to analyze proactive cyber intelligence system approaches in detecting, identifying, and responding to cyberattack patterns against banking institutions; evaluate the implementation of adaptive security systems in addressing escalating threats; and examine the integration of strategic and progressive information protection policies. Using a qualitative approach through literature review and expert interviews, the analysis is conducted based on four core concepts: Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR). This research also integrates Cyber Kill Chain-based attack mapping and foresight planning using event, trend, driver, and scenario planning methods. The findings show that banking institutions still tend to be reactive, with weaknesses in early detection and the integration of intelligence into policy. The resulting strategic recommendations include strengthening CTI systems, formulating foresight-based adaptive policies, and developing short-, medium-, and long-term digital resilience roadmaps to build an adaptive, systematic, and sustainable cybersecurity posture for banking institutions.

Keywords: Cyber Threat Intelligence, Adaptive Security Framework, Strategic Intelligence, Cyber Resilience

Abstract

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List of Abbreviations

Abbreviation	Meaning	Explanation
AAL	Authenticator Assurance Level	Refers to the NIST concept of ‘Authenticator Assurance Level’ used in the context of cybersecurity or strategic intelligence in the banking sector.
AI	Artificial Intelligence	Refers to the use of artificial intelligence technology in the context of cybersecurity or strategic intelligence in the banking sector.
API	Application Programming Interface	Refers to another risky entry point, the mobile banking application and its application programming interface, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
APT	Advanced Persistent Threat tactics	Refers to advanced persistent threat tactics used in the context of cybersecurity or strategic intelligence in the banking sector.
ASF	Adaptive Security Framework	Refers to the Adaptive Security Framework concept used in the context of cybersecurity or strategic intelligence in the banking sector.
BCA	Bank Central Asia	Refers to Bank Central Asia, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
BCP	Business Continuity Plan	Refers to institutions that consistently test and update their business continuity plan, discussed in the context of cybersecurity or strategic intelligence in the banking sector.

Abbreviat ion	Meaning	Explanation
BCRIP	Business Continuity and Recovery Integration Plan	Refers to the Business Continuity and Recovery Integration Plan concept used in the context of cybersecurity or strategic intelligence in the banking sector.
BI	Bank Indonesia	Refers to Bank Indonesia, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
BIA	Business Impact Analysis	Refers to another key step — reformulating data protection policy based on risk appetite and the results of a Business Impact Analysis — discussed in the context of cybersecurity or strategic intelligence in the banking sector.
BIN	Badan Intelijen Negara (State Intelligence Agency)	Refers to the State Intelligence Agency (Badan Intelijen Negara), discussed in the context of cybersecurity or strategic intelligence in the banking sector.
BIS	Bank for International Settlements	Refers to the Bank for International Settlements, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
BRI	Bank Rakyat Indonesia	Refers to another significant event involving an alleged attack on Bank Rakyat Indonesia, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
BSI	Bank Syariah Indonesia	Refers to Bank Syariah Indonesia, discussed in the context of cybersecurity or strategic intelligence in the banking sector.

Abbreviat ion	Meaning	Explanation
BSSN	Badan Siber dan Sandi Negara (National Cyber and Crypto Agency)	Refers to the annual report released by the National Cyber and Crypto Agency (Badan Siber dan Sandi Negara), discussed in the context of cybersecurity or strategic intelligence in the banking sector.
CISO	Chief Information Security Officer	Refers to the Chief Information Security Officer role, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
CR	Cyber Resilience	Refers to the Cyber Resilience concept used in the context of cybersecurity or strategic intelligence in the banking sector.
CSF	NIST Cybersecurity Framework	Refers to the NIST Cybersecurity Framework, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
CSIRT	Computer Security Incident Response Team	Refers to the fact that many institutions have not yet established a Computer Security Incident Response Team, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
CTI	Cyber Threat Intelligence	Refers to the Cyber Threat Intelligence concept used in the context of cybersecurity or strategic intelligence in the banking sector.
CVSS	Common Vulnerability Scoring System	Refers to the Common Vulnerability Scoring System, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
DLP	Data Loss Prevention	Refers to Data Loss Prevention, discussed in the context of

Abbreviat ion	Meaning	Explanation
		cybersecurity or strategic intelligence in the banking sector.
DNS	Domain Name System	Refers to the Domain Name System, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
EDR	Endpoint Detection and Response	Refers to certain actors using AI to map detection gaps in endpoint detection and response systems, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
GPN	Gerbang Pembayaran Nasional (National Payment Gateway)	Refers to Bank Indonesia's oversight of national digital infrastructure such as the National Payment Gateway (Gerbang Pembayaran Nasional), discussed in the context of cybersecurity or strategic intelligence in the banking sector.
IDS	Intrusion Detection System	Refers to a communication channel designed to resemble normal network traffic so as not to trigger defense systems such as firewalls or an intrusion detection system, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
IIV	Infrastruktur Informasi Vital (Vital Information Infrastructure)	Refers to Indonesia as part of the Vital Information Infrastructure (Infrastruktur Informasi Vital), discussed in the context of cybersecurity or strategic intelligence in the banking sector.
IOC	Indicator of Compromise	Refers to the exchange of indicators of compromise, discussed in the context of

Abbreviat ion	Meaning	Explanation
		cybersecurity or strategic intelligence in the banking sector.
ISAC	Information Sharing and Analysis Center	Refers to the Information Sharing and Analysis Center, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
ISMS	Information Security Management System	Refers to the most urgent strategic effort being the comprehensive implementation of an Information Security Management System, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
LMS	Learning Management System	Refers to institutions also being advised to develop an internal Learning Management System-based training system, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
MII	PT Mitra Integrasi Informatika	Refers to the Cybersecurity Division Head of PT Mitra Integrasi Informatika, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
MLA	Mutual Legal Assistance	Refers to many digital crime service providers operating from jurisdictions that lack mutual legal assistance agreements, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
MTTD	Mean Time to Detect	Refers to a SOC focused on mean-time-to-detect efficiency, discussed in the context of cybersecurity or strategic intelligence in the banking

Abbreviat ion	Meaning	Explanation
		sector.
MTTR	Mean Time to Respond	Refers to Mean Time to Respond, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
NDR	Network Detection and Response	Refers to Network Detection and Response, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
NIST	National Institute of Standards and Technology	Refers to the National Institute of Standards and Technology, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
OJK	Otoritas Jasa Keuangan (Financial Services Authority)	Refers to the Financial Services Authority (Otoritas Jasa Keuangan), discussed in the context of cybersecurity or strategic intelligence in the banking sector.
OSINT	Open-Source Intelligence	Refers to data collection conducted using open-source intelligence methods, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
POJK	Peraturan Otoritas Jasa Keuangan (Financial Services Authority Regulation)	Refers to one of the key policies serving as a strategic foundation for strengthening information security systems in the banking sector, namely the Financial Services Authority Regulation (Peraturan Otoritas Jasa Keuangan), discussed in the context of cybersecurity or strategic intelligence in the banking sector.
PSE	Penyelenggara Sistem Elektronik (Electronic System Operator)	Refers to the Electronic System Operator (Penyelenggara Sistem Elektronik), discussed in the

Abbreviat ion	Meaning	Explanation
		context of cybersecurity or strategic intelligence in the banking sector.
RDP	Remote Desktop Protocol	Refers to the Remote Desktop Protocol, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
RTO	Recovery Time Objective	Refers to this strategy incorporating the integration of Recovery Time Objective targets, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
SDN	Software Defined Networking	Refers to adaptive machine-learning-based DDoS within a Software Defined Networking environment, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
SI	Strategic Intelligence	Refers to the Strategic Intelligence concept used in the context of cybersecurity or strategic intelligence in the banking sector.
SIEM	Security Information and Event Management	Refers to Security Information and Event Management, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
SKKNI	Standar Kompetensi Kerja Nasional Indonesia (Indonesian National Work Competency Standard)	Refers to the Indonesian National Work Competency Standard (Standar Kompetensi Kerja Nasional Indonesia), discussed in the context of cybersecurity or strategic intelligence in the banking sector.
SOAR	Security Orchestration, Automation, and Response	Refers to Security Orchestration, Automation, and Response,

Abbreviat ion	Meaning	Explanation
		discussed in the context of cybersecurity or strategic intelligence in the banking sector.
SOC	Security Operation Center	Refers to the establishment of a Security Operation Center, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
SOP	Standard Operating Procedure	Refers to a Standard Operating Procedure, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
SPBE	Technical Standards and Procedures for Building Electronic-Based Government System Applications	Refers to the Technical Standards and Procedures for Building Electronic-Based Government System Applications, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
TI	Information Technology	Refers to vulnerabilities also arising from dependence on third parties for the provision of information technology, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
TIS	Tanggap Insiden Siber (Cyber Incident Response)	Refers to the National Cyber Incident Response Competency Standard (SKKNI Tanggap Insiden Siber), discussed in the context of cybersecurity or strategic intelligence in the banking sector.
TLS	Transport Layer Security	Refers to many banks still relying on limited encryption such as the Transport Layer Security protocol, discussed in the context of cybersecurity or

Abbreviation	Meaning	Explanation
		strategic intelligence in the banking sector.
TTP	Tactics, Techniques, and Procedures	Refers to Tactics, Techniques, and Procedures, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
UEBA	User and Entity Behavior Analytics	Refers to updating the security infrastructure baseline by incorporating User and Entity Behavior Analytics capabilities, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
VPN	Virtual Private Network	Refers to a Virtual Private Network, discussed in the context of cybersecurity or strategic intelligence in the banking sector.
XXE	XML External Entity	Refers to an XML External Entity, discussed in the context of cybersecurity or strategic intelligence in the banking sector.

Chapter 1. Introduction

1.1 Research Background

In May 2023, the Indonesian public was shocked by a LockBit 3.0 ransomware attack that struck Bank Syariah Indonesia, crippling digital services and causing a data leak of 1.5 terabytes (Olavsrud, 2023). A year earlier, Bank Indonesia fell victim to the Conti ransomware, which targeted the internal network system of one of its regional offices and successfully stole tens of thousands of internal files (Bitdefender, 2022). In addition, large-scale data breach incidents were also reported at Bank Jatim and BCA (Anuar et al., 2022; Kompas, 2025), while Bank Rakyat Indonesia faced a surge in phishing cases that caused significant losses among customers (Tangka & Lompouliu, 2024). These types of attacks — ransomware, phishing, and data breaches — represent the most significant strategic threats to the national banking sector.

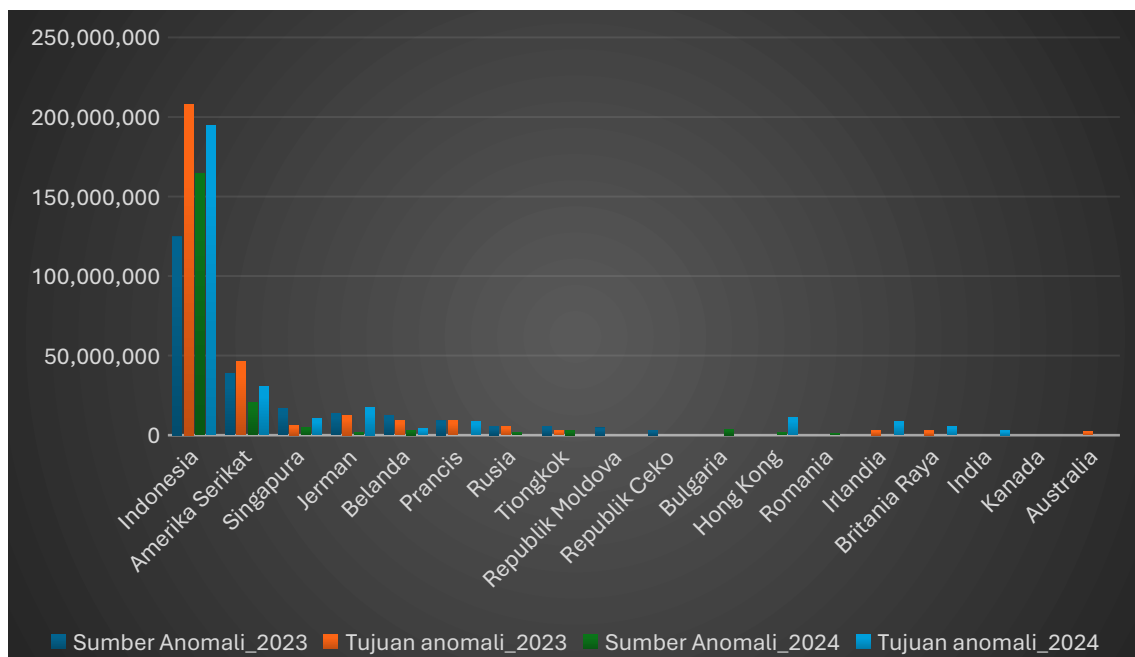


Figure 1.1. Sources and Targets of Indonesian Cyber Attack Anomalies

Source: Adapted from the Indonesia Cyber Security Landscape 2023–2024

Based on the annual report released by the National Cyber and Crypto Agency (BSSN), Indonesia has consistently ranked among the highest globally in the volume of

cyber anomalies, both as a source and as a target of digital anomaly activity, for two consecutive years. In 2023, 124,646,656 anomalies originated from Indonesia, while 208,061,734 anomalies were detected targeting domestic digital systems (BSSN, 2023). This trend continued in 2024, with a significant increase in anomalies originating from Indonesia to 164,432,795, although the number of incoming anomalies decreased slightly to 194,562,390 (BSSN, 2024). This surge in anomaly activity indicates that the national digital ecosystem is not only a vulnerable target of attacks, but also a potential medium for spreading cyberattacks to various foreign jurisdictions.

This phenomenon indicates the existence of compromised domestic digital devices that have subsequently been misused as botnets or distribution channels for malicious cross-border activity, including servers, endpoints, and applications operating within strategic infrastructure environments. In this context, this condition creates an urgent need to strengthen control systems over affected devices, particularly those related to strategic national sectors such as financial and banking services, which the government has designated as part of the Vital Information Infrastructure (IIV) (BSSN, 2024).

Furthermore, the distribution pattern of anomalies in 2024 shows new dynamics in the global threat landscape. Countries such as the United States, Singapore, and Germany remain consistently in the top five, both as contributors and recipients of digital anomalies. However, the emergence of countries such as Hong Kong, Ireland, and India in the top ten marks a broader and more widespread geographic expansion of the attack surface. In addition, countries such as Australia, the United Kingdom, and Canada also recorded significant volumes as destinations of anomaly activity, indicating that attack vectors are no longer concentrated in certain regions but have spread across various areas with heterogeneous digital structures.

On the sectoral side, BSSN's official report also underlines that the government sector remained the entity most affected by cyberattacks throughout 2023, accounting for 71% of all identified incidents (BSSN, 2023). The financial and banking sector followed in second place, contributing 12% of incidents. Although its incident count is not quantitatively dominant, this sector has a high level of vulnerability and broad systemic risk, given its strategic position in maintaining national economic stability, managing sensitive data, and sustaining public trust in financial institutions. Attacks on banking systems have the potential to cause serious disruption to digital transactions, leaks of

customers' personal data, and significant financial and reputational losses. These risks can place direct pressure on institutional operational continuity, weaken public perception, and shake investor and regulator confidence in the resilience of Indonesia's financial system.

The increasingly complex escalation of cyber threats shows that a reactive approach is no longer sufficient, necessitating an adaptive, strategic-intelligence-based defense system. In this context, the application of the Adaptive Security Framework (Lopes & Oliveira, 2015) becomes highly relevant for building a flexible, responsive protection mechanism capable of real-time threat detection. Furthermore, the concept of Strategic Intelligence (Walsh, 2017) is needed to formulate institutional policies responsive to long-term, multidimensional threat dynamics. On the other hand, to ensure service continuity after an incident and accelerate operational recovery, the Cyber Resilience approach (Linkov & Kott, 2019) becomes the main foundation for building a robust and sustainable cyber resilience system, particularly in the financial sector, which is highly vulnerable to attacks with systemic impact.



Figure 1.2. Number of Cyberattacks on Indonesian Banking (2021–2024)

Source: Adapted from reports by Infobanknews (2021), OJK (2022), BSSN (2023–2024), Antara News (2023), and Tempo (2024).

The data presented in the chart shows a significant escalation in both the scale and intensity of cyberattacks on banking institutions in Indonesia from 2021 to 2024. In 2021, more than 1.6 billion cyber traffic anomalies targeting the national financial sector were recorded. Most of these anomalies were related to suspicious activity and indications of attempts to exploit system vulnerabilities (Infobanknews, 2021; BSSN, 2023). However, in the following year, this figure decreased drastically to around 403 million attacks. This decrease, while seemingly positive at first glance, does not fully reflect an improvement in the effectiveness of security systems. Rather, differences in the methodological approaches used by relevant institutions, such as OJK, to classify and report incidents also influenced the officially recorded number of attacks (OJK, 2022).

This trend changed dramatically in 2023, coinciding with a series of major cyber incidents, including the LockBit 3.0 ransomware attack on Bank Syariah Indonesia (BSI), as well as data breach incidents affecting other financial institutions. During this period, the recorded number of cyberattacks surged again to reach 1.9 billion, reflecting the level of digital system vulnerability in the banking sector that has not yet been optimally addressed (Antara News, 2023; Olavsrud, 2023).

The peak of the surge in cyberattacks occurred in 2024, with an estimated 4 billion attacks targeting national banking services and information systems. This figure is the highest in the past four years and represents significant pressure on the digital resilience of the financial sector (Tempo, 2024). This sharp increase correlates with the growing integration of digital banking services and the expansion of open banking systems, which in turn broadens the attack surface that can be exploited by malicious actors.

Overall, this data confirms that Indonesia's banking sector is facing an increasingly complex and intense cyber threat landscape. This condition demands a strategic response through the implementation of adaptive security approaches, supported by a proactive

cyber intelligence system, and the strengthening of progressive information protection policies integrated with the ever-evolving dynamics of digital threats.

This increased intensity of cyberattacks affects not only technical aspects but also threatens economic stability, public trust, and national digital resilience. Given these conditions, this research is highly important to conduct, in order to examine how an intelligence analysis approach can function in recognizing attack patterns, providing early warning, and formulating more strategic and targeted mitigation strategies. This research specifically limits its focus to four main categories of cyberattacks, with banking institutions operating in Indonesia as its object.

The conceptual framework of this research refers to four main approaches: Cyber Threat Intelligence, Adaptive Security Framework, Strategic Intelligence, and Cyber Resilience. Each has a complementary function: Cyber Threat Intelligence (Tounsi & Rais, 2018) functions to systematically identify threat patterns and characteristics; Strategic Intelligence (Walsh, 2017) is used to analyze long-term risks and potential strategic impacts for financial institutions; Adaptive Security Framework (Lopes & Oliveira, 2015) helps assess the flexibility and responsiveness of security systems to dynamic threats; while Cyber Resilience (Linkov & Kott, 2019) is used to strengthen early prevention against subsequent cyberattack threats.

The research gap in this study stems from the continued dominance of theoretical and technical approaches in the study of cyberattacks, which have not deeply addressed the context of Indonesia's banking sector as part of the Vital Information Infrastructure (IIV) that is highly vulnerable to digital threats. No research has been found that simultaneously maps phishing, ransomware, and data breach patterns based on real incidents at national banking institutions, so understanding of cyber risk remains general and not contextualized. The absence of a predictive strategic intelligence framework to support early prevention represents an important gap yet to be filled. This research therefore presents an integrated approach combining four core concepts — Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR) — to address the need for a cyber intelligence system reflective of national conditions. The analysis is directed at actually mapping cyberattack patterns in the Indonesian banking sector for the 2021–2024 period. In addition, a foresight-based scenario planning approach is applied to measure institutions' adaptive readiness to face

changes in the threat landscape. The resulting strategies are reflective, proactive, predictive, and systematic, aimed at strengthening national digital resilience sustainably.

This research thus addresses academic and practical needs in examining the effectiveness of intelligence-based cyber defense strategies in Indonesia's banking sector. Through a qualitative approach and literature-based case mapping, this study is expected to contribute to strengthening a banking security system that is more adaptive, resilient, and integrated with the evolving cyber threat landscape.

1.2 Problem Statement and Research Questions

1.2.1. Problem Identification

This research examines various cyberattack incidents that occurred within Indonesia's banking environment between 2021 and 2024. Its focus covers cross-institutional studies, including Bank Indonesia, Bank Syariah Indonesia, and BCA — banks that are well documented as victims of phishing, ransomware, and data breach attacks.

All the cases analyzed are drawn from real events, not simulations. This enables the analysis to be based on empirical data and an intelligence framework. Using a qualitative approach, this research examines attack patterns, the defense strategies applied by each institution, and analyzes the long-term consequences for the stability of banking institutions and public trust.

This study does not focus on technical aspects, but rather explores further how such technical data can be translated into intelligence strategies to support the formulation of adaptive and sustainable cybersecurity policy for the banking sector.

1.2.2. Strategic Intelligence Focus

This research is built upon four main conceptual frameworks: Cyber Threat Intelligence (Tounsi & Rais, 2018), Strategic Intelligence (Walsh, 2017), Adaptive Security Framework (Lopes & Oliveira, 2015), and Cyber Resilience (Linkov & Kott, 2019). Each concept provides a complementary analytical dimension.

1. Cyber Threat Intelligence is used to understand attack patterns as well as the tactics and techniques used by perpetrators.
2. Strategic Intelligence provides a framework for assessing the relationship between cyberattacks and policy decision-making.
3. Adaptive Security Framework serves to evaluate institutions' readiness to respond and adapt to a dynamic threat landscape.
4. Cyber Resilience is used to measure how robust banking institutions are in withstanding and recovering from cyber incidents.

By integrating these four approaches, this research is expected to produce an in-depth analysis that not only explains the forms of attacks, but also assesses the response strategies applied, and the extent to which those approaches can guarantee the continuity and operational security of banks.

1.2.3. Research Questions

To address the objectives of this research, three main questions are formulated as follows:

1. How can an adaptive security approach be implemented by banking institutions in Indonesia to address the continuously evolving dynamics of cyber threats?
2. How can a proactive cyber intelligence system support early detection of and response to cyberattack patterns in the national banking sector?
3. How can progressive and integrated information protection policy be strengthened to improve the digital resilience of banking institutions amid escalating cyber risk?

1.3 Research Objectives

This research aims to provide a strategic and comprehensive understanding of cyberattack threats against Indonesia's banking sector during the 2021–2024 period, and how an intelligence approach can be effectively used to respond to those threats. The research objectives are formulated to address the three main aspects outlined in the problem statement, consistent with the conceptual framework used: Cyber Threat Intelligence, Adaptive Security Framework, Strategic Intelligence, and Cyber Resilience.

Specifically, this research aims to:

1. Analyze proactive cyber intelligence system approaches in detecting, identifying, and responding to cyberattack patterns targeting banking institutions.
2. Analyze the implementation of adaptive security approaches in addressing the escalation of cyber threats against Indonesia's banking sector.
3. Analyze the integration of progressive and strategic information protection policy within banking cybersecurity systems.

With these three main objectives established, this research is expected to make meaningful contributions in two main domains. First, from an academic standpoint, this study aims to expand the scope of scientific inquiry and deepen theoretical development in the field of strategic intelligence and cybersecurity, particularly within the financial services sector. Second, in the practical dimension, the results of this research are expected to serve as a strategic reference for banking institutions and relevant authorities in formulating cyber risk mitigation policies that are more responsive, intelligence-based, and relevant to the continuously evolving dynamics of digital threats. With this approach, the output of this research is not only analytical and theoretical in nature, but also carries concrete applied value in strengthening the resilience of the national banking sector against potential future cyberattacks.

1.4. Research Limitations

This research is temporally limited to the 2021–2024 period, with a spatial focus on banking institutions in Indonesia. This limitation is intended to ensure that the cases examined remain contextually relevant and can illustrate the contemporary cyber threat landscape. The categories of phishing, ransomware, and data breach attacks were selected because they have been identified as the most significant threats in various national and international reports (ENISA, 2023; BSSN, 2023), giving them academic validity as the main typologies in cybersecurity studies of the financial sector.

1.5 Significance of the Research

This research develops an analytical model based on Cyber Threat Intelligence, Adaptive Security Framework, Strategic Intelligence, and Cyber Resilience to enrich the study of strategic intelligence, cybersecurity, and the financial sector. This research also fills a gap in local literature, which has limited strategic approaches, and supports the strengthening of graduate-level curricula. In practical terms, its results provide an empirical-data-based analysis of cyber readiness for the formulation of policy, operational standards, and early-prevention guidelines against cyberattacks. This research thereby strengthens the strategic and sustainable digital resilience of the national banking sector.

1.6 Scope of the Study

This research focuses on the 2021 to 2024 period, during which the acceleration of digitalization in the national banking sector occurred alongside an increase in the volume and complexity of cyberattacks. BSSN reports indicate that more than 1.6 billion cyber traffic anomalies were detected throughout 2021, with the financial sector among those most affected (BSSN, 2021). Incidents such as the LockBit 3.0 ransomware attack on Bank Syariah Indonesia in 2023 underscore the need to strengthen national digital economic resilience (Olavsrud, 2023). The government responded through Presidential Regulation No. 82/2022, while OJK strengthened cybersecurity regulation through POJK No. 11/POJK.03/2022.

In terms of geographic scope, this research focuses on Bank Indonesia and Bank Syariah Indonesia, which are classified as Vital Information Infrastructure (IIV) under Presidential Regulation No. 82/2022. The author analyzes the main forms of attack — phishing, ransomware, and data breaches — with a focus on operational, financial, and reputational impact on institutions. This study uses the frameworks of Cyber Threat Intelligence (Tounsi & Rais, 2018), Adaptive Security Framework (Lopes & Oliveira, 2015), Strategic Intelligence (Walsh, 2017), and Cyber Resilience (Linkov & Kott, 2019). This research is directed toward supporting the formulation of early-prevention strategies against cyberattacks in the national banking sector.

1.7 Organization of the Thesis

This research, titled “Intelligence Analysis of Cyberattacks on the Security of Banking Institutions in Indonesia,” is organized into five main chapters that are logically and thematically integrated. The general organization is outlined as follows:

Chapter I: Introduction

This chapter serves as the foundational introduction to the issues raised in the research, beginning with a presentation of actual cyberattack cases occurring in the national banking sector. It then explains the problem statement and research questions, research objectives, academic and practical significance, the scope limitations of the study (in terms of time, region, and theme), the research method used, and the overall organization of the thesis. This chapter provides the initial framework that unifies the direction and logic of the research as a whole.

Chapter II: Literature Review

This chapter presents the conceptual and theoretical foundations of the analysis in this research. The main concepts used include Cyber Threat Intelligence, Adaptive Security Framework, Strategic Intelligence, and Cyber Resilience. It also presents relevant prior literature, along with an analysis of the gaps this research aims to fill.

Chapter III: Research Methodology

The third chapter explains the methodological approach used, namely a qualitative method with a literature study technique. It describes the sources and types of data, interviews, secondary data collection, and the analysis method used. All of these are linked to the strategic intelligence approach that forms the basis for analyzing threat patterns and policy responses to cyberattacks in the banking sector.

Chapter IV: Research Results and Discussion

This chapter presents the results of the analysis of the four main categories of cyberattacks — phishing, ransomware, and data breaches — targeting banking institutions in Indonesia. The discussion focuses on mapping attack characteristics, analyzing system weaknesses, and evaluating the institutional strategies and intelligence-based policies applied in response. The analytical approach is directly linked to the conceptual framework outlined in the previous chapter.

Chapter V: Conclusions and Recommendations

The final chapter summarizes the key findings of this research, directly linked to the problem statement and research objectives. The conclusions are synthesized, and recommendations are provided to strengthen early detection systems, adaptive defense strategies, and the integration of intelligence approaches into banking cyber risk management in Indonesia. These recommendations are directed at supporting national digital security policy more strategically and sustainably.

Chapter 2 LITERATURE REVIEW

2.1. Literature Study

This research draws on six book-length sources representing each of the four main concepts, namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR). For CTI, the book written by Lee (2022) presents a comprehensive account of the stages of the cyber threat intelligence cycle, from data collection through to the delivery of intelligence products. On the ASF concept, the reference by Tiller (2017) emphasizes the need for a security approach that is adaptive in nature, adjusting to continuously evolving threats in real time. This book also highlights the integration of methods such as the cyber kill chain into cyber defense systems that are responsive to the dynamics of digital attacks.

Meanwhile, the Strategic Intelligence framework is reinforced by three book references: Prunckun (2010), who outlines the stages of analysis in the strategic intelligence cycle to support evidence-based decision-making; Lindgren and Bandhold (2009), who describe the use of scenario planning as a strategic planning tool for confronting future uncertainty; and Saritas (2016), who introduces the systemic foresight methodology approach for identifying trends, drivers of change, and constructing roadmaps. For the CR concept, the book written by Linkov and Kott (2019) offers a comprehensive perspective on the importance of digital resilience, focusing not only on preventive aspects but also on an institution's ability to withstand and recover from cyber incidents. These six books play an important role in building the theoretical foundation of the research and strengthening the analysis of the strategic readiness of the banking sector in facing digital threats.

2.2. Prior Studies

In constructing the theoretical foundation and conceptual framework of this research, 58 scientific articles were analyzed and classified into four main groups, namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR). Of this total, the CTI concept is represented by 13 articles published between 2011 and 2025. These articles show the evolution of understanding of cyber threat intelligence, which was initially technical and reactive in

nature, developing toward a strategic approach that supports decision-making at the organizational level. The issues discussed include threat taxonomies, information-sharing mechanisms, collaborative platform architectures, and decision-making models based on technical and contextual indicators.

The ASF concept is the category with the largest literature contribution, namely 26 articles published between 2016 and 2024. A range of approaches is discussed in these publications, from the integration of machine learning and MAPE-K-loop-based architectural design, to dynamic risk-governance frameworks. This body of literature emphasizes the importance of building adaptive security systems capable of transforming responsively to the dynamics of the threat landscape, and of supporting automatic detection and mitigation processes. Thus, the accumulated body of study strengthens the analytical framework for evaluating the effectiveness of security-system adaptivity in the risk-laden and digitally complex banking sector.

2.2.1. Cyber Threat Intelligence

The table of prior studies related to the Cyber Threat Intelligence (CTI) concept summarizes 13 scientific articles published between 2011 and 2025. These publications consistently show that CTI plays a crucial role in shaping proactive cyber security systems. The CTI approach enables institutions to identify and detect patterns of digital attack early, to analyze the tactics and techniques used by threat actors, and to map threat indicators based on actual available data. In addition, the conceptual contribution of the reviewed literature reinforces the urgency of integrating CTI into managerial and operational decision-making frameworks, including the need for interoperability in information exchange and cross-institutional collaboration in sharing threat intelligence. Thus, CTI is positioned not merely as a technical detection tool, but also as a strategic instrument for understanding, analyzing, and anticipating attack patterns that can systemically threaten the national banking system.

Table 2.1. List of Prior Studies Related to the Cyber Threat Intelligence Concept

No	Source	Category	Research Objective	Methodology	Results and Conclusion
1	Tounsi, W.,	<i>Cyber Threat</i>	Reviews the	Systematic	CTI consists

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	& Rais, H. (2018). A survey on cyber threat intelligence: Definition, sources, and sharing mechanisms .	<i>Intelligence</i>	definitions, sources, and sharing mechanisms in Cyber Threat Intelligence (CTI) and provides a classification framework usable in operational security environments.	literature study of scientific publications and industry practice related to CTI, combined with conceptual analysis.	of four levels of intelligence (strategic, tactical, operational, technical) and requires support from automated systems and inter-institutional collaboration to share threat information efficiently.
2	Mavroeidis, V., & Bromander, S. (2017). Cyber threat intelligence model: An evaluation of taxonomies	<i>Cyber Threat Intelligence</i>	Evaluates the use of taxonomies and ontologies in constructing CTI models to support security-system	Evaluation of various CTI taxonomy and ontology models and analysis of practical implementation in	CTI interoperability requires standardization of ontologies and data structures so that it can be effectively

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	and ontologies.		interoperability.	security environment.	shared across platforms and security organizations.
3	Schlette, S. (2025). The anatomy of intelligence-led cyber defense: A strategic approach to threat anticipation.	<i>Cyber Threat Intelligence</i>	Describes elements of CTI as part of an intelligence-based defense strategy in the digital financial sector.	Case-study and conceptual approach based on a classic intelligence framework applied within cyber security systems.	CTI helps accelerate the detection of and response to attacks by integrating technical indicators with context-based decision-making systems.
4	Wagner, C., Dulaunoy, A., Iklody, A., & Wagener, G. (2019). MISP: The	<i>Cyber Threat Intelligence</i>	Develops and evaluates an open-source platform for sharing CTI between	Software system design and study of the deployment of the MISP platform in	Platforms such as MISP enable real-time collaboration, improving threat-

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	design and implementation of a collaborative threat intelligence sharing platform.		organizations.	national and international security organizations.	detection efficiency through the open exchange of technical data and attack indicators.
5	Cascavilla, A., Conti, M., & Dargahi, T. (2021). Cyber threat intelligence: A multi-vocal literature review.	<i>Cyber Threat Intelligence</i>	Compiles the results of academic studies and industry reports to form a comprehensive perspective on CTI practices and challenges.	Multi-vocal literature review of academic publications, industry white papers, and global threat reports.	CTI is not only defensive but also offensive in nature, through dark-web monitoring, cybercrime taxonomies, and the formation of contextual risk indicators.

No	Source	Category	Research Objective	Methodology	Results and Conclusion
6	Guarascio, M., Peres, D., & Costante, E. (2022). A collaborative CTI platform for detecting distributed cyber threats.	<i>Cyber Threat Intelligence</i>	Designs a distributed CTI platform to detect threats that spread across organizations collaboratively.	Design and evaluation of the ORISHA system as a CTI-based intrusion detection platform within a multi-institution environment.	CTI-based collaboration accelerates cross-organizational cyber-attack detection and improves the accuracy of digital threat analysis in the financial sector.

No	Source	Category	Research Objective	Methodology	Results and Conclusion
7	Goel, S., Hong, Y., & Papadopoulos, T. (2022). Bridging CTI and executive decision-making: An integrated framework.	<i>Cyber Threat Intelligence</i>	Examines the integration of CTI into managerial decision-making processes within financial organizations.	Case study of financial institutions and development of an integrative framework linking technical intelligence and strategic policy.	The gap between technical data and decision-making can be bridged through threat visualization and CTI-based strategic communication.
8	Al-Muhaisen, H., & Almuhaideb, A. (2023). Unified cyber threat intelligence framework for financial institutions.	<i>Cyber Threat Intelligence</i>	Constructs a unified CTI framework encompassing a threat knowledge base, automated detection system, and data visualization for financial	Design of the CTI framework with simulation and functional validation within the financial ecosystem.	This framework is effective in improving organizational readiness for attacks by combining intelligence insight with real-time decision-support

No	Source	Category	Research Objective	Methodology	Results and Conclusion
			institutions.		systems.
9	Ainslie, D., de Boer, P. T., & Scholten, H. (2023). Organizational adaptation to threat intelligence in cyber security.	<i>Cyber Threat Intelligence</i>	Evaluates the application of the classic intelligence cycle within the security policy of financial institutions.	Empirical study of institutional structures and processes for absorbing CTI information into security systems.	CTI needs to be internalized into an organization's security strategy and policy, not merely viewed as raw technical data.
10	Alaeifar, L., Tho, N., & Ramezani, M. (2024). Legal and ethical considerations in cyber threat intelligence	<i>Cyber Threat Intelligence</i>	Explains the legal and ethical principles in the CTI-sharing mechanism between financial institutions.	Study of international regulatory documents and analysis of industry practice in threat-data management.	The success of CTI depends on a strong legal framework and trust, as well as data protection and encryption in information

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	sharing.				exchange.
11	Cascavilla, A., Conti, M., & Dargahi, T. (2021). Cyber threat intelligence: A multi-vocal literature review.	<i>Cyber Threat Intelligence</i>	Comprehensively reviews the development of research and industry practice within the CTI domain.	Literature review with a multi-source approach, covering scientific articles and industry reports since the 2010s.	CTI plays an important role in shaping proactive and predictive security systems against cyber attacks.
12	Hutchins, Cloppert, & Amin (2011). Intelligence-Driven Computer Network Defense Informed by Analysis of Adversary	<i>Cyber Threat Intelligence</i>	Systematically maps the stages of a cyber attack to support intelligence-based defense.	Development of the conceptual 'Cyber Kill Chain' model based on real attack case studies.	There are seven attack stages that can be intervened for prevention. This model is relevant for analyzing real incidents and identifying points of

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	Campaigns and Intrusion Kill Chains				weakness in bank security.
13	NIST (2012). Guide for Conducting Risk Assessments (NIST SP 800-30 Revision 1)	<i>Cyber Threat Intelligence</i>	Provides a risk-analysis approach based on threat, vulnerability, and impact.	Formulation of the qualitative model Risk = Threat × Vulnerability × Impact for information-security risk management.	This model helps map and prioritize risk based on the state of threats, system weaknesses, and potential consequences. Used to strengthen risk analysis in the banking sector.

According to (Schlette, 2025), CTI is an effort to provide evidence-based information regarding various forms of cyber threats, equipped with relevant context to support mitigative decision-making. The main elements of CTI include technical indicators, standardized data structures, sharing platforms among stakeholders, and incident-handling protocols aligned with the intelligence cycle. In the context of digital financial systems, this approach strengthens situational awareness and accelerates the detection process for potential organized attacks.

As put forward by (Mavroeidis & Bromander, 2017), CTI contains in-depth information about the actors, methods, objectives, and techniques of an attack used, as well as the timing and scope of the cyber incident in question. Their study also underscores the challenges of system interoperability and the importance of using consistent ontologies to guarantee data integration across platforms. Its relevance to the banking world lies in the urgency of using open standards to share attack indicators, in order to promote a collectively oriented early-warning system.

Research by (Tounsi & Rais, 2018) contributes an important conceptual framework for understanding the dimensions of CTI based on its level, namely strategic, tactical, operational, and technical. Their main emphasis lies on technical intelligence, which requires automated, real-time mechanisms to manage short-lived information such as hashes, IP addresses, or malicious domains. In the banking sector, this approach is vital for confronting phishing attacks and banking malware that change rapidly.

A study by (Ainslie et al., 2023) highlights the importance of Cyber Threat Intelligence (CTI) in supporting adaptive security decision-making. CTI enables organizations to leverage the latest threat information to adjust their security configurations quickly and relevantly. For the banking sector in Indonesia, CTI integration—for example from the National Cyber and Crypto Agency (BSSN) or bodies such as the Information Sharing and Analysis Center (ISAC)—can improve the ability to predict and detect the constantly evolving *modus operandi* of attacks. Thus, an intelligence-based approach becomes one of the important foundations for building an effective adaptive security system.

The Cyber Kill Chain model developed by Lockheed Martin is one of the most widely used intelligence-based approaches for systematically analyzing cyber-attack patterns. This model describes the sequential stages traversed by an attacker, from the initial

information-gathering phase through to execution of the final objective, such as data theft or system takeover. Specifically, this framework consists of seven main phases: reconnaissance (surveillance of the target), weaponization (designing malware based on the vulnerability found), delivery (delivering the malware to the target), exploitation (leveraging the vulnerability to gain access), installation (planting malicious tools in the system), command and control (remote control of the system), and actions on objectives (execution of the attack's goal, such as data theft or sabotage) (Hutchins, Cloppert, & Amin, 2011).

Each phase in this model not only represents a technical step taken by an attacker, but also opens space for defense teams to detect and thwart the attack before it reaches a critical point. For example, if reconnaissance activity or malware delivery can be recognized earlier through certain technical indicators, then subsequent phases such as installation and control can be proactively prevented. Therefore, the Cyber Kill Chain functions not only as a description of an attack, but also as an anticipatory framework that enables the application of an adaptive, intelligence-based defense-in-depth.

In the context of this research, the Cyber Kill Chain model is adopted to systematically map actual cyber-attack incidents in the Indonesian banking sector. By mapping the stages of attacks that occurred in each real case onto the Cyber Kill Chain phases, the researcher can identify points of failure or lapses in a bank's defense system. This information is very important in answering the first research question, namely how the technical pattern of an attack unfolds and where its vulnerabilities lie. Furthermore, this approach makes an analytical contribution to the formulation of a more structured, data-based strategy for strengthening cyber security systems. In this regard, the Cyber Kill Chain reinforces the argument that a detailed understanding of an attacker's *modus operandi* is a crucial first step in building the cyber resilience of financial institutions.

One approach commonly used to analyze information-security risk is the quantitative-conceptual model formulated as $\text{Risk} = \text{Threat} \times \text{Vulnerability} \times \text{Impact}$, as introduced by the National Institute of Standards and Technology (NIST, 2012). This formula explicitly asserts that the level of risk faced by an organization is determined by three interrelated main components: threat, vulnerability, and impact. Each factor makes a significant contribution to shaping the overall magnitude of risk.

A threat refers to the potential occurrence of an event that can cause loss to a system or digital asset. In the context of the banking sector, threats can arise from various sources, such as malicious actors (hackers, cyber criminals, or deviant insiders), systemic disruptions, natural disasters, and technical damage. Vulnerability, on the other hand, reflects the degree of weakness in an organization's security system — whether from unpatched software, weaknesses in system configuration, or human factors such as low awareness and security training among bank employees. Impact, meanwhile, refers to the negative consequences that arise when a threat successfully exploits a vulnerability, including potential financial loss, breaches of personal data, damage to institutional reputation, disruption of critical services, and regulatory sanctions.

This model is commonly used in qualitative risk-assessment approaches, in which risk is considered high when all three components are equally at a high level. Conversely, a reduction in any one component — for example by closing an existing vulnerability — can significantly lower total risk. In the context of this research, this formula is used as a conceptual tool for structuring the analysis of cyber-security vulnerabilities in the banking sector, particularly in identifying priorities for system strengthening.

Applying the Threat–Vulnerability–Impact framework enables the researcher to conduct a systematic evaluation through three stages. First, mapping specific threats relevant to the banking ecosystem, such as trends in financial malware, spear phishing targeting bank staff, or the activity of APT groups. Second, an in-depth analysis of internal vulnerabilities, whether from the information-technology architecture, inconsistent security policies, or human and internal-process weaknesses. Third, an assessment of potential impact, such as estimated losses due to a breach of the payment system or disruption to interbank clearing services.

By integrating these three components, the research can produce the most critical risk priorities to be handled strategically. This approach also enables the formulation of more precise system-strengthening recommendations — whether through reducing exposure to threats (for example through improved cyber intelligence and cooperation with law enforcement), closing vulnerability gaps (through technology and training), or mitigating impact (such as implementing contingency plans and security insurance). Thus, this model helps ensure that all aspects of risk management are not treated separately, but considered comprehensively as an integrated system. In this study, that framework is used

to strengthen the answer to the second research question, namely how the level of vulnerability of the banking system can be analyzed and addressed through a structured, risk-based intelligence approach.

Further, (Wagner et al., 2019) examines the collaborative aspect of CTI management, emphasizing the need for a framework that enables real-time exchange of threat information between organizations. This study reflects the importance of collaboration, as exemplified by FS-ISAC within the global financial ecosystem. For national banking institutions, this approach provides justification for forming a joint CTI forum as a collective response to the escalation of cyber threats.

Meanwhile, (Cascavilla et al., 2021) broadens the CTI perspective by combining the results of academic studies and industry reports, through a multi-vocal literature-review approach. They construct a taxonomy of cybercrime activity and emphasize the importance of monitoring the dark web and malware repositories as an intelligence source that can help banks map emerging risks. This approach affirms that CTI is not only defensive but also offensive in the context of early detection based on the digital-crime context.

A significant contribution is also given by (Guarascio et al., 2022) through the design of the ORISHA system, a collaborative platform for detecting intrusions based on distributed CTI. This study demonstrates that interconnection between cross-organizational monitoring systems can improve the accuracy and speed of cyber-attack detection. Its implication for the banking sector is the importance of developing a mutually integrated cyber-monitoring ecosystem, in order to form an intelligence-based collective defense.

On the strategic side, (Goel et al., 2022) emphasizes that CTI does not rest solely on technical data, but must also become a foundation in managerial decision-making and information-security policy. They show that the gap between technical data and policy is often an obstacle to responding to cyber incidents holistically. In the context of banking institutions, using CTI as an input for strategic decision-making becomes a necessity for creating organizational resilience.

An integrated framework model was developed by (Al-Muhaisen & Almuhaideb, 2023), who structure CTI into three main components: a threat knowledge base, a

detection system, and data visualization. This structure is considered capable of providing a complete picture of the threat situation, supporting rapid data validation, and encouraging evidence-based decision-making in the financial sector. This framework can serve as a reference for building an internal cyber-intelligence center at banking institutions.

A study by (Ainslie et al., 2023) reveals a gap in current CTI practice, namely the dominance of technical approaches that are poorly aligned with organizational policy planning. They propose that the classic intelligence cycle needs to be re-adapted into the institutional framework, so that CTI can be internalized into an institution's strategy and policy. Within the framework of this thesis, this approach reinforces the urgency of viewing CTI not merely as a technical tool, but also as a long-term risk-management support system.

According to (Alaeifar et al., 2024), it is important to build a CTI-sharing mechanism that is not only technically effective, but also compliant with legal, ethical, and regulatory principles. This study suggests the development of a threat-data-sharing system that is private, encrypted, and trust-based. For national banking institutions, this study provides a conceptual foothold for forming a collaborative CTI network among banks, regulators, and other policymakers.

2.2.2. Strategic Intelligence

The table of prior studies on the Strategic Intelligence (SI) concept consists of nine scientific articles published between 2016 and 2025. All of these references make an important contribution to understanding the role of strategic intelligence in supporting long-term planning processes, scenario formulation, and risk-based decision-making in the context of digital security. These articles emphasize that SI does not merely present raw information, but filters, processes, and interprets data to generate predictive insight usable by the policymakers of financial organizations. In the context of this research, SI becomes a crucial element in strengthening the function of early detection of escalating digital threats through a scenario-based approach and system-vulnerability analysis, thereby supporting the resilience of national banking institutions against systemic and sustained cyber attacks.

No	Source	Category	Research	Methodolog	Results and
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Table 2.2. List of Prior Studies Related to the Strategic Intelligence Concept

In the Handbook of Intelligence Analysis, Hank Prunckun offers a fundamental understanding of intelligence as a discipline with two main sides, namely as a form of knowledge and as a set of systematic activities. Intelligence as knowledge refers to the result of information processing that then forms a strategic understanding of a particular entity, event, or situation. This knowledge is structured and in-depth in nature, and aims to support decision-making at various levels, especially at the strategic level (Prunckun, 2010).

On the other hand, intelligence as activity comprises a work cycle consisting of the stages of planning, data collection, processing, analysis, and the delivery of information to stakeholders. It is through this mechanism that intelligence can present relevant and timely information, whether for tactical, operational, or strategic needs. At the strategic level, intelligence is aimed at providing a projective picture of the future, including identifying potential threats and vulnerabilities, and providing an early-warning system that can strengthen the resilience of an institution's policy and security (Prunckun, 2010).

Prunckun's conceptual contribution becomes an important theoretical foundation for formulating intelligence approaches in the public-policy sector, including in cyber security and banking. In the context of this research, Prunckun's thinking is used to answer the third research question relating to intelligence strategy and policy innovation in the banking sector. By adhering to this intellectual framework, banking intelligence strategy can be built methodologically and systematically, so that it is able to proactively address security challenges and strengthen decision-making processes based on data and intelligence analysis.

Foresight, in the context of this research, is used as a supporting, strategic framework, not as the main concept. The foresight approach is positioned as a systematic, scenario-based method of thinking aimed at designing anticipatory policy for confronting the uncertainty and complexity of the future (Lindgren & Bandhold, 2009). Its relevance grows all the more important given the dynamic, cross-sectoral character of cyber threats, which cannot always be effectively addressed through a conventional, reactive approach.

Therefore, foresight plays a role in strengthening the function of Strategic Intelligence (SI), particularly in formulating adaptive and prospective cyber-security policy.

The foresight model referenced in this research adopts the Systemic Foresight Methodology framework (Saritas, 2016), which consists of five main stages: event, trend, driver, scenario planning, and road map. All of these stages are applied in an integrated manner in order to produce a comprehensive mapping of the dynamics of cyber threats in the national banking sector. The initial stage involves identifying real events and trends of increasing attacks (event and trend), followed by an analysis of the driving factors influencing the resilience of the system (drivers). Based on the results of this exploration, a scenario quadrant is compiled as a tool for mapping an anticipatory policy direction (scenario planning). The final stage, in the form of a road map, is formulated as a form of medium- to long-term early-prevention strategy recommendation, in line with the principles of strategic-intelligence-based decision-making (Walsh, 2017).

Thus, foresight in this research does not only serve as an additional analytical tool, but also as a predictive instrument that supports the strengthening of institutional resilience. This approach enables the formulation of a more adaptive cyber-security policy, not only in response to actual conditions, but also to various possible disruptive threats in the future.

Strategic intelligence is part of an intelligence system that functions to provide a comprehensive understanding of threats, trends, and the dynamics of the strategic environment that can affect an organization's long-term sustainability. In the context of banking cyber security, this approach helps financial institutions anticipate digital threats, build an adaptive risk-mitigation framework, and formulate policy based on strategic projections.

According to (Tounsi & Rais, 2018), strategic intelligence differs from a technical approach because it places greater emphasis on the analysis of macro tendencies, such as changes in attack patterns or shifts in threat-actor motivation, which are then used as a basis for organizational decision-making. This knowledge enables management not only to react to incidents, but also to design more comprehensive and visionary long-term policy.

In another study, (Gentry & Gordon, 2019) note that strategic intelligence functions as a communication channel between intelligence analysts and senior decision-makers, in

which an intelligence report is not only informative in nature, but also contains alternative scenarios, risk analysis, and applicable policy recommendations. This function is highly relevant in the banking sector, which is very vulnerable to operational disruption caused by large-scale cyber attacks.

A study by (Gentry, 2023) underscores the importance of strategic early warning in the cyber domain. Although the technical challenges and dynamics of cyber actors are highly complex, an intelligence approach can still be used to identify the early signals of an attack. This is possible if an institution is able to develop a data-collection system integrated with trend analysis and anomaly indicators, so that mitigation action can be taken proactively.

Meanwhile, (Zhuang et al., 2021) developed a strategic cyber-warning model based on four components: vulnerability, threat, intrusion, and anomaly detection. This model emphasizes the importance of inter-unit collaboration, parallel monitoring of attack vectors, and the integration of data from various sources as the keys to the success of an early-detection system. In the digital-bank environment, this approach will strengthen predictive capability for anticipating the escalation of security incidents.

Regarding technological development, (Lim, 2016) argues that big-data analytics capability is one of the main drivers for strengthening strategic intelligence. By analyzing very large volumes of data, an organization can identify hidden patterns that serve as early markers of a threat. However, the author also emphasizes that data interpretation must still take strategic context into account so that the results of the analysis do not lead to misleading conclusions.

(Trim & Lee's research, 2022) proposes an integrative approach between artificial-intelligence (AI)-based threat modeling and understanding of the socio-cultural dimension of the attacker. Synergy between the algorithmic approach and human intelligence enables improved institutional resilience against various forms of cyber attack. Within the banking environment, this strategy will provide strong support for increasing the speed and precision of intelligence-based decision-making.

The contribution of (Walsh, 2017) lies in emphasizing the importance of governance in strategic-intelligence analysis through a scientific approach. He suggests that an intelligence product should not be based solely on the assumptions or intuition of the analyst, but should also be equipped with rigorous qualitative or quantitative

methodology, in order to improve validity and reliability as a basis for strategic recommendations to policymakers.

In an empirical review, (Jebril et al., 2020) conclude that mastery of strategic intelligence contributes significantly to an organization's competitive advantage, especially when integrated with risk-management and information-security policy. In the banking sector, the success of this integration will directly affect institutional reputation and customer trust, as well as operational stability amid the rising tide of digital threats.

A study by (Al-Muhaisen & Almuhaideb, 2023) formulates a framework that combines technical threat intelligence with strategic analysis within a single integrated system. They divide the CTI structure into three layers: a threat database, an automated detection system, and a visual display based on risk metrics. This model is highly relevant for supporting strategic decision-making based on actual, real-time data.

Lastly, (Aziz & Daryanto, 2025) present a case study on the use of strategic intelligence to detect cybercriminal activity through the mapping of digital communication on the dark web. Their findings show that the use of intelligence-based observation methods successfully identified threat actors before an attack occurred, and produced effective cross-institutional coordination in risk mitigation.

2.2.3. Adaptive Security Framework

The table of prior studies discussing the Adaptive Security Framework (ASF) concept summarizes 26 scientific articles published between 2016 and 2024. These publications generally reflect increasing global attention to the need for a dynamic and responsive cyber-security system, particularly in confronting ever-evolving attacks. This review covers the development of security architecture based on risk-based adaptation, the application of the MAPE-K-loop principle, the integration of machine learning for early detection, and the formulation of security policy adaptive to actual risk context. In the context of this research, the existence of ASF reinforces the argument for the need to transform the banking institution's defense approach from a reactive to a proactive model, through strengthening a systemically integrated predict-prevent-detect-respond mechanism. ASF also serves as a methodological foundation for designing a security system capable of adapting to changes in threats, technological dynamics, and institutional-governance needs in the financial sector.

Table 2.3. List of Prior Studies Related to the Adaptive Security Framework Concept

No	Source	Category	Research Objective	Methodology	Results and Conclusion
1	Kure et al. (2022). Integrating risk management and threat intelligence for adaptive cybersecurity	<i>Adaptive Security Framework</i>	Proposes integrating risk management and threat intelligence to create a security system that is adaptive in responding to the dynamics of cyber attacks.	Theoretical study with a qualitative approach and literature review related to risk management and threat intelligence.	The integration of risk and threat intelligence can form an efficient adaptive cyber-defense system for anticipating and responding to attacks.
2	Melaku (2023). Simplifying adaptive cybersecurity governance for financial institutions	<i>Adaptive Security Framework</i>	Offers a simple and adaptive cyber-security governance model based on continuous risk updates, particularly in the financial sector.	Conceptual analysis and institutional policy study on the application of adaptivity within cyber-security systems.	A simple, adaptive governance model is effective in responding to cyber threats and strengthening an institution's defense capability.
3	Calvo & Beltrán (2022).	<i>Adaptive Security</i>	Designs an adaptive,	System-architecture	The MAPE-K system is

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	Adaptive MAPE-K based security architecture in banking	<i>Framework</i>	MAPE-K-based security architecture capable of adjusting security controls in real time based on risk evaluation.	design based on MAPE-K principles; simulation testing on digital banking infrastructure.	able to improve system responsiveness to continuously changing threats through risk-evaluation-based adaptation.
4	Cherif et al. (2022). Adaptive multi-factor authentication for fraud prevention in banking	<i>Adaptive Security Framework</i>	Develops an adaptive multi-factor authentication system that adjusts the level of verification based on the user's profile and transaction context.	Design of the authentication system and evaluation of the adaptive algorithm based on transaction context and risk profile.	The adaptive authentication system has proven to reduce fraud risk by contextually strengthening the layers of security.
5	Caliaberah et al. (2020). Open-source	<i>Adaptive Security Framework</i>	Creates an open-source-software-	Development of open-source	Open-source-based early

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	early detection system for ransomware in banking		based ransomware-detection system capable of recognizing suspicious behavior before data encryption occurs.	software system; simulative testing against ransomware-attack scenarios.	detection is able to prevent data encryption by ransomware and strengthen early defense.
6	Aslam et al. (2022). Machine learning-based DDoS mitigation in SDN environments	<i>Adaptive Security Framework</i>	Builds an adaptive DDoS-attack-mitigation model using a machine-learning approach within SDN network infrastructure.	System-mitigation modeling and simulation within an SDN network; evaluation of the model's adaptivity performance.	The adaptive DDoS-mitigation model improves the effectiveness of protecting digital banking services against high-capacity attacks.
7	N'Douba (2024). Adaptive authentication	<i>Adaptive Security Framework</i>	Applies NIST guidance for adaptive authentication	Design of an adaptive authentication	The application of adaptive authentication

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	using NIST AAL in banking security systems		in protecting banking data, particularly in confronting unauthorized access.	framework using NIST's AAL parameters and scenario-based risk system testing.	n improves the protection of sensitive data by resetting security levels based on suspicious activity.
8	Asmar & Tuqan (2024). Continuous learning systems for adaptive cybersecurity	<i>Adaptive Security Framework</i>	Shows the importance of a security system capable of continuous learning in improving resilience against new threats.	Literature review and machine-learning simulation in developing an automatic detection-and-response system.	An adaptive learning system enables an automatic response to new threats without excessive manual intervention.
9	Santiago & Nery (2023). Counterintelligence integration in	<i>Adaptive Security Framework</i>	Integrates a counter-intelligence strategy into an adaptive	Integrative analysis of counter-intelligence and cyber-	Counter-intelligence integration strengthens predictive

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	adaptive cyber defense		cyber-defense framework to improve proactive detection and prevention.	security literature; formulation of an adaptive theoretical framework.	capability and early prevention within an adaptive cyber-security system.
10	Kelsey (2016). Gartner's continuous compromise model for adaptive security	<i>Adaptive Security Framework</i>	Introduces Gartner's continuous-compromise model to underpin an adaptive, ongoing cyber-security approach.	Conceptual review of the Gartner report; study of the model's application in financial-institution security systems.	The continuous-compromise model is effective as a basis for designing a layered, flexible defense within the digital banking environment.
11	A Context-Aware Adaptive Security Model Based on	<i>Adaptive Security Framework</i>	Analyzes an adaptive security approach based on the	System design based on the MAPE-K cycle;	The MAPE-K cycle enables the automatic adjustment

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	MAPE-K for Digital Environments (Lara et al., 2019)		MAPE-K cycle and context-awareness for digital organizations.	evaluation of adaptation to the organizational environment context.	of a system to a dynamic digital environment.
12	Risk-based Adaptive Security Model for Cybersecurity (Calvo & Beltrán, 2022)	<i>Adaptive Security Framework</i>	Applies the Risk-based Adaptive Security (RiAS) model to adjust security controls based on real-time risk.	Design of a three-layer RiAS architecture with automated, risk-context-based decision-making.	RiAS is able to improve security effectiveness based on actual risk context.
13	Towards Adaptive Cybersecurity Governance in Financial Institutions (Melaku, 2023)	<i>Adaptive Security Framework</i>	Examines a simple yet responsive adaptive-security governance model for financial-sector	Theoretical study and conceptual design of a risk-management strategy and adaptive	Simple, risk-oriented policy adaptivity is very important for the sustainability of a bank's

No	Source	Category	Research Objective	Methodology	Results and Conclusion
			organizations.	policy.	cyber security.
14	Machine Learning-Driven Adaptive Response to Cyber Threats (Asmar et al., 2024)	<i>Adaptive Security Framework</i>	Develops a machine-learning algorithm to strengthen an adaptive response to new attack patterns.	Development of a machine-learning-based system and evaluation of the adaptive response.	Machine learning independently and efficiently strengthens the system against new threats.
15	CTI-Supported Adaptive Cybersecurity Decisions in Financial Sector (Ainslie et al., 2023)	<i>Adaptive Security Framework</i>	Assesses the integration of Cyber Threat Intelligence in supporting adaptive security decisions quickly and accurately.	Policy analysis and CTI integration through case studies and a security-adaptation framework.	CTI plays an important role in adjusting security systems to current threats.
16	Operational Resilience and Adaptive Security in	<i>Adaptive Security Framework</i>	Examines the assume-breach principle and	Policy-document analysis and	The assume-breach principle becomes an

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	Global Finance (Crisanto et al., 2023)		operational resilience as a reference for adaptive security policy in the global financial sector.	comparative study of the application of global resilience principles.	important foundation for adaptive security policy at both global and local banks.
17	Cybersecurity Governance Challenges in Indonesian Banks (Sujono & Waluyo, 2023)	<i>Adaptive Security Framework</i>	Identifies the obstacles to cyber-security implementation in Indonesian banking and suggests adaptive solutions based on technology and governance.	National survey and evaluation of technical and institutional obstacles in bank-security implementation.	Bank Indonesia still experiences a significant gap in infrastructure and human resources for adaptive security.
18	AI-Based Monitoring Systems for Dynamic Banking Threats (Raharjo &	<i>Adaptive Security Framework</i>	Analyzes banks' need for an AI-based real-time monitoring system in	Expert interviews and needs analysis for an adaptive monitoring system in	A real-time monitoring system is highly needed and becomes the foundation

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	Ismail, 2023)		confronting dynamic digital threats.	the banking sector.	for strengthening early detection.
19	Designing Adaptive Security Systems with Threat-Aware Detection (Kurniawan & Sari, 2022)	<i>Adaptive Security Framework</i>	Proposes a security system based on early detection and parameters adaptive to the level of cyber threat.	Development of an early-detection framework using a threat-aware-system approach.	A threat-aware system is able to adaptively and in real time improve the efficiency of a bank's protection.
20	Evaluating Adaptive Organizational Cybersecurity Structure (Nugroho, 2021)	<i>Adaptive Security Framework</i>	Evaluates the readiness of a bank's cybersecurity organizational structure to respond to incidents with an adaptive system.	Evaluation of the organizational structure and case study of bank incident-response speed.	A responsive security-organization structure accelerates the mitigation of digital incidents.
21	A Maturity Model for	<i>Adaptive</i>	Develops a cyber-security	Construction of a	Management support

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	Adaptive Cybersecurity in Banking Sector (Fitriani & Astari, 2024)	<i>Security Framework</i>	maturity-evaluation model based on management support for adaptivity.	maturity model and empirical validation of an institution's actual condition.	greatly affects the effectiveness of implementing cyber-security adaptation.
22	Integrating SIEM and CTI for Adaptive Incident Detection (Pranata et al., 2023)	<i>Adaptive Security Framework</i>	Examines the effectiveness of integrating SIEM and threat intelligence in accelerating adaptive incident detection and response.	Experiment on a SIEM system and measurement of digital-incident response time.	The integration of SIEM and threat intelligence improves the efficiency of incident detection and mitigation.
23	Risk Threshold-Based Security Architecture in Core Banking (Hasibuan, 2020)	<i>Adaptive Security Framework</i>	Designs a security architecture based on a transaction-risk threshold for core-	Technical architectural design and simulation of the efficiency of	A risk-threshold-based model can reduce fraud and minimize data leakage.

No	Source	Category	Research Objective	Methodology	Results and Conclusion
			banking systems.	transaction-based risk regulation.	
24	AIDS: Adaptive Intrusion Detection System for Banking (Stiawan et al., 2021)	<i>Adaptive Security Framework</i>	Develops a machine-learning-based adaptive intrusion-detection system for bank transaction data.	Testing of the AIDS model using an Indonesian digital banking transaction dataset.	AIDS shows high accuracy in detecting new attacks and improving system resilience.
25	Adaptive Cyber Alliance Framework for National Finance (Alimudin & Handayani, 2023)	<i>Adaptive Security Framework</i>	Proposes an adaptive-cyber-alliance framework to strengthen response coordination between financial institutions and regulators.	Construction of an adaptive collaboration framework among national financial-system entities.	Adaptive collaboration between institutions is important for repelling large-scale attacks and aligning a national response.
26	NIST (2018). Framework for Improving Critical Infrastructure	<i>Adaptive Security Framework</i>	Constructs a cyber-security-risk-management framework	Development of the conceptual 'Cyber Kill Chain'	There are seven attack stages that can be intervened

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	Cybersecurity		based on five main functions.	model based on real attack case studies.	for prevention. This model is relevant for analyzing real incidents and identifying points of weakness in bank security.

The massive digital transformation within the banking system has led to increased exposure to continuously developing cyber threats. In responding to these dynamics, the Adaptive Security Framework (ASF) approach becomes highly relevant because it provides a flexible, responsive defense mechanism capable of adjusting its security strategy in real time to constantly changing threats.

According to (Kure et al., 2022), conventional security models tend to be static and unable to respond quickly to complex, continuously evolving cyber-attack patterns. They propose integrating risk management and threat intelligence to form a security system capable of adaptively detecting, analyzing, and responding to changing situations. This concept becomes an important theoretical basis for this research because it aligns with the intelligence-based approach used to analyze threats to the banking sector.

Further, (Melaku, 2023) argues that the complexity of the cyber-security framework in many organizations actually hinders implementation effectiveness. He suggests a simple yet adaptive cyber-security governance model, based on continuously updated risk assessment, so that institutions can adjust their policies and security mechanisms to changes in threat and technology conditions. This framework is relevant to the context of

this research because it offers an agile structure for banks to comprehensively strengthen their cyber-defense capability.

From a technical-architecture perspective, (Calvo & Beltrán, 2022) designed an adaptive model based on the MAPE-K principle (Monitor, Analyze, Plan, Execute over a Knowledge base), which enables a security system to adjust control configuration based on continuously ongoing risk evaluation. This system allows for the automatic strengthening of security controls when an increase in risk is detected, so that responses to threats such as phishing or system infiltration can be carried out more effectively.

In the realm of fraud prevention, (Cherif et al., 2022) developed a system that combines adaptive multi-factor authentication with a machine-learning-based anomaly-detection algorithm. This system automatically adjusts the level of verification based on the user's profile and transaction context. This is important in the banking sector, where attacks that exploit social engineering such as phishing can be prevented by automatically stepping up the authentication process when suspicious activity is detected.

Research by (Caliabera et al., 2020) focuses on countering ransomware in the banking environment. They built an early-detection system based on open-source software capable of identifying ransomware behavior in its early phase before data encryption occurs. This approach shows that an adaptive security architecture can function as an early-warning system, which is very important for protecting sensitive financial data from irreparable damage.

On the side of protecting service availability, (Aslam et al., 2022) developed an adaptive-machine-learning-based DDoS mitigation system within a Software Defined Networking (SDN) environment. This model automatically updates network-traffic rules based on traffic-pattern analysis, so that it is able to minimize the impact of DDoS attacks on online banking services.

From the perspective of access control and data-breach prevention, (N'Douba, 2024) uses NIST guidance on Authenticator Assurance Level (AAL) to apply adaptive authentication. When the system detects an anomaly, the level of authentication is automatically increased to protect sensitive information. In a banking context, this mechanism strengthens the protection of user credentials and prevents unauthorized access to accounts or systems.

An approach emphasizing continuous learning is stressed by (Asmar & Tuqan, 2024), who describe that machine-learning-based security systems have the potential to become increasingly adaptive over time. With the ability to continuously update its detection model based on the latest attack patterns, this system is able to improve an institution's resilience against dynamic and complex threats.

Complementing these approaches, (Santiago & Nery, 2023) highlight the importance of integrating a counter-intelligence strategy into an adaptive security system. They emphasize that future cyber-defense systems must not only be responsive, but also have predictive capability to identify and prevent attacks before they occur. In this research, that view reinforces the argument that adaptive defense underpinned by strategic intelligence is a primary foundation for cyber resilience in the banking sector.

The Adaptive Security Framework emphasizes the importance of a dynamic cyber-security approach capable of adapting to constantly changing threat patterns. Unlike a conventional approach, which is generally static and relies only on an initial configuration, the adaptive framework assumes that the possibility of a security breach is a certainty, so that every organization needs to treat its system as being in a state of continuous compromise (Gartner, 2016, in Kelsey, 2016). With this "continuous compromise" line of thinking as a basis, banking institutions are required to build a cyber-security system that is not only structurally strong, but also flexible and responsive to every change in situation.

The implication of this approach is that a bank cannot simply apply an initial-prevention mechanism, but must also have the ability to detect threats in real time, respond to incidents quickly, and consistently adjust security policy and configuration based on the latest risk evaluation. In this regard, the adaptive cycle proposed by Gartner — namely "predict – prevent – detect – respond" — provides a comprehensive approach for confronting advanced cyber attacks (Kelsey, 2016). This cycle ensures that if one layer of defense fails to prevent an attack, the next layer can immediately take over, while the system continues to learn from every event as a form of continuous improvement.

In the banking context, the main function of an adaptive security framework is to support system resilience, namely the ability to keep delivering service to customers even while facing an attack, while simultaneously adjusting the defense mechanism to be stronger going forward (Crisanto et al., 2023). This framework requires the integration of

various strategic capabilities such as threat-intelligence-based prediction, preventive efforts through security controls, detection capability through system monitoring and intrusion detection, and rapid response to incidents that occur (Calvo & Beltrán, 2022). This functionality is vital in the financial industry, given that the banking sector is one of the primary targets of cyber attacks that can cause large financial losses and damage public trust and broad economic stability.

Therefore, the adaptive approach provides protection that is not only reactive but also proactive and layered, for example through periodically updating firewall rules based on the latest threat patterns or by adjusting the level of user authentication to a particular transaction's risk profile (Calvo & Beltrán, 2022). In general, an adaptive security system acts like a living entity capable of adjusting its security policy and configuration to the risk conditions currently underway.

In addition, this framework also promotes the use of analytics technology, threat intelligence, and machine learning in sharpening threat detection. In digital banking systems, where the volume of transaction data is very large, applying this technology enables the early detection of anomalies or new attack patterns. The Adaptive Security Framework provides a continuous feedback mechanism, in which every incident or near-miss is used as material for evaluation to refine the existing detection system (Ainslie et al., 2023). Adaptivity in security policy also means that the system can contextually increase sensitivity or even restrict access based on the escalation of global threats or suspicious behavior detected on a customer's account (Lara et al., 2019).

With all of these mechanisms combined, an adaptive security system plays an important role in strengthening the cyber resilience of the banking sector — not merely preventing attacks from occurring, but also ensuring that the organization is able to respond effectively, minimize impact, and grow stronger from every disruption it experiences.

The Adaptive Security Framework concept has undergone significant development alongside the increasing complexity and sophistication of cyber threats. This review raises a number of key pieces of literature from experts considered to represent the state of the art of the adaptive security framework and relevant for application in Indonesia's banking sector.

First, the report from (Gartner, 2016, in Kelsey, 2016) is one of the early milestones that systematically introduced adaptive security architecture. That report emphasizes the importance of adopting the assumption that an organization's system has already been, or will soon be, compromised, so that the response to threats must be continuous and proactive in nature. This thinking is used as a theoretical basis in this research because it drives a paradigm shift from a reactive approach toward a continuously adaptive approach, highly relevant for the dynamic banking sector.

Next, a study by (Lara et al., 2019) makes an important contribution by examining an adaptive security approach based on the MAPE-K cycle (Monitor–Analyze–Plan–Execute–Knowledge). They emphasize the importance of context-awareness and the system's ability to automatically adjust to a changing environment. This reference is chosen because it offers a strong conceptual basis that cyber security must be built upon an autonomous feedback mechanism, enabling a system to independently evaluate and adjust its security policy.

The Risk-based Adaptive Security (RiAS) model developed by (Calvo & Beltrán, 2022) is a concrete example of implementing a risk-based adaptive approach. RiAS is designed with a three-layer architecture — namely measurement, decision-making, and adaptation — which enables the system to monitor an asset's context in real time, calculate the actual level of risk, and then automatically adjust security controls. This approach shows that a security system need not be uniform, but can be adjusted to a specific and changing risk scenario. This reference is considered representative because it demonstrates a real form of implementation highly relevant to the banking sector, which faces rapidly shifting risk.

The National Institute of Standards and Technology (NIST) developed an integrated framework known as the NIST Cybersecurity Framework (CSF), designed to help organizations systematically manage and reduce cyber-security risk. This framework encompasses a set of internationally recognized standards, guidelines, and best practices that can be widely applied, including in sectors classified as critical infrastructure such as banking (NIST, 2018).

The CSF divides the cyber-security-management approach into five core functions, namely: Identify, Protect, Detect, Respond, and Recover. The Identify function focuses on mapping the institutional context, identifying key assets, and analyzing vulnerability

and risk; Protect covers the implementation of protective measures to keep important systems and services protected; Detect refers to an organization's ability to recognize security incidents quickly and accurately; Respond contains incident-handling procedures to limit impact; and Recover covers the restoration of business functions and the strengthening of a system's resilience post-incident (NIST, 2018).

In addition to these five core functions, the NIST CSF also organizes various control categories and sub-categories as technical and policy guidance for compiling an information-security plan appropriate to the organization's level of risk. This framework is flexible and can be adapted for various types of institutions, including banking, which face increasingly complex digital-security challenges.

In the context of this research, the NIST CSF is used as an analytical tool for evaluating the impact of cyber attacks and identifying the need to strengthen the national banking system's security. This framework supports the first and second research questions, namely regarding the effectiveness of the security system during an incident and the strategy for strengthening the system going forward. Through the NIST CSF approach, the researcher can assess the extent to which the existing cyber-security system meets global best-practice standards, and identify gaps in specific functions that have not been optimal. For example, if an incident reveals a delay in detection (the Detect function) or unpreparedness in responding (the Respond function), then the CSF provides a systematic basis for evaluating and improving that aspect.

Thus, the application of the NIST CSF in this study not only provides a methodological basis for assessing the readiness and weaknesses of the cyber-security system in the banking sector, but also serves as a foothold for compiling policy recommendations based on a structured, operational-resilience-oriented framework.

The thinking of (Melaku, 2023) also extends the scope of adaptivity into the aspect of organizational governance. In the framework developed, adaptivity is not merely a matter of technology or systems, but must also become part of an organization's overall strategy and policy. This framework emphasizes the importance of a flexible strategic security direction, continuous risk management, and optimal use of security resources. This study is relevant to the Indonesian context, because it shows that adaptive governance is needed so that banks can transform in the face of continuously developing threats.

The role of smart technology in adaptive security is explained in depth by (Asmar et al., 2024), who show how the application of machine-learning algorithms such as anomaly detection and user-behavior analytics can improve a system's responsiveness to new attack patterns. This study is considered important because it offers an innovative solution, namely enabling a security system to learn automatically from digital transaction data, so that an adaptive response can be carried out without direct human intervention. This is very beneficial for financial institutions in Indonesia, which face limited human resources and an increasing volume and variety of cyber attacks.

Furthermore, research by (Crisanto et al., 2023) conducted under the Bank for International Settlements (BIS) shows that second-generation cyber-security regulation now adopts the assume-breach principle and focuses on operational resilience. This finding indicates that global financial regulators are beginning to push banks not only to strengthen prevention, but also to improve incident-detection and response capacity. In the Indonesian context, a similar policy direction is beginning to be adopted by the Financial Services Authority (OJK) and Bank Indonesia, through strengthening incident-management provisions and periodic-mitigation-testing requirements. This reference reinforces the position that the adaptive approach is not merely a scientific concept, but has become a policy standard within the international banking sector.

A domestic study by (Sujono & Waluyo, 2023) shows that around 60% of banks in Indonesia still face challenges in meeting established cyber-security standards. Limitations in technological infrastructure and a shortage of expert personnel are the main obstacles to implementing effective security policy. This finding confirms a gap between theory and practice in the field. Therefore, the thinking of the experts reviewed in this section becomes highly relevant as a source of solutions. Adaptive approaches — whether based on automation, artificial intelligence, or risk-based governance — can serve as strategic alternatives for addressing resource constraints and improving the efficiency of security investment.

Overall, the thinking of experts reviewed over the 2021–2024 period shows a significant paradigm shift in cyber security, particularly in the banking sector. There is a consensus that a static approach is no longer adequate for confronting the dynamics of modern threats. Instead, a security system must be adaptive, agile, and based on data and intelligence. The references used in this review not only reflect the state of the art in the

field of adaptive security, but also provide a strong theoretical foundation for analyzing and formulating solutions appropriate to conditions in Indonesian banking.

The application of an adaptive security approach becomes increasingly urgent when linked to a number of cyber-attack incidents that have targeted the Indonesian banking sector in recent years. One notable incident occurred in December 2021, when Bank Indonesia (BI) became the target of a ransomware attack. In that event, the hacker group Conti reportedly succeeded in infiltrating BI's system and encrypting a number of data (Nangoy et al., 2022). Even so, BI reported that mitigation steps had been taken such that public services continued to run normally and the recovery process could proceed promptly. BI's success in countering the impact of this attack indicates the importance of detection and response capacity — two main elements emphasized in the adaptive security framework.

Assuming that the system had already been, or was in the process of being, infiltrated, BI appears to have developed an incident-response structure and recovery scenarios that could be activated quickly. This approach aligns with the principle advocated by (Kelsey, 2016) and supported by the view of international regulators (Crisanto et al., 2023), namely the need to shift from post-incident response toward continuous and proactive handling before an attack spreads.

A more critical situation occurred in May 2023, when Bank Syariah Indonesia (BSI) experienced a system disruption lasting several days due to a LockBit 3.0 ransomware attack. The perpetrator group was reported to have successfully extracted around 1.5 terabytes of data, including sensitive information from 15 million customers and employees, which was then published after ransom-payment negotiation attempts failed (Chakravarti, 2023). This incident shows that a cyber attack can cause total operational paralysis and pose a risk of massive data leakage. The non-material impact of this attack, such as the loss of customer trust and the decline in institutional reputation, was no less significant.

Within the Adaptive Security Framework, the BSI case shows the potential weakness of early-detection capability and unpreparedness in effectively responding to an incident. The amount of data successfully stolen indicates a possible failure at both the prevention and detection stages. This opens up the possibility that the monitoring system was not yet able to identify suspicious activity in real time, or that the isolation and automatic-control

mechanism had not been operating optimally. If the security system applied had been thoroughly adaptive, then the system should have been able to detect anomalies such as a spike in data-encryption activity and automatically activate a mitigation scenario, such as disconnecting the connection or activating a fail-safe backup.

In addition, the adaptive framework also emphasizes the importance of post-incident learning. In this context, the event experienced by BSI becomes an important lesson for all financial institutions in Indonesia to immediately strengthen their security posture, among other things by proactively conducting threat hunting and tightening disaster-recovery mechanisms, as described by (Ainslie et al., 2023).

Threats to the banking sector have also emerged in other forms, for example through hacking claims by certain individuals or groups. One example is the emergence of the figure "Bjorka" in the 2022–2023 period, who claimed to have obtained access to data belonging to Indonesian institutions, including data belonging to 4.9 million Bank Central Asia (BCA) customers. Although BCA denied the truth of this claim and ensured that customer data remained safe, this phenomenon nonetheless caused public unease (Kompas, 2025). From an adaptive-security perspective, incidents of this kind underscore the importance of intelligence capability and an early-warning system capable of monitoring data-leak claims on online forums. BCA's swift step in issuing a public clarification also reflects one form of adaptive response, namely information transparency and proactive handling to preserve stakeholder trust.

2.2.4. Cyber Resilience

The table of prior studies raising the Cyber Resilience (CR) concept covers 10 scientific articles published between 2019 and 2025. In general, these publications emphasize the importance of building an institution's digital resilience through a holistic approach that focuses not only on attack prevention, but also on recovery and operational continuity post-incident. These articles underscore the role of business-continuity-planning strategy, real-time incident-response capability, and the integration of CTI and institutional policy in creating a resilient security ecosystem. In the context of this research, CR contributes to enriching the analytical framework regarding the readiness of Indonesian banking institutions to confront systemic-scale cyber disruption, while also reinforcing the urgency of strengthening the adaptive and collaborative aspects in responding to the increasingly complex dynamics of digital threats.

Table 2.4. List of Prior Studies Related to the Cyber Resilience Concept

No	Source	Category	Research Objective	Methodology	Results and Conclusion
1	Organizational Culture and Cyber Resilience (Bada & Nurse, 2020)	<i>Cyber Resilience</i>	Analyzes the role of organizational culture in shaping cyber resilience beyond the technical aspect.	Literature review and empirical study of public- and private-sector organizations.	Employee awareness and routine training are important elements in supporting a resilient security system.
2	Governance and Cyber Resilience in Financial Sector (Von Solms & Van Niekerk, 2021)	<i>Cyber Resilience</i>	Examines the link between risk-based governance and an organization's resilience capacity.	Conceptual analysis and case study of financial-sector organizations.	Integrating resilience indicators into a managerial performance system improves incident response and recovery.

No	Source	Category	Research Objective	Methodology	Results and Conclusion
3	A Multidimensional Framework for Cyber Resilience (Woods & Böhme, 2021)	<i>Cyber Resilience</i>	Develops a four-dimensional cyber-resilience framework: resist, absorb, recover, adapt.	Development of a theoretical model and simulation on digital organizations.	Adaptive capacity is the most crucial indicator in determining an institution's post-incident resilience.
4	Inter-Agency Collaboration and Cyber Resilience (Amin et al., 2022)	<i>Cyber Resilience</i>	Investigates the effect of cross-institutional coordination effectiveness on improving cyber resilience.	Institutional-network analysis and study of financial-sector regulation in Indonesia.	Collaboration between institutions becomes an important foundation for strengthening national cyber defense.
5	Incident Simulation Modeling for Financial Sector Resilience (Stiawan et al., 2020)	<i>Cyber Resilience</i>	Develops a cyber-incident simulation model to improve the readiness of the financial sector.	Development of a system-based scenario model and simulative experiment.	Periodic exercises and response simulations improve collective awareness and responsiveness to threats.
6	Integrating Cyber	<i>Cyber Resilience</i>	Analyzes the implementation	Regulatory analysis and	Many banks still focus on

No	Source	Category	Research Objective	Methodology	Results and Conclusion
	Resilience in Banking Policies (Dewi & Mahardika, 2024)		n of cyber-resilience principles in bank policy in Indonesia.	case study of national bank internal policy.	administrative compliance and have not yet substantively integrated the resilience aspect.
7	Business Continuity Planning in Indonesian Digital Banks (Hendrayana, 2023)	<i>Cyber Resilience</i>	Examines the importance of real-time decision-making within a business-continuity plan for the digital banking sector.	Quantitative study of the readiness of digital-recovery systems at banks in Indonesia.	Most banks do not yet have a system capable of restoring service within 4 hours post-incident.
8	Crisis Communication and Public Trust in Cyber Incidents (Putra & Wibowo, 2023)	<i>Cyber Resilience</i>	Analyzes the link between crisis-communication strategy and the restoration of public trust post-cyber-attack.	Qualitative study based on interviews and analysis of a bank's post-incident communication policy.	Transparency and speed of communication greatly influence the process of restoring a financial institution's reputation.

No	Source	Category	Research Objective	Methodology	Results and Conclusion
9	Second Generation Cybersecurity Regulation and Resilience (Crisanto et al., 2023)	<i>Cyber Resilience</i>	Reviews the operational-resilience approach within second-generation cyber-security regulation.	Analysis of global-regulator policy and empirical study of the financial sector.	Global regulators are now pushing financial institutions to adopt the compromise assumption and focus on detection and response capacity.
10	Trends in Cyber Incidents and an Evaluation of Indonesia's Financial-Sector Response (Simorangkir et al., 2025)	<i>Cyber Resilience</i>	Analyzes the increase in cyber incidents in Indonesia and the effectiveness of financial institutions' responses.	Quantitative study based on incident data from BSSN and financial-institution reports.	A 43% rise in incidents indicates an urgent need for a more adaptive and resilient security system.

Furthermore, adaptivity also implies the need for continuous evaluation of system vulnerability. Incidents of this kind push banks to reconsider the effectiveness of their security controls, for example whether their data-exfiltration-monitoring system is already adequate, or whether an improvement in sensitive-data encryption is needed.

In the national context, the adaptive approach must also take into account the applicable legal and regulatory framework. The Government of Indonesia, through regulations such as Bank Indonesia Regulation No. 23/6/PBI/2021 on cyber security for payment-system operators, and Law No. 27 of 2022 on Personal Data Protection, has provided a strong legal basis for supporting the implementation of adaptive security. However, implementation challenges remain significant. Data from the National Cyber and Crypto Agency (BSSN) show that the number of cyber incidents affecting the banking sector in 2023 rose by 43% compared to the previous year (BSSN, 2023, in Simorangkir et al., 2025). This figure reflects the escalation of domestic threats alongside the acceleration of financial-industry digitalization. The economic consequences of such attacks also cannot be ignored. For example, the attack on BSI was reported to include a ransom demand of up to Rp300 billion, not including recovery costs and the potential loss from the misuse of customer data. Therefore, the application of an adaptive security framework is no longer an option, but an urgent strategic necessity for banking institutions in Indonesia.

Through an analysis of the BI, BSI, and alleged BCA leak cases, it can be seen that institutions that have adopted adaptive principles tend to be more resilient in responding to attacks and maintaining service continuity. Conversely, institutions that have not adequately developed an adaptive system risk experiencing significant disruption when confronted with an attack. The Adaptive Security Framework can serve as a solution for closing this gap by holistically integrating technology, procedure, and cyber intelligence. Thus, banks can build a cyber-security system that is not only resilient and flexible, but also able to maintain public trust and service continuity, while also supporting the stability of the national financial system.

A number of recent empirical studies further reinforce the urgency of applying an adaptive security framework within the context of national financial-system digitalization. Research by (Raharjo & Ismail, 2023) identifies that the readiness of commercial banks to confront dynamic cyber threats requires a security-decision-making capability based on up-to-date, real-time data. This study also emphasizes that an artificial-intelligence-based monitoring system is an urgent need, particularly for proactively detecting suspicious activity on digital banking services such as mobile-banking applications, which are considered to carry a high level of vulnerability. Meanwhile, findings from (Kurniawan & Sari, 2022) reveal that most banks in Indonesia

still rely on reactive protection mechanisms, and so have not yet been able to respond quickly to the escalation of threats. Therefore, they recommend strengthening early-detection capability and developing a threat-aware security system, namely a system that can automatically adjust protection parameters according to the level of threat, supported by policy-based adaptive systems.

Other research by (Nugroho, 2021) shows that adaptive security is not solely related to the technological aspect, but also encompasses institutional readiness in responding to rapidly developing threats. Their findings indicate that many national banking institutions still lack an agile and responsive cyber-security organizational structure, so that they are slow in taking mitigative action when an incident occurs. Therefore, the integration of technology, risk management, and leadership in cyber security becomes an important element in forming a comprehensive adaptive security architecture.

In terms of measuring security maturity, a study by (Fitriani & Astari, 2024) develops a maturity-model-based evaluation framework for assessing adaptive implementation in the banking sector. The evaluation results show that most banks are at the 'initiating' to 'developing' stage, and that implementation progress is highly dependent on top-management support. Therefore, strengthening adaptive risk governance is viewed as the main foundation for creating sustainable cyber resilience.

An important contribution also comes from (Pranata et al., 2023), who analyze the link between the use of a Security Information and Event Management (SIEM) system and the effectiveness of adaptive incident response at state-owned banks. This study finds that banks that have integrated SIEM with a threat-intelligence feed are able to accelerate incident detection by up to 35% faster, while also automating the process of blocking suspicious access. This finding confirms that a security system based on data integration and automated response is a key element of system adaptivity.

In the context of protecting core-banking systems, (Hasibuan, 2020) proposes a security architecture based on risk-level thresholds, which adjusts security controls to the context of transaction risk. He emphasizes that a static security approach is no longer relevant, especially for systems with extensive digital interconnection. This is important because a number of fraud and credential-theft cases in Indonesia are often caused by weak contextual security. A contribution in the field of adaptive intrusion detection is presented by (Stiawan et al., 2021), who developed a machine-learning-based adaptive

intrusion detection system (AIDS). This system has been tested on Indonesian banking transaction data and proven able to identify attacks, including zero-day attacks, with an accuracy level above 90%. This innovation shows that an adaptive cyber-security environment needs to integrate a self-learning-environment mechanism as part of the system.

Besides the technological aspect, institutional collaboration is also an important part of the adaptive approach. A study by (Alimudin & Handayani, 2023) highlights the weakness of the cyber-intelligence information-exchange system among institutions in Indonesia, including between banks, regulators, and technology providers. They propose developing the concept of an adaptive cyber alliance that enables more flexible coordination and collective response to shared threats. On the regulatory side, (Dewi & Mahardika, 2024) note that many banks have not fully integrated adaptive security principles into their compliance practice with domestic regulation, including in the implementation of the OJK Circular Letter on Cyber Resilience. Therefore, they recommend developing compliance-automation systems that can automatically adjust to changes in provisions and external threat conditions, so that adaptive principles can be implemented thoroughly.

Finally, the human-resources aspect cannot be ignored either. Research by (Hendrayana, 2023) shows that training that adaptively develops cyber situational awareness has a positive impact on the readiness of digital-bank security personnel. Banks that regularly conduct training based on actual threat scenarios have proven to be better prepared to confront real attacks. This finding confirms that adaptivity is determined not only by systems and policy, but also by organizational culture and the readiness of human resources to respond to situations in a timely manner.

The ransomware attack on Bank Syariah Indonesia in 2023, which caused digital-service disruption for several days and a leak of data from more than 15 million customers, shows that the national banking sector does not yet have adequate cyber resilience for confronting large-scale cyber attacks. In line with this, (Linkov et al., 2019) assert that cyber resilience requires not only an organization's ability to prevent attacks, but also to ensure operational continuity during an incident, as well as the ability to recover quickly and efficiently after an attack occurs. This view is highly relevant in the

context of Indonesian banking, given that many financial institutions do not yet have an integrated recovery procedure that can be immediately executed in a crisis situation.

Research conducted by (Bada & Nurse, 2020) broadens the understanding of cyber resilience by emphasizing that, besides technical systems, the aspect of organizational culture also plays a central role. They argue that employee training and improved situational awareness of potential threats are main pillars in building solid digital resilience. In the Indonesian context, weak internal cyber-security literacy is often a vulnerable point, even when technical infrastructure has been adequately prepared. Meanwhile, a study developed by (Von Solms & Van Niekerk, 2021) highlights the importance of risk-based governance in strengthening an organization's resilience capacity. They show that organizations that have integrated resilience indicators into their performance-management system and strategic decision-making have a better ability to respond to and recover from cyber incidents. This is an important note for many banks in Indonesia that have not yet fully incorporated resilience parameters into their key performance indicators.

From a more comprehensive perspective, (Woods & Böhme, 2021) develop a four-dimensional cyber-resilience framework, namely resistance, absorptive capacity, recovery process, and adaptation. This model places an organization's ability to absorb and adjust to digital disruption as a key resilience indicator. Therefore, in this research, the variables of intelligence strategy and security-policy flexibility become the two main aspects used to measure the level of resilience of bank security systems in Indonesia. A study from (Amin et al., 2022) also makes an important contribution by showing that cyber resilience does not rest solely on technology, but also depends heavily on the effectiveness of cross-institutional coordination. This finding shows that collaboration between banks and regulators such as OJK and BSSN has not yet been optimal in handling cross-domain cyber incidents. Therefore, this research takes the position that strengthening a resilient defense system requires a collaborative and integrated approach.

Next, (Stiawan et al., 2020) formulate a scenario-based incident-response model that can be used to strengthen the readiness of financial institutions in confronting sudden cyber attacks. Through periodic simulation, organizations are expected to build a collective response and raise awareness across all elements of the institution. This model becomes important in the Indonesian context, particularly because the attack on BSI

revealed the absence of an effective early-warning system. In the policy realm, (Dewi & Mahardika, 2024) note that the implementation of cyber-resilience principles in banks' internal policy remains uneven. Although Bank Indonesia Regulation No. 23/6/PBI/2021 has provided a framework for cyber security, its implementation at most institutions still focuses on administrative-compliance aspects rather than substantive resilience strengthening. Therefore, this research also considers the effectiveness of the regulatory framework in genuinely supporting system resilience.

On the operational side, (Hendrayana, 2023) emphasizes that real-time decision-making capability in disaster recovery and business-continuity planning is an important component in building the resilience of the banking sector. Based on 2022 data, more than 60% of digital banks in Indonesia did not yet have a service-recovery mechanism able to function optimally within less than four hours after an incident. This fact forms the basis of the argument in this research for developing a more proactive and resilient resilience-policy design.

Finally, the contribution of (Putra & Wibowo, 2023) reinforces the urgency of reputation management as part of digital resilience. They state that in the context

of a post-incident situation, public trust is greatly determined by the transparency and effectiveness of a financial institution's crisis-communication strategy. Therefore, cyber resilience in this research is also understood not only as technical capacity, but also as encompassing a social dimension such as trust management and the reconstruction of a banking institution's reputation in Indonesia.

2.3. State of the Art, Research Gap, and Novelty

2.3.1. State of the Art

This research rests on four main conceptual approaches, namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR). Each of these four approaches represents a key stage in the management of cyber-attack risk, from the early detection of threats, adaptation to the dynamics of attacks, strategic intelligence-based decision-making, through to the continuous strengthening of a banking institution's resilience. All four were chosen to provide an integrated analysis of the dynamics of cyber attacks targeting the Indonesian

banking sector, so as to not only understand incidents technically, but also to formulate an adaptive and resilient institutional response.

Based on the results of an analysis of the 62 references of prior study compiled in Chapter 2, no research was found that explicitly integrates these four approaches into a single systematic analytical flow. Most prior studies tend to discuss Cyber Threat Intelligence, Adaptive Security Framework, Strategic Intelligence, and Cyber Resilience separately and partially, without building a thorough, logical linkage from the initial detection stage through to the achievement of institutional resilience. In addition, existing studies also have not yet combined these four approaches with the foresight method to strengthen the aspect of medium- to long-term risk prediction and the formulation of a threat-projection-based prevention strategy.

In response to this gap, this research formulates an integrative approach by constructing a sequential CTI → ASF → SI → CR analytical flow, combined with the application of foresight as an additional analytical tool. The foresight approach is used to identify the tendencies in threat development, trace structural driving factors, construct various future scenarios, and design an adaptive-security-strategy roadmap relevant to the national banking context. Thus, this research offers a new contribution in the form of a conceptual update not yet found in prior literature, while also enriching the cyber-resilience strategy of the Indonesian banking sector through a more reflective, anticipatory, and intelligence-based perspective.

2.3.2. Research Gap

Based on the results of a review of the 62 references in Chapter 2, several research gaps can be identified that form an important basis for the construction of this research. First, the concept of Cyber Threat Intelligence (CTI) in various studies still focuses on the technical aspect of threat-data collection and analysis, without clearly linking it to the process of strategic decision-making at the institutional level. Second, the Adaptive Security Framework (ASF) approach that has been developed to date places greater emphasis on strengthening the technical infrastructure of system adaptation, but has not yet contextualized it as an intelligence-based and sustainable security-policy adaptation mechanism. Third, the application of Strategic Intelligence (SI) still tends to be limited to the national-defense domain and has not been widely used as an instrument for

formulating a technical-information-based banking-sector cyber-security strategy. Fourth, the Cyber Resilience (CR) approach in prior studies is generally positioned merely as part of disaster recovery or business continuity, without being systematically linked to the processes of early detection, dynamic adaptation, and intelligence-based decision-making.

In addition, no prior research was found that comprehensively integrates the four main approaches (CTI, ASF, SI, and CR) into a single, structured analytical workflow, while combining them with the foresight method to strengthen the predictive dimension in formulating a cyber-attack-prevention strategy. This conceptual gap indicates the urgency of building a new approach able to cohesively link the technical, adaptive, strategic, and resilient aspects, in order to more reflectively and anticipatorily strengthen the cyber resilience of the Indonesian banking sector against the increasingly complex escalation of digital threats.

2.3.3. Novelty

This research develops an approach by integrating four main conceptual frameworks, namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR), into a single structured and continuous analytical pathway. These four frameworks are arranged in the analytical sequence CTI → ASF → SI → CR to build continuity between the process of threat detection, security-system adaptation, strategic-policy formulation, and the achievement of institutional resilience against cyber attacks in the banking sector.

In addition, this research introduces the integration of the foresight method into the analytical framework, which was not found in the 62 references of prior study analyzed. The foresight method is applied with reference to the Systemic Foresight Methodology (Saritas, 2016) and the scenario approach in Scenario Planning: The Link Between Future and Strategy (Lindgren & Bandhold, 2009), to strengthen predictive capability in analyzing and anticipating future cyber-attack risk. Thus, foresight is used to identify threat trends, project potential impact, and design an adaptive long-term risk-mitigation strategy.

This research also expands the application of Strategic Intelligence into the banking sector, which was previously predominantly applied in the field of national defense (Walsh, 2017). Through this approach, this research offers a new contribution by constructing an integrated, adaptive, intelligence-based cyber-security analytical framework, in order to strengthen the national banking sector's resilience against the increasingly complex and dynamic escalation of digital threats.

2.4. Conceptual Framework

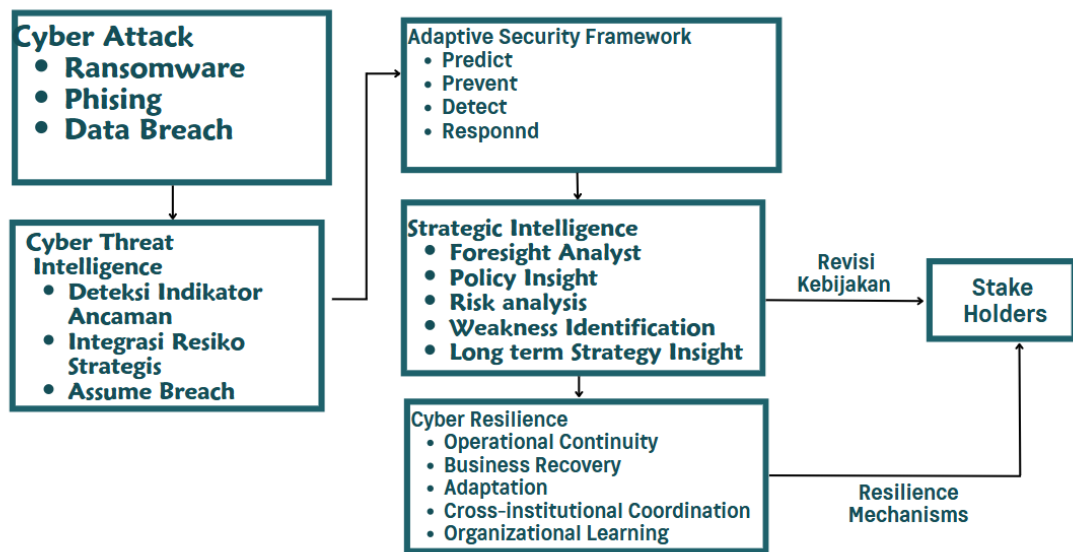


Figure 2.1. Conceptual Framework for Intelligence Analysis of Cyber Attacks in the Indonesian Banking Sector

Source: Adapted and synthesized from (Tounsi & Rais 2018), (Lopes & Oliveira 2015), (Walsh 2017), and (Linkov & Kott 2019)

The conceptual framework in this research is constructed to depict the logical and systematic flow in analyzing cyber attacks against Indonesian banking institutions through a comprehensive intelligence-based approach. The starting point of this

framework is the phenomenon of the increasing intensity and precision of cyber attacks, such as ransomware, phishing, and data breaches, which are increasingly complex and threaten the stability of the national financial system.

The initial response step to this threat escalation is carried out through the use of Cyber Threat Intelligence (CTI) as put forward by (Tounsi & Rais, 2018). CTI plays a role in detecting attack indicators, integrating technical and contextual risk, and applying the assume-breach principle — namely assuming that the system has already been infiltrated and that proactive detection efforts are needed from the outset. This process becomes an important foundation for strengthening vigilance and readiness in confronting continuously evolving cyber threats.

The information resulting from CTI mapping then becomes a key input for the Adaptive Security Framework (ASF) as developed by (Lopes & Oliveira, 2015). ASF aims to build a dynamic and adaptive security system through four main stages: predict, prevent, detect, and respond. With this approach, the system not only responds after an attack occurs, but also has predictive capability in confronting new, as-yet-unrecognized attack patterns.

The results of processing CTI and ASF are then used in Strategic Intelligence (SI) analysis, which links technical data with strategic insight to support long-term security-policy decision-making. In this context, SI is focused on identifying policy weaknesses, institutional-risk analysis, and the formulation of strategic insight through a foresight-analyst approach — namely a study based on future scenarios, trends, and a measurable, anticipatory digital-defense roadmap (Walsh, 2017).

The SI process produces recommendations that are relevant and can be directly translated into policy revisions, to be used by stakeholders for the comprehensive strengthening of the security system. This becomes a prerequisite for the formation of Cyber Resilience (CR) capacity, which is the final stage of this framework.

As formulated by (Linkov & Kott, 2019), Cyber Resilience encompasses an institution's ability to maintain operational continuity, carry out efficient service recovery, and adapt to the escalation of threats through cross-institutional coordination and continuous organizational learning. With integrated resilience mechanisms in place, the system is able to absorb disruption, respond to incidents without disrupting core services, and transform both tactically and strategically.

Overall, this framework shows the integrated flow $CTI \rightarrow ASF \rightarrow SI \rightarrow CR$, which is not only reactive but also strategic and proactive. This approach forms a solid cyber-intelligence foundation for continuously and sustainably addressing the challenges of national banking digital security, based on real data, responsive policy, and institutional readiness

Chapter 3 Research Method

3.1. Research Approach

This research adopts a qualitative approach in order to comprehensively examine the dynamics of cyber attacks specifically targeting the banking sector in Indonesia, with an emphasis on analyzing the resulting impact and the prevention strategies that have been implemented by the relevant institutions. This approach was chosen because it enables the researcher to explore a complex phenomenon thoroughly, capture the subjective perspectives of stakeholders, and understand the social and institutional background shaping the response to cyber threats (Creswell & Creswell, 2018). The complexity of cyber attacks is influenced not only by technical factors, but is also closely linked to the development of digital technology, institutional dynamics, and the direction of national policy in the field of information security. Through this approach, the analysis is directed at examining the strategic factors, security-management structures, and institutional policies that play an important role in strengthening the cyber resilience of the national financial sector.

The research methodology is constructed by integrating four main conceptual frameworks, namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR), designed to complement one another in answering the overall research questions. CTI is used to identify attack patterns and the characteristics of threat vectors based on empirical evidence (Tounsi & Rais, 2018), while ASF plays a role in evaluating the flexibility and adaptive capability of the security system in the face of threat escalation through the Monitor–Analyze–Plan–Execute cycle approach (Lopes & Oliveira, 2015). SI is used to examine how intelligence information is translated into institutional strategic decision-making (Walsh, 2017), and CR is used to evaluate an institution's capacity to sustain services, carry out recovery, and continuously adapt to cyber disruption (Linkov & Kott, 2019). To complement the analysis, this research also integrates a foresight approach based on five main stages — event, trend, driver, scenario planning, and roadmap (Saritas, 2016) — to enrich the predictive dimension in identifying attack patterns, projecting long-term impact, and designing an adaptive prevention strategy. With this framework, the research is directed

toward building an analysis that is not only adaptive and reflective, but also able to respond to the escalating complexity of cyber threats in the national banking sector.

3.2. Type of Research

The type of research used in this study is qualitative. This research is designed to examine in depth the experiences, meaning-making, and interpretations of stakeholders in the Indonesian banking sector regarding the phenomenon of cyber attacks over the 2021–2024 period. The main focus of the research is directed at uncovering how strategic actors in the banking sector understand patterns of cyber attack such as phishing, ransomware, and data breach, assess the impact of attacks on the institution's operational, financial, and reputational aspects, and identify the effectiveness of the prevention efforts that have been applied. The choice of the phenomenological research type is based on the consideration that this approach enables a comprehensive exploration of the essence of informants' subjective experience (Creswell & Poth, 2018). Through phenomenology, the researcher can understand the stakeholders' in-depth perceptions of the cyber-attack dynamics they have experienced, which cannot be fully uncovered through a purely quantitative approach or technical observation alone. This type of research is considered more relevant than a case study or grounded theory, given that this research does not focus on constructing new theory or an in-depth exploration of a single specific institution, but rather aims to produce a cross-institutional understanding of cyber threats in the banking sector (Neubauer, Witkop, & Varpio, 2019).

By using the phenomenological research type, this research is expected to present an analysis grounded in the real experience of stakeholders, so that it can formulate a strategy for strengthening cyber resilience in the Indonesian banking sector, including the development of adaptive early-prevention measures for confronting the future escalation of cyber-attack threats.

This research is focused on exploring patterns of cyber attack such as phishing, ransomware, and data breaches, which strategically threaten the stability of the national banking sector. The focus of the study is directed specifically at conventional banking institutions in Indonesia and at Bank Indonesia as the central-bank authority. The main objective of this research is to identify the range of attacks that occurred over the 2021–2024 period, analyze the impact of the attacks on the operational, financial, and

reputational aspects of banking institutions, and examine the various forms of early-prevention effort that have been applied to strengthen the cyber resilience of the national financial sector in responding to the increasingly complex escalation of digital risk. Thus, this research is expected to make an analytical contribution to strengthening the cyber-protection capability of the banking sector in Indonesia, while also enriching the development of a mitigation strategy adaptive to the dynamics of future threats.

This research focuses specifically on two main institutions within the national financial system, namely Bank Indonesia (BI) and Bank Syariah Indonesia (BSI), as the objects of study. Bank Indonesia was chosen because its function as the central bank makes it a main pillar in maintaining macroeconomic stability, the integrity of the payment system, and public trust in the national financial sector. This strategic position means that any potential compromise of BI's cyber security carries major systemic consequences; a failure to maintain digital resilience at the central-bank level could reinforce a perception of vulnerability among other conventional banks in Indonesia. Meanwhile, Bank Syariah Indonesia was chosen based on its status as the largest sharia banking institution in the country, which operationally and systemically has a broad scope of influence. The ransomware-attack incident that occurred at BSI in 2023, which caused service disruption and a large-scale data leak (Olavsrud, 2023), confirms that even large institutions still face real vulnerability to cyber threats. The selection of these two institutions reflects the research objective of examining how cyber attacks affect the operational, financial, and reputational aspects across various spectrums of banking institutions in Indonesia, and of examining the early-prevention measures that have been undertaken to strengthen the cyber resilience of the national banking sector.

3.3. Data Collection Method

The interview method in this research was conducted in depth using a semi-structured approach to elicit richer, more contextual information. The interviews involved four main informants selected based on their expertise and relevance to cyber-security issues in the national banking sector. These informants comprised a white hat hacker active in reporting system vulnerabilities, a representative from the State Intelligence Agency (BIN) working in the field of cyber intelligence, a national cyber-security expert, and a

cybersecurity Division Head from PT Mitra Integrasi Informatika (MII), a company that provides digital-security solutions for banking institutions such as Bank Indonesia, BSI, and BNI. Through these interviews, primary data was obtained that enriches the understanding of cyber-attack patterns, response mechanisms, and institutional readiness in confronting the escalation of digital risk.

3.3.1 Primary Data

The source of the primary data in this research was obtained through in-depth interviews with a number of strategic stakeholders. The informants involved include representatives from Bank Indonesia (BI) and Bank Syariah Indonesia (BSI) as representations of national banking institutions, an academic with expertise in the field of cyber security, and a white hat hacker experienced in identifying and reporting information-system vulnerabilities to the relevant institutions. In addition, a representative from the State Intelligence Agency (BIN) was also involved to provide a perspective on the dynamics of threats and cyber-defense strategy from a national-intelligence point of view.

3.3.1.1 List of Informants

In this research, primary data was collected through semi-structured in-depth interviews with four key informants representing strategic, technical, and practical perspectives on national banking cyber security. These informants comprised a national cyber-security expert (Dr. Pratama Persadha), an official from BIN's Deputy for Cyber Intelligence, a security practitioner from a banking-technology vendor (Hilal Akbar, PT MII), and a white hat hacker from the JadiHacker community. Each interview was designed to elicit a uniform set of data needs, adjusted to the context of each informant's expertise, such as an analysis of attack patterns based on the Cyber Kill Chain, the effectiveness of Cyber Threat Intelligence (CTI), the institution's adaptive readiness, and inter-institutional response and coordination mechanisms. The interviews were conducted online via Zoom and WhatsApp from late May to early June 2025.

Table 3.1. List of Informants

No	Informant	Number	Data Needs	Time & Date Conducted
1	Dr. Pratama Dahlian Persadha, S.Sos. MM Chairman, CISSReC Cyber Security and Communication Research Institute	1 person	1. Explore critical views on the increasingly complex trends and patterns of cyber attacks in the banking sector, including the stages of the Cyber Kill Chain. 2. Assess the effectiveness of policy and the implementation of security systems based on risk adaptation and the latest technology. 3. Examine the role of Cyber Threat Intelligence and the challenges of its integration into national decision-making systems. 4. Evaluate the quality of regulation and cross-sector coordination in supporting cyber resilience. 5. Identify potential ways to strengthen digital resilience through a foresight approach and intelligence-based policy.	Via Zoom Meeting, on 3-06-2025
2	Dani (alias) Deputy for Cyber Intelligence, BIN	1 person	1. Explore strategic and national perspectives on cyber-attack patterns threatening the banking sector, including the stages of the Cyber Kill Chain. 2. Assess the readiness of the detection system and inter-institutional coordination in responding to systemic-scale cyber attacks. 3. Examine the integration of Cyber Threat Intelligence into public policy and strategic decision-making. 4. Evaluate the effectiveness of the national framework, such as the establishment of CSIRT and the implementation of POJK. 5. Explore the state-intelligence challenges in strengthening the	Via WhatsApp chat, on Friday 30-05-2025

No	Informant	Number	Data Needs	Time & Date Conducted
			early-warning system for the digital financial sector.	
3	Hilal Akbar, S.Kom CyberSecurity Division Head, PT Mitra Integrasi Informatika	1 person	1. Explore the implementation of technical security systems within the banking environment, including responses to ransomware and phishing, including the stages of the Cyber Kill Chain. 2. Assess the readiness of banking institutions to build an early-detection system and adaptive-security architecture. 3. Examine the use of CTI and Security Information and Event Management (SIEM) in supporting security decision-making. 4. Evaluate the synergy between security vendors and banking institutions in responding to digital attacks. 5. Identify the challenges in cross-system security orchestration and the strengthening of the incident-response process.	Via Zoom Meeting, on Friday 30-05-2025
4	Julio (alias) White Hat Hacker (JadiHacker community)	1 person	Explore experience and technical perspective on common cyber-attack patterns targeting the banking sector, including the stages of the Cyber Kill Chain. Assess the effectiveness of the early-detection and adaptive-response systems applied by banking institutions. Identify the role of and obstacles to the implementation of Cyber Threat Intelligence (CTI) at the technical and tactical level. Examine vulnerability-reporting practices and institutional response to vulnerability disclosure.	Via Zoom Meeting, on Friday 30-05-2025

No	Informant	Number	Data Needs	Time & Date Conducted
			Explore views on security collaboration between ethical hackers and financial institutions.	

Interview Guide

The interview guide was compiled based on nine main questions designed to elicit informants' perceptions regarding the four main correspondence variables, namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR). All of these questions were given to the four informants with a uniform structure, but with the depth of follow-up adjusted flexibly to each informant's context and competence. This approach was intended to obtain comprehensive strategic, technical, and policy data, and to support validity through source triangulation.

Table 3.2. List of Interview Guide Items

Correspondence Variable	Question	Informant
Cyber Threat Intelligence (CTI)	In your view, what are the common patterns of cyber attack occurring in the banking sector?	All four informants were given the same nine main questions, covering the aspects of CTI, ASF, SI, and CR. The questions were designed to elicit strategic, technical, and policy views related to banking cyber resilience. Although the question structure is the same,
Cyber Threat Intelligence (CTI)	What is your view on the current implementation of Cyber Threat Intelligence (CTI) at Indonesian banking institutions?	
Adaptive Security Framework (ASF)	In your view, to what extent has an adaptive security system been implemented within the national banking environment?	
Adaptive	How effective is the early-detection	

Correspondence Variable	Question	Informant
Security Framework (ASF)	system currently in use in confronting attacks such as ransomware and phishing?	follow-up depth was adjusted to each informant's background. This approach ensures that the data obtained is contextual, comprehensive, and supports the triangulation of the analysis.
Strategic Intelligence (SI)	How do you view the role of strategic intelligence in the security decision-making process in the banking sector?	
Strategic Intelligence (SI)	What are the main challenges in integrating strategic intelligence into public or corporate policy?	
Strategic Intelligence (SI)	How can a scenario-based strategy strengthen an institution's readiness to confront cyber threats?	
Cyber Resilience (CR)	In your view, how strong is the digital resilience of the Indonesian banking sector in confronting systemic cyber attacks?	
Cyber Resilience (CR)	What efforts need to be made to strengthen incident response and post-attack recovery at banking institutions?	

3.3.2. Secondary Data

The secondary data source was collected through a review of documents and literature related to policy, regulation, and relevant empirical studies. The main focus of the review is directed at regulations issued by the Financial Services Authority (OJK), particularly POJK No. 11/POJK.03/2022 on the Implementation of Information Technology by Commercial Banks and POJK No. 38/POJK.03/2016 on the Implementation of Risk Management in the Use of Information Technology by Commercial Banks, as well as through a review of relevant academic journals. The literature reviewed includes the results of prior research on cyber attacks in the banking sector, the concepts of Cyber Threat Intelligence, Adaptive Security Framework, Strategic Intelligence, and Cyber

Resilience, and studies related to cyber-security policy in the financial sector. The focus of the analysis is directed at enriching theoretical understanding and supporting the interpretation of the primary-data findings obtained in this research.

3.4. Data Collection Technique

This research was conducted using a qualitative method; data collection in this research was carried out through two main methods, namely semi-structured interviews and document study. Data collection in this research was carried out through two main methods, namely semi-structured interviews and document study. This type of interview was chosen because it provides a directed set of questions, while still giving informants the flexibility to elaborate on their views, experiences, and perspectives more broadly. This method is considered effective for exploring social dynamics and the complexity of experience, which often cannot be fully uncovered through a structured-interview approach. The semi-structured interview offers a balance between systematic structure and narrative exploration, making it highly suitable for research into social and strategic phenomena such as cyber-security issues in the banking sector (Alshenqeeti, 2018).

The secondary data in this research was obtained through a document study of various relevant written sources, including official regulations such as POJK No. 11/POJK.03/2022 on the Protection of Vital Information Infrastructure in the Financial Services Sector, POJK No. 38/POJK.03/2016 on the Implementation of Risk Management in the Use of Information Technology by Commercial Banks, and academic journals discussing cyber-attack issues in the national banking sector.

The use of document study was chosen because it is able to provide a normative and evidentiary basis important for strengthening the analysis of the primary findings. Document analysis in qualitative research functions to broaden understanding of the social, historical, and policy context underlying the phenomenon under study (O'Leary, 2017).

3.5. Data Processing

The data analysis in this research was carried out through six main stages arranged following the principles of reflexive thematic analysis and combining inductive and deductive approaches simultaneously (Braun & Clarke, 2006), and reinforced by a hybrid thematic-analysis approach (Fereday & Muir-Cochrane, 2006). To maintain the credibility and integrity of the analysis results, the principles of trustworthiness in thematic analysis were also adopted in line with the guidance of (Nowell et al., 2017).

1. The first stage in the data-processing process is to carry out initial coding of the interview transcripts. At this stage, the researcher marks pieces of data relevant to the research focus and assigns code labels representing a particular meaning or concept. This process is conducted openly, allowing for the emergence of a variety of codes reflecting the complexity of the informants' answers. Coding was carried out systematically across all transcripts, and the results became the basis for identifying deeper themes in the subsequent stage.
2. Identification and Development of Themes. This stage involves identifying and developing themes. The researcher analyzes the relationships and patterns of meaning among the codes; a theme is understood as a meaningful pattern relevant to the research questions. This process is iterative in nature, allowing for the revision and refinement of themes, including splitting overly broad themes or merging overlapping ones. The criteria of internal coherence and external distinctiveness between themes serve as the main reference at this stage (Nowell et al., 2017). The themes developed are linked to the research's conceptual framework to strengthen their theoretical relevance.
3. Clustering Based on the CTI, ASF, SI, and CR Framework. The final themes that have been identified are then grouped into four broad categories in accordance with the research's conceptual framework. Each theme is mapped into the Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), or Cyber Resilience (CR) cluster, based on its fit with the operational definition of each respective concept (Tounsi & Rais, 2018; Lopes & Oliveira, 2015; Walsh, 2017; Linkov & Kott, 2019). For example, a theme regarding the identification of attack indicators is clustered under CTI, a theme regarding post-incident security-system adaptation falls under ASF, a theme related to security-strategy coordination falls under SI, and a theme regarding post-attack recovery capability falls under CR. This strategy ensures that the

findings are well structured within the analytical framework that was designed from the outset of the research.

4. **In-depth Interpretation and Meaning-Making.** After the themes are grouped into clusters, the researcher carries out an in-depth interpretation of each theme and cluster. Interpretation was conducted using a descriptive-analytical approach, exploring the meaning behind each theme, the relationships between themes, and their practical and theoretical implications for the context of cyber attacks in the Indonesian banking sector. Each theme is analyzed not only based on empirical data, but is also linked to relevant literature to strengthen the validity of the interpretation. The researcher also examines cross-cluster linkages, for example how weaknesses in threat intelligence (CTI) can affect organizational resilience (CR). Direct quotations from participants are used to illustrate and enrich the meaning-making of each theme (Braun & Clarke, 2006).
5. **Validation of Findings and Triangulation.** To increase the credibility of the research results, a validation strategy is applied through the use of triangulation. Source triangulation is carried out by comparing data obtained from various sources, such as interviews, documents, and field observation. Method triangulation is applied where data was collected through a combination of techniques, such as interviews and focus group discussions (FGD). In addition, theory triangulation is used by analyzing the data based on the four main conceptual frameworks (CTI, ASF, SI, and CR) to enrich the interpretive perspective. Where the implementation of triangulation is limited, data validity is still maintained by adopting the principles of trustworthiness such as member checking, an audit trail, peer debriefing, and researcher reflexivity (Noble & Heale, 2019).
6. **Drawing Conclusions.** Overall, this data-analysis process integratively combines a bottom-up, data-based approach with a top-down, theory-based approach. All findings are systematically mapped within the CTI, ASF, SI, and CR framework. Validation is carried out through triangulation and other credibility strategies to ensure results that are accurate, comprehensive, and academically accountable. This approach is in line with the principles of contemporary qualitative analysis, which emphasize flexibility, reflexivity, and rigor in understanding complex phenomena (Nowell et al., 2017; Braun & Clarke, 2019).

3.6. Data Analysis

This research uses a thematic data-analysis method guided by the predetermined conceptual framework, namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR). The choice of thematic analysis is based on its flexibility and transparency in identifying and organizing patterns within qualitative data (Braun & Clarke, 2019). This analysis is carried out through a series of systematic phases, but in practice is iterative and reflective in nature, rather than rigidly linear, so that the researcher can move back and forth between stages according to the needs of the analysis (Braun & Clarke, 2019). Thus, throughout this process, the CTI, ASF, SI, and CR conceptual framework will function as the main theoretical lens for understanding and interpreting the research findings.

These four main concepts become the top-level thematic categories. First, Cyber Threat Intelligence (CTI) refers to evidence-based knowledge regarding ongoing or newly emerging cyber threats. In this research, CTI covers information such as Indicators of Compromise and patterns of tactics or the modus operandi of attackers that can be identified from empirical data. Second, the Adaptive Security Framework (ASF) represents the principle of adaptive security management, namely the need to dynamically adjust a defense strategy in responding to continuously developing threats (Lopes & Oliveira, 2015). This category encompasses themes related to a system's or organization's capacity to update security controls and iteratively apply incident response.

Next, Strategic Intelligence (SI) relates to the strategic use of information to support long-term decision-making processes (Walsh, 2017). Within this framework, SI covers an institution's long-term perspective in designing security policy, as well as building cross-institutional collaboration in confronting the dynamics of cyber threats. Finally, Cyber Resilience (CR) foregrounds a system's ability to withstand, recover from, and adapt after experiencing a cyber attack. This approach emphasizes that an attack is a certainty, so the main focus is on efforts to restore critical functions and increase post-incident adaptation, not merely on preventive action (Linkov & Kott, 2019).

In its application, this conceptual framework will become the reference for compiling the main themes. However, the analysis process is not fully deductive in nature. The researcher still leaves room for new, inductively arising findings, so that this analytical approach combines deductive and inductive methods simultaneously. A hybrid approach

of this kind is believed to enrich the research's rigor, since the theoretical structure provided by deductive analysis is combined with sensitivity to the empirical findings generated by the inductive approach (Fereday & Muir-Cochrane, 2006, cited in Proudfoot, 2023). Through this strategy, the analysis results are expected to represent a balance between a solid theoretical framework and the authentic validity of field data.

The data analysis in this research was carried out through six main stages arranged following the principles of reflexive thematic analysis and combining inductive and deductive approaches simultaneously (Braun & Clarke, 2006), and reinforced by a hybrid thematic-analysis approach (Fereday & Muir-Cochrane, 2006). To maintain the credibility and integrity of the analysis results, the principles of trustworthiness in thematic analysis were also adopted in line with the guidance of (Nowell et al., 2017).

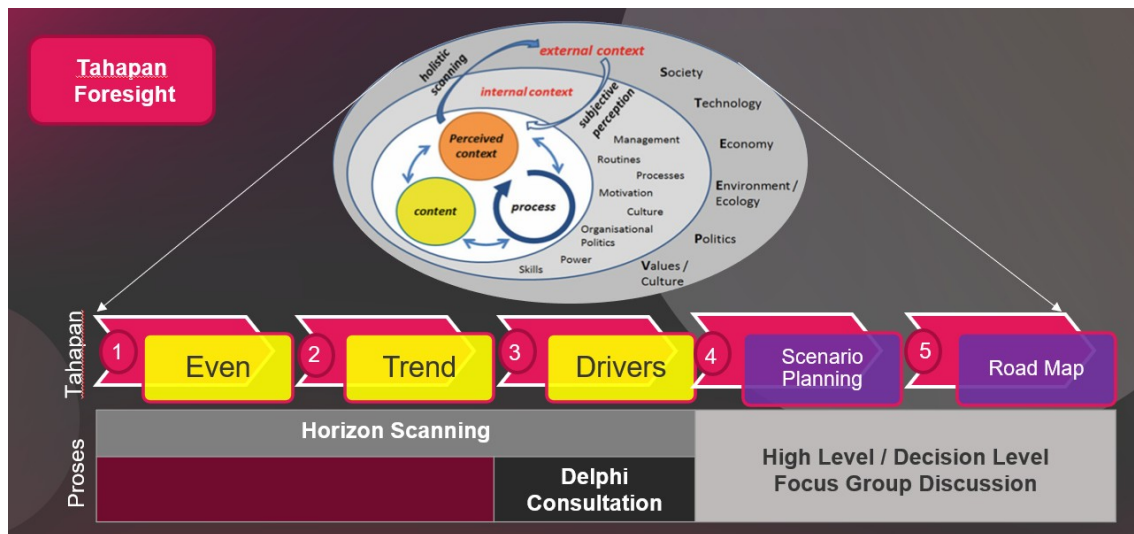


Figure 3.1. Stages of Foresight

Source: Adapted from Saritas, 2016

To strengthen the main conceptual framework used in this research — namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR) — the foresight approach is integrated as an analytical instrument playing a strategic role in deepening the interpretation of findings and broadening the predictive scope in data processing. Foresight does not function as the main approach, but rather as a relevant analytical tool for increasing the capacity to anticipate threats, particularly in formulating medium- to long-term cyber-attack-prevention policy in the banking sector.

The foresight model adopted refers to the systemic approach (Systemic Foresight Methodology) as formulated by (Saritas, 2016), which involves five main stages: identification of events (event), analysis of tendencies (trend), tracing of structural driving factors (drivers), scenario construction (scenario planning), and the design of a policy roadmap (roadmap). All of these stages are applied in an integrated manner to answer the third research objective, namely analyzing the efforts to prevent cyber attacks against banking institutions. Within this framework, foresight strengthens the predictive dimension of Strategic Intelligence (Walsh, 2017), deepens the mapping of cyber-attack patterns (Tounsi & Rais, 2018), supports the assessment of security-system adaptivity

(Lopes & Oliveira, 2015), and drives the long-term strengthening of institutional resilience (Linkov & Kott, 2019).

Table 3.3. Foresight Stages in Strengthening the Four Basic Concepts of the Research

Foresight Stage	Description	Objective
Event	<i>An event refers to an actual occurrence that has taken place or is ongoing, and is relevant to the topic being examined within the foresight framework</i>	Identifying real incidents and the escalation of cyber attacks against the banking sector as a basis for empirical analysis.
Trend	<i>A trend is a direction of change that is repetitive or continuously developing over time, observable through data, policy, or the behavior of particular actors. Within the foresight framework, a trend acts as a bridge between past events and possible future shifts</i>	Examining patterns and tendencies of threat, policy, and institutional response developing within the digital financial sector.
Driver	<i>A driver is a main driving force that shapes or accelerates change within a system. In foresight, a driver can be technological, regulatory, social, economic, or geopolitical in nature, and is often a determining factor in constructing various future scenarios</i>	Determining the key factors that drive banking cyber-security vulnerability or resilience.
Scenario	<i>Scenario planning is a</i>	Formulating possible future

Foresight Stage	Description	Objective
Planning	<i>systematic process for constructing alternative pictures of the future based on a combination of the drivers that have been analyzed. This stage does not aim to predict the future with certainty, but to explore the various possibilities (what-if scenarios) that could occur if certain variables move in a different direction or intensity</i>	directions (resilience or crisis) based on the institution's level of readiness to confront cyber threats.
Roadmap	<i>A roadmap is the final stage in the foresight process, presenting a phased action plan based on the scenarios that have been formulated. A roadmap sets out concrete steps to be taken in the short, medium, and long term, in order to anticipate, mitigate, or capitalize on the possible futures that have been mapped.</i>	Compiling short-, medium-, and long-term strategic recommendations to strengthen the national banking sector's defense system and digital resilience.

Analysis Using Scenario Planning

Figure 3.2 displays the scenario-planning quadrant used to map possible futures based on two main axes representing the critical variables in this research, namely the resilience capability of banking institutions (Driver X) and the attack capability of cyber actors (Driver Y). Each quadrant depicts one of four possible scenarios developed, the things that may or may not occur, and the strategic response that needs to be prepared in confronting development challenges and digital-security priorities. This framework

serves as a visual aid for exploring the spectrum of risk and adaptive opportunity within the context of national banking cyber resilience.

Figure 3.2. Strategic Scenario Quadrant

Source: Adapted from Lindgren & Bandhold, 2009

The X axis in this scenario quadrant represents the level of adaptive readiness of the cyber-security system, determined by three main dimensions: readiness of security technology (including early detection and Zero Trust), regulatory flexibility (the ability of policy to adapt to threats), and strategic digital leadership (the role of leadership in directing cyber-security transformation) (Kindervag, 2010; POJK No. 11/POJK.03/2022; Westerman, Bonnet, & McAfee, 2014). Institutions with a strong combination of these three elements are positioned on the X+ side, while institutions with limited adaptive capacity are on the X– side.

The Y axis in this foresight quadrant depicts the level of potential and projected cyber threat expected to affect the banking sector over the medium- to long-term time horizon. This dimension is formed from the interpretation of historical cyber-attack trends, weak signals, and socio-technological dynamics indicating a tendency toward future threat escalation. Factors such as increasing dependence on digital services, the annual cycle of financial activity, and the not-yet-optimal mitigation of systemic vulnerability are the main indicators used in constructing this projection (Gentry, 2023; Olavsrud, 2023).

At the Y– pole, a condition with a relatively stable and controllable level of threat potential is shown, whereas the Y+ pole represents a situation with high threat potential — with a tendency toward increasing aggressiveness, volume, and complexity of attack. This dimension reinforces an anticipatory, rather than reactive, foresight approach

oriented toward the formulation of a cyber-security policy that is adaptive, inclusive, and resilient over the long term.

Within the foresight quadrant framework developed in this research, Scenario 1 depicts the most concerning situation, namely when a banking institution is at a low level of adaptive readiness (X–), while the potential for cyber threat shows a high tendency toward escalation (Y+). This condition indicates that although warning signals and escalatory trends have been identified through horizon scanning and expert interviews, the institution's capability to respond technologically, regulatorily, and through digital leadership has not been adequate. This unpreparedness opens a serious gap for increasing systemic-disruption risk and delay in incident mitigation.

Conversely, Scenario 2 reflects the ideal condition hoped to be achieved, in which the institution's internal adaptive readiness is already at a high level (X+), and the potential for cyber threat is also in an increasing status (Y+). In this situation, the institution has succeeded in building a security system that is structurally and strategically adaptive, supported by the readiness of early-detection technology, responsive regulatory flexibility, and digital leadership oriented toward continuous transformation. The response undertaken is not only reactive, but also anticipatory and collaborative across institutional actors, which ultimately strengthens the strategic position in maintaining national cyber resilience (Westerman, Bonnet, & McAfee, 2014; Kindervag, 2010).

Scenario 3, on the other hand, reflects a latent yet risky situation, namely when both adaptive readiness (X–) and threat potential (Y–) are at a low level. Although it appears stable on the surface, this condition can trigger the illusion of security, or a false sense of security, in which the absence of external pressure causes the institution to tend toward stagnation in innovation and the strengthening of its security system. If a sudden surge in attack were to occur, an institution in this scenario would be highly vulnerable because it lacks a solid systemic foundation in terms of both adaptation and institutional synergy (Gentry, 2023).

Meanwhile, Scenario 4 reflects a situation in which an institution's adaptive readiness has developed quite well (X+), but the perception or projection of threat potential remains low (Y–). In this configuration, although adequate mitigation tools and policy are internally available, the low level of external pressure can give rise to overconfidence and a decline in the intensity of inter-institutional collaboration. This can cause the prevention

strategy that has been built to fail to develop optimally and to lose its innovative momentum (Olavsrud, 2023).

Through the mapping of these four scenarios, foresight is used as a strategic tool for formulating an early-prevention policy that is more reflective, predictive, and grounded in long-term intelligence. Thus, this approach supports the strengthening of national financial institutions' readiness to confront the uncertainty of cyber threats in a more structured and systemic manner.

Chapter 4 Results and Discussion

Chapter 4 presents the findings and in-depth analysis based on the primary and secondary data collected during the research process. This chapter consists of two main parts. The first part (Section 4.1) describes the research findings on banking cyber resilience strategy through three main approaches, namely proactive cyber intelligence, a security system adaptive to threat escalation, and a structured information-protection policy. This analysis is built from the mapping of attack patterns, an understanding of system vulnerability, and the institutional response to cyber challenges. Meanwhile, the second part (Section 4.2) adopts a horizon-scanning-based foresight method to identify relevant events, trends, and drivers, then develops strategic scenarios and a policy roadmap as a projection of institutional readiness to confront cyber attacks in the future. This structure is designed to provide a comprehensive and integrative picture in answering the main focus and objective of the research.

4.1. Overview & Research Findings

Attacks that were previously probabilistic — that is, striking randomly or targeting common vulnerabilities — have now shifted toward targeted attacks that are directed and based on a target system's specific weaknesses. This transformation has even entered a more advanced phase, in which artificial-intelligence-powered attacks are beginning to be used by attackers to automate exploitation through Ransomware-as-a-Service (RaaS) schemes, deepfake-based social engineering, and modular, polymorphic exploit-kit and payload-delivery modules. In confronting these dynamics, a purely reactive approach is no longer sufficient, because it only responds to attacks after the incident. A proactive digital-security strategy is needed, through strengthening cyber threat intelligence (CTI) for early detection, and an adaptive approach so that the system can dynamically adjust to the continuously evolving threat landscape.

This research obtained primary data through in-depth interviews with four key informants representing the strategic and technical spectrum within the national cyber-resilience ecosystem. First, a representative from the Deputy for Cyber Intelligence of the State Intelligence Agency (BIN), who provided views on national synergy and cross-sector intelligence-coordination mechanisms. Second, a white hat hacker from the JadiHacker community who explained in detail the attack techniques and points of vulnerability in the banking system that have thus far been under-detected. Third, the Division Head of Cyber Security at PT Mitra Integrasi Informatika — a company that serves as a digital-security vendor for national banking institutions — who outlined the implementation challenges in defense architecture, SIEM integration, and the need for governance strengthening. Finally, Dr. Pratama Persadha, an expert and Chairman of the

Cyber Security and Communication Research Institute (CISSReC), who provided an analysis of public policy, human-resource readiness, and regulatory challenges in building a resilient national security system. These four informants were involved to validate the literature findings that had been mapped within the conceptual framework of Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR), thereby enabling a comprehensive analysis of the banking sector's readiness to confront systemic and continuously evolving cyber risk.

4.1.1 Banking Cyber Resilience Strategy Through an Intelligence, Adaptive Security, and Information Protection Approach

The increasing frequency and complexity of cyber attacks against banking institutions in Indonesia marks the urgency of formulating a digital resilience strategy that is not only responsive, but also foregrounds a proactive and thorough approach. This approach encompasses the integration of cyber intelligence for early detection, the application of a security system adaptive to the dynamics of attack, and a structured, cross-functionally integrated information-protection policy. All three become crucial elements in building a cyber-security system able to respond to threats comprehensively from both the technical and institutional-governance aspects. In this section, the discussion is focused on analyzing how cyber intelligence can recognize attack patterns before data exfiltration occurs, how a flexible security framework is applied to manage continuously developing risk, and how information policy is designed to sustain the resilience of financial institutions on an ongoing basis. This exposition is relevant to the three main foci of this research, namely the effectiveness of detection mechanisms, adaptive capacity, and policy accountability in supporting the continuity of banking-sector cyber security.

4.1.1.1. A Proactive Cyber Intelligence Approach in Confronting Cyber Attacks on the Banking Sector

The classification of interview results was carried out by mapping the findings based on the four main concepts in the research's conceptual framework, namely Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR). Each finding was analyzed through thematic coding

reflecting the substantive elements of each category. The initial codes obtained depict a diversity of information ranging from threat-pattern detection, adaptive response to attack, strategic security planning, through to an institution's resilience against systemic cyber attacks. The table below summarizes the relationship between the initial codes, conceptual category, and main theme identified in the analysis.

Table 4.1. Classification of Initial Codes and Main Themes by Concept

Source: Researcher's Data

Initial Code	Category	Main Theme
<i>Technical Level, Threat Indicator, Operational Level, Strategic Level, Tactical Level, Threat Intel Sharing, Intel Lifecycle Process, Malware Exploit Profiling, Threat Actor Attribution, Threat Intel Tools.</i>	Cyber Threat Intelligence	Early Detection and Projection of Cyber Threat Patterns
<i>Detect, Predict, Prevent, Respond.</i>	Adaptive Security Framework	Adaptive Response to Threat Escalation
<i>Event, Trend, Driver, Institutional Readiness, Leadership Intel, Risk Based Policy, Scenario Planning, Strategic Planning, Roadmap</i>	Strategic Intelligence	Integration of Strategic Projection into Cyber Security
CTI and Strategic Policy Integration, Resilience to AI-based Ransomware, Bank Operational Resilience, Incident Recovery and Response	Cyber Resilience	Digital Resilience Against Systemic Attacks

The code and category structure developed in this research reflects a cross-concept integration between Cyber Threat Intelligence (CTI), Adaptive Security Framework (ASF), Strategic Intelligence (SI), and Cyber Resilience (CR). Each framework is classified into operational elements reflecting its function in building a proactive, adaptive, and strategic cyber-security system. CTI centers on the process of early detection and mapping of threat patterns through technical indicators, actor profiles, and

a thorough intelligence cycle. Meanwhile, ASF is focused on a dynamic response to threat escalation, through the stages of detection, prediction, prevention, and response. SI, on the other hand, plays a role in integrating future projection into cyber-security policy through a foresight approach, while CR strengthens institutional resilience through policy integration, recovery capacity, and resistance to systemic disruption.

In an implementation context, the interconnection between these codes shows that banking institutions' cyber resilience cannot be built partially, but must occur through functional synergy between the main concepts. For example, the integration of threat indicators from CTI, such as threat indicator and malware exploit profiling, becomes an important input for the predict and prevent stages in ASF, which is then used as a basis for formulating risk-based policy within the SI framework. Further, the role of CR strengthens the entire cycle by emphasizing the importance of rapid recovery and institutional adaptation, particularly in confronting systemic threats such as AI-based ransomware. Thus, this code structure not only represents a theoretical framework, but also shows the functional relationships underlying the construction of a comprehensive, contextual cyber-security model for confronting the increasingly sophisticated threats in the Indonesian banking sector.

Cyber-Attack Pattern Analysis Based on the Cyber Kill Chain

To more systematically understand the technical stages of a cyber attack, this research uses the Cyber Kill Chain model as an analytical framework for mapping the attack patterns occurring at banking institutions in Indonesia. This model is used to identify the critical phases of each attack, from the initial reconnaissance stage through to final execution against the target. The data analyzed was obtained through triangulation from four main informants: a representative from the State Intelligence Agency (BIN), PT Mitra Integrasi Informatika (PT MII), a white hat hacker, and a national cyber expert. Validation was carried out based on the frequency of confirmation of each stage in the Cyber Kill Chain by the informants. The results of this triangulation are presented in Table 4.2 below.

Table 4.2. Informant Triangulation Against the Cyber Kill Chain Analysis

Source: Researcher's Data Processing Results

Cyber Kill Chain Stage	BIN	PT MII	White Hat Hacker	Cyber Expert	Validation
Reconnaissance	–	✓	✓	✓	Three Sources
Weaponization	–	✓	✓	–	Two Sources
Delivery	–	–	✓	–	One Source
Exploitation	–	✓	✓	–	Two Sources
Installation	–	✓	✓	✓	Three Sources
Command and Control (C2)	–	✓	✓	✓	Three Sources
Action on Objectives	✓	✓	✓	–	Three Sources

The Reconnaissance Stage (Initial Surveillance)

In the Cyber Kill Chain model developed by (Hutchins 2011), the reconnaissance stage marks the first phase of a cyber-attack sequence and forms the main foundation for the attacker to design a comprehensive attack strategy. At this stage, the threat actor seeks to gather technical and non-technical information to map the target and identify potential gaps that can be exploited. Data collection is carried out using open-source intelligence (OSINT) methods and other passive approaches, so as not to trigger the target institution's cyber-defense detection system. The information gathered covers internal network structure, user accounts, application architecture, through to the configuration of the banking institution's information-system services.

In the context of a phishing attack, the attacker generally targets the personal and organizational information of bank employees, especially those who have administrative access to internal systems. This process is carried out through browsing official websites, annual reports, and professional social media such as LinkedIn. Tools

such as Hunter.io and email permutators are used to build a database of valid institutional email accounts. The bait email compiled by the attacker is then made to resemble official communication in order to increase the chances of successful social engineering. According to Julio, a white hat hacker, "attackers often disguise themselves as a bank's legitimate vendor or business partner, and even insert accurate organizational information based on crawling of the public domain." This statement shows that attackers do not only rely on technical techniques, but also exploit psychological gaps and the victim's communication habits.

Further, Dr. Pratama Persadha, a cyber-security expert from CISSReC, explains that the reconnaissance process often takes place over a long duration, even before the attack truly begins. He asserts that, "what is often not realized is that attackers have very long amounts of time to conduct a survey. They are patient, they monitor, they observe, they see who plays an important role, then they map out a way in — it could be through email, through a partner, through a weak device." This shows that the reconnaissance process is carried out gradually and systematically, with the aim of building an in-depth understanding of the organization's weak points.

Meanwhile, in a data-breach scenario, the threat actor focuses on the technical infrastructure aspect and weak points in the information system. Reconnaissance is carried out by scanning open ports, identifying active servers, and exploring vulnerable endpoints using tools such as Shodan, Censys, and Nmap. Their main target is systems running old, unpatched software, or connected to third-party services that have security gaps. According to Hilal Akbar, Division Head at PT Mitra Integrasi Informatika, "hackers often target systems running old technology or unprotected by anomaly-detection systems." He also adds that another risky entry point is mobile-banking applications and application programming interfaces (APIs) directly connected to the bank's core system.

This explanation is reinforced by Dr. Pratama, who states that, "what is most exploited now is precisely APIs that aren't strictly audited. Through an API, access can penetrate the main system, even if the endpoint isn't monitored. Banks widely use APIs to third parties, and that's the most vulnerable gap." This view asserts that the reconnaissance

stage has evolved from mere network mapping toward a comprehensive survey of the digital ecosystem, including open-service architecture vulnerable to exploitation.

In the context of a ransomware attack, the reconnaissance phase is carried out in depth and in a structured manner against a bank's entire digital environment, from internal systems and user accounts through to network architecture and Active Directory. Attackers use tools such as Masscan and ZMap to detect sensitive ports, and to identify vulnerabilities in Remote Desktop Protocol (RDP) services, Virtual Private Network (VPN), and unprotected endpoints. According to Dani, an official at the Deputy for Cyber Intelligence of the State Intelligence Agency, "attackers generally know a bank's internal system even before they get in, because the mapping process is carried out through a botnet or malware already present outside the main perimeter beforehand."

In line with this, Dr. Pratama adds that, "new ransomware types are smarter now, they monitor the network first, they study who has the highest privilege, then they send the payload. So it's not blind firing, but very well planned." This strategy indicates that before deploying a payload, attackers ensure a complete understanding of the most strategic access pathways and the points in the system with the highest impact. Their main target is a system storing critical data, including database servers and core banking systems, so that the resulting damage will have a broad impact on the institution's operations.

Thus, the reconnaissance phase in phishing, data breach, and ransomware attacks against banking institutions shows an escalation in the complexity of the methods used. Reconnaissance is no longer limited to the technical aspect, but covers social elements, organizational behavior, and the dynamics of digital integration between systems. This process becomes the key factor in determining the success of a subsequent attack and provides an early intervention opportunity if recognized early by a proactive cyber-intelligence system. Therefore, strengthening the security system must include the ability to detect reconnaissance activity in real time, through monitoring of data-traffic anomalies, an audit of public information exposure, and detection of suspicious probing activity.

The Weaponization Stage

After the attacker has obtained a complete map of the target through the reconnaissance stage, they will enter the weaponization phase, namely the process of compiling a digital attack instrument based on the information that has been gathered. Within this framework, a weapon does not merely refer to malware, but to a combination of malicious payload that has been tailored to appear legitimate and not trigger the security detection system when sent to the target. The aim of this stage is to produce an attack vector capable of passing through the initial security layer and causing an undetected system compromise.

In the case of phishing, the weaponization stage is generally realized through the compilation of a bait email tailored to the individual characteristics or position of the prospective victim. Personal and structural information previously obtained — for example from the institution's official site, professional social media such as LinkedIn, or organizational repositories — is used to design a message with a specific work context. In practice, the attachment is packaged in a format such as .docx, .xlsm, or .pdf embedded with a malicious macro, or a link to a spoofed login page mimicking the bank's authentic interface. Julio, a white hat hacker, states that "attackers usually compose their email like a meeting invitation, a payment invoice, or a cooperation letter. Inside it, malware has already been embedded that activates when the file is opened." In addition, techniques such as typosquatting are used to fake the sender's domain so that it appears identical to the official email address, thereby misleading the victim both visually and psychologically.

In a data-breach attack, this stage involves developing exploit code targeting a previously identified vulnerability in the information system. The attacker's focus is on exploiting gaps in the Application Programming Interface (API), server misconfigurations, or the presence of legacy systems that have not been patched. Exploitation tools such as SQL injection, XML External Entity (XXE), and remote code execution are used to plant malicious instructions into the system's communication pathway. Hilal Akbar reveals that "several major attacks happened because of an API gap that was left open, then forced commands were inserted to read files or access user tokens." A cyber weapon in this stage is not merely malware in the traditional sense, but a digital command able to be used to access or extract sensitive data without detection.

Meanwhile, in a ransomware attack, the weaponization phase covers the creation of a payload in the form of a locker ransomware or crypto-ransomware that can be automatically activated once it successfully reaches the target system. Attackers often modify existing variants such as LockBit, Conti, or Ryuk — so that they are compatible with the target system and able to evade signature-based defense systems. Additional functions such as persistence, obfuscation, and command-and-control (C2) are also embedded to maintain control over the infected system. Dani, an official of BIN's Deputy for Cyber Intelligence, asserts that "attackers usually have already tailored their ransomware so it's not detected. They also have a command center to schedule when the files get encrypted." This reflects that the weaponization stage is designed specifically and not carried out generically.

Support for this observation is also given by Dr. Pratama Persadha, cyber-security expert and Chairman of the CISSReC Cyber Research Institute. He explains that "a lot of ransomware now uses polymorphic techniques, so every victim can be given a different form. Some spread via document, some via open port, depending on the weakness present." With polymorphism capability, malware can dynamically change its code structure to evade detection by digital-signature-based defense software, thereby complicating both detection efforts and forensic analysis.

An analysis of these three attack patterns shows that the weaponization stage is undergoing a significant transformation toward a more contextual and sophisticated approach. Not only technical gaps are being exploited, but also vulnerabilities in the working patterns and digital communication of banking institutions. Therefore, Cyber Threat Intelligence (CTI)'s capability to detect indicators of weaponization, such as file-structure anomalies, unusual Domain Name System (DNS) traffic, and deviant internal email-distribution patterns, becomes essential in building a proactive and adaptive defense system.

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The Delivery Stage

After the weaponization phase has successfully produced a malicious payload ready to be activated, the cyber attacker will enter the delivery stage, namely the process of distributing the attack instrument to the previously mapped target. The success of an attack at this stage is heavily influenced by the selection of a delivery vector appropriate

to the target's characteristics, and by the effectiveness of the evasion mechanism in avoiding detection by network security systems. This stage becomes a critical point because it marks the start of direct interaction between the attacker and the targeted system.

In a phishing scenario, payload distribution is carried out by sending a spear-phishing email specifically designed to resemble the internal communication pattern of a banking institution. The email content usually contains a format familiar to the victim, such as a virtual meeting invitation, a password-change notification, or a customer transaction confirmation. The attachment sent is often a .docx, .xlsm, or .pdf file with an active macro, or contains a hyperlink leading to a spoofed login page resembling the bank system's authentic page. Julio, a white hat hacker, explains that "usually after the bait email is sent, the attacker just waits for the victim to click the file or log in to the fake page. Once that happens, access opens immediately." This delivery technique exploits the target's work routine to lower vigilance and increase the chance of successful payload execution.

In the context of a data-breach attack, the delivery method is not always explicit, but rather through the insertion of malicious commands into a legitimate communication channel, such as an API request, a digital form, or a vulnerable backend service. The attacker uses this pathway to send the payload in the form of a hidden instruction disguised as normal system traffic. Hilal Akbar, Division Head at PT Mitra Integrasi Informatika, reveals that "in an attack on an API, the attacker sends a request that looks ordinary, but inside it there's a hidden command. The system doesn't realize, then it's executed." Thus, the delivery stage in a data-breach scenario shows a tendency to exploit application-layer vulnerability that is difficult to distinguish from regular system activity.

For a ransomware attack, the delivery mechanism is carried out through various more complex channels, including insertion via an email attachment, a drive-by-download scheme from a compromised site, and exploitation of open ports such as Remote Desktop Protocol (RDP) and Virtual Private Network (VPN). In many cases, the attacker exploits a botnet or malware already implanted in the network beforehand to deliver a modified payload. Dani from the State Intelligence Agency explains that "there's also entry through an open RDP port, after which the attacker just sends the ransomware remotely, using a command center." This indicates an evolution in ransomware distribution methods, from

a manual approach into an automated model integrated with a command-and-control (C2) system.

Further, Dr. Pratama Persadha adds that today's payload-delivery methods have developed into a much less conventional form. He states that, "sometimes now ransomware is inserted via a fake internal-application update, or a file uploaded to an internal file-sharing service. The system trusts it because it comes from within." This statement asserts that internal distribution vectors have now become a major challenge, because a system tends to trust data traffic originating from the internal environment, thereby opening opportunities for attacks arising from the manipulation of an insider delivery vector.

Overall, the delivery stage in all three types of attack — phishing, data breach, and ransomware — shows an increase in precision and camouflage capability. The distribution pathway is no longer simple, but has merged with a bank's work-communication infrastructure and internal information systems. Therefore, strengthening the early-detection system based on anomaly-based intrusion-detection systems, and implementing multi-factor authentication and monitoring of internal file-sharing activity, becomes highly crucial to prevent the successful process of payload delivery from advancing to the exploitation stage.

The Exploitation Stage

The exploitation phase is a crucial stage in the cyber-attack cycle, in which the malicious payload previously sent to the target is activated to exploit a gap in the security system. This activation becomes the starting point for deeper penetration into the system, aimed at gaining control, disabling defense mechanisms, or opening access to high-value digital resources. The success of this stage depends heavily on the precision in selecting an appropriate vulnerability, and on the payload's ability to evade detection by defense tools such as antivirus, firewall, and intrusion-detection systems.

In the case of phishing, the exploitation process occurs immediately after the victim takes action on the attachment or hyperlink sent through the manipulative email. This interaction triggers the execution of a malicious script, such as a macro in a document, that quietly implants malware into the user's system. This malware can open a reverse-shell connection, run a keylogger function, or extract credentials for further system access. According to Julio, a white hat hacker, "once the file is opened or the fake login

page is filled out, the system is compromised right away. The attacker can get in, monitor, even control the victim's system without them realizing it." This shows that an attack at this stage is not directly destructive, but relies on a hidden and manipulative infiltration technique.

In a data-breach attack, the exploitation stage is carried out through the execution of an exploitation command that infiltrates through a previously identified communication channel, such as an application-programming-interface (API) interface or a web form lacking adequate validation. Once that command is run by the target system, the attacker can access sensitive data such as authentication tokens, internal system configuration, or even a customer database. Hilal Akbar asserts that "when an API doesn't validate input properly, the attacker can use a certain command to directly access a file or grab data behind the scenes." An attack at this stage often proceeds without a conspicuous trace because it mimics legitimate data traffic within the system.

In a ransomware scenario, exploitation is usually carried out by immediately activating the encryption payload after successfully entering the target system. This process involves automatic execution that encrypts important files, disables backup features, and causes thorough operational disruption. Dani from the State Intelligence Agency (BIN) states that "once the ransomware is active, it immediately locks all important files, sometimes even disabling the security system so recovery isn't possible." Attackers also often exploit living-off-the-land techniques, using a system's built-in tools such as PowerShell or WMI to carry out system manipulation covertly and minimize detection.

Further, Dr. Pratama Persadha adds that today's ransomware variants have evolved with the ability to conduct a lateral scan of the internal network before executing encryption, so that they can reach files on other servers or accounts with privileged access. He explains, "ransomware now isn't just active on one computer, but can spread to other systems first. So before encryption, it maps out everything important first." This shows that exploitation does not target only a single point of weakness, but is directed toward a thorough takeover of critical digital infrastructure.

In general, the exploitation stage of all three types of attack shows that attackers no longer rely on a single technique, but design an adaptive, layered strategy. Attacks do not merely exploit technical gaps, but also exploit the complexity of the system architecture

and existing digital-interaction patterns within the banking work environment. To respond to this challenge, a detection system is needed that is not only signature-based, but behavioral-analytics-based, able to recognize deviant behavior, along with an incident-response system integrated with internal security policy and relevant external regulation.

The Installation Stage

After the exploitation stage has successfully opened initial access to the target system, the attacker will continue to the installation phase, namely the process of implanting a control instrument or malware into the system to solidify their presence on the network. This stage aims to create a persistent, hard-to-detect access pathway, and enables the attacker to maintain control over the target system in the long term. Installation techniques are generally carried out stealthily, by disguising the malicious file as part of a normal system process, so it is not detected by conventional security solutions.

In the context of a phishing attack, installation occurs when a user clicks a malicious attachment or link in a spear-phishing email, which then triggers the download and execution of malware. This malicious software typically copies itself into a system directory or creates a registry entry to ensure it is active every time the system is turned on. Julio, a white hat hacker, explains that "attackers usually plant the file somewhere that isn't routinely checked, it can even be disguised as a system file. Once the system restarts, it's active again right away." This indicates that the attacker has studied the structure of the target's operating system and exploited operational gaps often ignored by users.

In a data-breach case, installation is carried out through the insertion of a backdoor or web shell into a server that has previously been exploited. By planting a backdoor, the attacker can obtain repeated access to the system without needing to go through a penetration process from the start. Hilal Akbar, Division Head at PT Mitra Integrasi Informatika, states that "once they've managed to get in through the API, the attacker usually installs a backdoor inside the system, sometimes using a filename identical to a system file, so it's not noticed." This practice allows the attacker to remain in the system covertly and to carry out data-extraction activity or configuration manipulation over the long term.

Meanwhile, in a ransomware attack, the installation process is more complex because it involves several additional components such as a loader, a persistence mechanism, and a connection to a command-and-control (C2) server. These components are used to schedule data encryption, disable the backup system, and evade detection by defense systems. Dani from BIN confirms that "ransomware now doesn't attack directly, but installs a control agent first to monitor the system, then schedules the encryption time. It can be a matter of days before the payload is actually run." This approach shows that attackers use a time-delayed-activation strategy to avoid triggering system alarms and to broaden the scope of compromise before launching the main attack.

In addition, Dr. Pratama Persadha emphasizes that modern ransomware variants have adopted a fileless-installation technique, namely a method of implanting malware directly into random-access memory (RAM) without writing a physical file to disk. He states that, "now there's ransomware that doesn't leave a file trace. It lives in RAM, and stays active until its mission is complete. This is what makes it harder and harder to trace." This approach makes installation activity not only time-efficient, but also very difficult to detect through traditional forensic-analysis methods, particularly on systems lacking a real-time memory-activity logging feature.

Overall, the installation stage across all three types of attack shows a tendency for attackers to create a long-term foothold within the victim's system. Installation is not only aimed at maintaining access, but also supports subsequent attacks such as access-privilege escalation, system control, through to lateral distribution. Therefore, banking institutions need to strengthen their detection capability through the implementation of an endpoint detection and response (EDR) system, memory-based protection, and active monitoring of changes to the registry and system directories. Early detection of an indication of installation becomes crucial for breaking the attack chain before the attacker reaches the next stage in the kill chain.

The Command and Control (C2) Stage

After the installation process has successfully implanted malware into the target system, the next stage in the cyber-attack cycle is Command and Control (C2), namely the crucial phase in which the attacker establishes a covert communication channel with the infected device. Through this channel, the attacker actively obtains remote control,

allowing them to issue additional commands, extract data in stages, and carry out the further stages of the cyber operation adaptively. This communication channel is designed to resemble normal network traffic so that it does not trigger defense systems such as a firewall or intrusion-detection system (IDS).

In a phishing-attack scenario, the C2 stage functions to maintain an active connection between the victim's system and the control server held by the attacker. Once the malware is active, the victim's device automatically establishes a connection to the central server to receive new commands or update the attack function. Julio, a white hat hacker, reveals that "once the malware is active, it immediately opens a channel to the command center, then waits for instructions. It can be told to grab data, install a new program, or monitor the victim's activity." This approach shows that the attack is controlled in real time and can be modified according to the situation, such as taking over camera software, recording keystrokes (a keylogger), or extracting additional credentials.

In a data-breach case, C2 communication is used to facilitate covert data exfiltration, often through a modified protocol or an encrypted communication pathway, such as HTTPS, DNS tunneling, or a covert-channel method. According to Hilal Akbar, "sometimes attackers use DNS so they're not detected. The data is drip-fed, sent slowly through a channel that appears legitimate." This strategy shows how attackers disguise malicious activity behind a protocol regarded as legal by the security system, so that the traffic is not automatically blocked.

In the context of a ransomware attack, the C2 stage plays a role in carrying out the encryption process, delivering the decryption key, and managing negotiations related to the ransom demand. The attacker can activate the encryption command remotely, monitor the success of the attack, and determine when to deliver the ransom note. Dani from BIN asserts that "communication to the central server is what lets them monitor who's hit, how many files are encrypted, and when they'll send the ransom demand." In addition, this channel is also used to distribute malware updates, or to add a new module quietly into the victim's system.

Dr. Pratama Persadha states that C2 architecture has now evolved from a conventional, centralized model into a far more resilient decentralized structure. He explains that "C2 now widely uses platforms that are hard to trace. Some use Telegram,

some are also hosted on cloud services commonly used by offices, so they don't stand out." This evolution shows a shift from a fixed server toward peer-to-peer (P2P) infrastructure or the use of anonymizer cloud services, which significantly complicates tracking and mitigation efforts by security parties.

Overall, the Command and Control stage reflects how attackers maintain dominance over the target system through a hidden and adaptive communication channel. If not actively monitored, this activity can continue undetected over the long term, particularly when network traffic is very well disguised. Therefore, financial institutions need to strengthen network-behavior-analysis-based detection systems, expand threat-hunting practice, and implement automatic-blocking systems based on C2 indicators sourced from an up-to-date cyber-intelligence database (Tounsi & Rais, 2018).

The Actions on Objectives Stage

The closing phase in the Cyber Kill Chain cycle is Actions on Objectives, namely the stage in which the threat actor realizes the strategic goal of the cyber attack after obtaining thorough control over the target system. Unlike the previous stages, which focused on exploration or initial penetration, this phase is directed entirely toward causing real impact — whether in the form of theft of high-value information, sabotage of operational systems, thorough encryption of important data, or manipulation of the attacked institution's digital assets. Thus, this phase becomes the realization of the accumulation of technical, social, and tactical processes since the start of reconnaissance.

In a phishing scenario, the final action taken by the attacker is generally the takeover of authentication credentials, infiltration of the internal financial system, and further distribution of the malicious software. Once the victim's identity has been successfully compromised, the attacker gains the ability to carry out illegal transactions, access or modify customer data, and distribute malware into the institution's internal network. Julio states that "at the final stage, the attacker can use the victim's account to log into the bank system, access the database, or even use it to spread the attack to coworkers." This clarifies that phishing does not only function as an initial step, but also as an entry point enabling the scale of the attack to expand across the entire organizational system.

For a data-breach type attack, the Actions on Objectives phase is realized through large-scale data exfiltration aimed at sensitive data such as customer personal-identity information, financial-transaction history, through to internal documents of high strategic value. Hilal Akbar underscores that "usually the attacker already knows what they want to take from the very start. They immediately grab the important data, then send it to an external server using a staged method so it's not detected." This strategy shows that the attacker already has an explicit target and carries out the data-extraction operation using a stealthy method, so as to be able to evade traditional security-system detection.

Meanwhile in a ransomware attack, action at this stage covers thorough encryption of files and the system, along with the delivery of a ransom note to the victim. The attacker designs the attack so that it not only disables operational function, but also deletes recovery pathways such as backup and restore points, thereby placing the victim in a negotiation position. According to Dani from BIN, "at this stage, the attacker has already locked the system, sent the ransom message, and is monitoring whether the victim responds. Sometimes they also copy the data before it's encrypted, for extra pressure." This tactic depicts a double-extortion scheme, in which the victim is confronted with the threat of data leakage and system paralysis simultaneously.

Dr. Pratama Persadha adds that the Actions on Objectives strategy is increasingly complex because attackers have adopted a low-and-slow approach, namely carrying out their action gradually and covertly to maximize impact while avoiding detection. He explains that "now they don't immediately destroy or steal, but they map out first which asset is most critical, then carry out their action. Some wait for the moment, for example during an audit, or when the system is busy." A model of operation like this reflects the technical maturity and professionally designed maneuvering of cyber-attacker groups.

Overall, the Actions on Objectives stage reflects the realization of the entire cyber strategy that has been built since the earliest phase of the attack. The resulting effect can produce long-term consequences for the banking system, including service disruption, significant financial loss, a decline in public trust, and potential legal action. Therefore, the banking sector's cyber-security system must be designed not only to detect and respond early, but also to support the long-term recovery and learning process as part of strengthening cyber resilience (Linkov & Kott, 2019).

Integration of CTI and Detection Frameworks as a Reinforcement for Cyber-Kill-Chain-Based Attack-Pattern Analysis

The level of banking institutions' readiness to confront the escalation of cyber attacks is directly influenced by the effectiveness of applying a proactive cyber-intelligence approach. In a continuously developing digital landscape rife with vulnerability, Cyber Threat Intelligence (CTI) plays an important role as an early-detection mechanism for various attack indicators, even before reaching the exploitation phase. This approach rests on prediction and risk analysis sourced from technical, tactical, and strategic data, thereby going beyond the traditional, tendentially reactive response pattern (Tounsi & Rais, 2018).

Julio, a white hat hacker active in testing various institutions' security systems, states that although there is not yet a uniform national standard, a number of banks in Indonesia have selectively adopted various CTI platforms. He explains:

"Some tools commonly used in a strategic environment and beginning to be adopted by the banking sector include MISP for sharing indicators of compromise, Darktrace for AI-based anomaly detection, and ThreatConnect and Anomali ThreatStream for tactical and strategic intelligence gathering. Internal honeypot systems for profiling phishing and malware, and using cryptographic algorithms to detect data exfiltration."

From this explanation, it can be seen that threat detection cannot be run separately between divisions, but must be integrated and automated. The implementation of these various tools aligns with the four-layer CTI framework developed by (Tounsi & Rais, 2018), namely at the strategic, tactical, operational, and technical level, which comprehensively supports the formulation of objective-data-based security policy.

Julio also emphasizes that the attack-analysis process is carried out in tiers and simultaneously. He states:

"Analysis is run simultaneously at three levels, namely tactical, operational, and strategic. This process involves the use of SIEM such

as Cyber Kill Chain, Elastic SIEM and IBM QRadar, as well as the MITRE ATT&CK framework for mapping tactics, techniques, and procedures. We also use Ghidra and IDA Pro for malware reverse engineering, and Zeek and Suricata to monitor lateral movement."

The combination of these tools and frameworks strengthens the contextual detection system, not only to identify indicators of compromise, but also to understand the pattern of attack escalation from the initial stage through to system takeover. The MITRE ATT&CK framework developed by (Center for Threat-Informed Defense, 2020) has been globally recognized as an operational taxonomy based on adversary behavior, and serves as a reference for developing behavior-based security systems.

In addition, Julio explains that testing of cyber-defense-system readiness is also carried out through real simulation by a red team. He explains:

"The red team carries out reconnaissance through OSINT tools such as Shodan and SpiderFoot. The payload is sent via spear phishing. Once inside, the attack moves to lateral movement and privilege escalation, then the data is encrypted and extracted using DNS tunneling. All of this activity is closely monitored and evaluated before being reported."

This methodology is highly consistent with the Cyber Kill Chain framework proposed by (Hutchins, Cloppert, & Amin, 2011), which divides the attack cycle into seven phases from reconnaissance through to actions on objectives. Each phase provides a mitigation opportunity if intervened in a timely and systematic manner.

Support for this view is also given by Hilal Akbar, as Division Head at PT Mitra Integrasi Informatika. In his professional capacity, Hilal has handled the LockBit ransomware incident at Bank Syariah Indonesia and played a role in strengthening the digital-defense system at Bank BNI and Bank Indonesia. He emphasizes that:

"Banks use various CTI tools to detect and prevent cyber threats such as phishing, ransomware, and data leaks. Some commonly used tools include Recorded Future, SOCRadar, HYAS Insight, and E-Crime Intelligence Tools."

Each tool has a distinct functionality: Recorded Future presents real-time, contextual intelligence; SOCRadar provides malware indicators and risky IP reputation; HYAS

Insight detects command-and-control server infrastructure; and E-Crime Intelligence Tools monitor dark channels such as the dark web and cybercrime forums. The use of these tools enables banking institutions to shift from a reactive system toward a more predictive, real-time-threat-intelligence-based approach.

In addition to tools, further analysis methods are also explained in detail by Hilal:

"To analyze cyber-attack patterns, banks use a combination of tools and techniques such as SIEM (Splunk and IBM QRadar), TIPs such as ThreatConnect, and forensic methods such as memory analysis, disk imaging, and reverse engineering."

This approach forms a threat-intelligence loop involving a process of detection, validation, and further analysis, subsequently used in formulating a response policy. This model also represents the Adaptive Security Framework as explained by (Lopes & Oliveira, 2015), which emphasizes the need for a flexible defense system that evolves in line with the development of vulnerabilities and changing attacker modus operandi.

Hilal also highlights the main reasons the banking sector is a top target for attacks:

"Three main factors why hackers attack banking institutions are the high financial value, highly sensitive customer data, and the complexity of digital systems that open up many gaps."

These factors show that risk arises not only from the technical side, but also from the complexity of banking digital architecture, including the use of legacy systems, limitations in third-party integration, and a security-awareness gap at various organizational levels. For that reason, threat mapping and vulnerability intelligence become inseparable elements of a CTI-based defense strategy.

Overall, the integration between the CTI approach, the use of SIEM and TIPs systems, simulative testing, and evidence-based risk governance becomes the main foundation for developing an adaptive and strategic cyber-defense system. This framework implicitly reflects the Strategic Intelligence concept that emphasizes the importance of data-based analysis and anticipation of long-term risk (Walsh, 2017).

The ability to detect a cyber attack early is a crucial element in strengthening a banking institution's digital resilience. In the context of the rising intensity of phishing, ransomware, and data-breach attacks, the role of cyber intelligence becomes increasingly significant as a proactive instrument, not only for responding to threats, but also for predicting and preventing the escalation of cyber risk. This approach aligns with the Cyber Threat Intelligence (CTI) framework, which emphasizes the importance of collecting, analyzing, and disseminating threat information in a layered, behavior-based manner (Tounsi & Rais, 2018).

Based on an interview with Dani (alias), an official from the Deputy for Cyber Intelligence of the State Intelligence Agency, it is explained that the institution applies a series of technical and strategic steps as part of a proactive intelligence approach to potential digital attacks, particularly in the financial sector. In answering how early detection of an attack is carried out, Dani states that:

"We use a number of methods and tools such as sandboxing, network-traffic monitoring, security-log analysis, and digital forensics. In addition, penetration testing is also carried out periodically to identify potential technical vulnerabilities before they are exploited by irresponsible parties."

This statement depicts that BIN's strategy adopts a defense-in-depth approach, designed to provide the ability to detect and respond to threats simultaneously and in layers. Implementation of tools such as sandbox and log analysis supports the application of behavior-based detection, while periodic penetration testing reflects the strengthening of the predictive-defense function. This aligns with the CTI principle that encourages proactive detection of indicators of compromise through continuous monitoring of at-risk systems.

In answering the question of how attack-pattern mapping is carried out, Dani further explains that:

"We combine various data sources, including open source intelligence (OSINT), in mapping attack patterns. This is important to understand the attacker's strategy, including the techniques and tactics used in evolving attacks."

The use of OSINT as part of the CTI-analysis process broadens the reach of intelligence, enabling the accumulation of data from cyber forums, the dark web, and social media. This approach fits the MITRE ATT&CK taxonomy framework, which is used to identify tactics, techniques, and procedures (TTP) based on threat-actor behavior (Center for Threat-Informed Defense, 2020). The integration of data from various sources enables the intelligence agency to form a comprehensive and accurate threat modeling, and supports a mitigation strategy responsive to the dynamics of the actual attack.

When asked why the banking sector is one of the main targets of cyber attacks, Dani explains that:

"Motivations for attacking banks are usually varied, ranging from economic gain such as a ransom demand, theft and exploitation of customer data, through to destructive aims to shake the stability of the national financial system."

This statement indicates that the motive behind attacks against banking institutions is not limited solely to economic interest, but can also be geopolitical and strategic in nature. This approach aligns with the Strategic Intelligence framework (Walsh, 2017), which explains that security risk needs to be seen not only from the technical side, but also from the dimension of actor, intent, and capability that is long-term in nature. Thus, strengthening cyber resilience in the financial sector requires the involvement of a multidimensional intelligence system able to reach the strategic layer of attacks aimed at shaking confidence in the national economic system.

Overall, the interview with Dani shows that the cyber-intelligence approach applied by BIN is holistic, predictive, and adaptive in nature. The strategy applied does not only focus on incident handling, but also covers early detection, threat modeling, and continuous multidimensional analysis. This directly answers the second research objective, namely identifying how a proactive cyber-intelligence system can support early detection and response to attack patterns in the banking sector. This approach shows the operational application of CTI principles as well as integration with strategic analysis for confronting increasingly complex and structured threats.

4.1.1.2. Application of an Adaptive Security Approach to the Escalation of Cyber Threats in the Banking Sector

The increasingly complex and hard-to-predict development of cyber attacks requires the national banking sector to no longer rely solely on a purely reactive defense approach. As part of vital information infrastructure, banking institutions are required to adopt a digital-security strategy that is adaptive and sustainable, so as to be able to recognize potential threats before escalation occurs, while also strengthening mitigation capacity and systematic incident response. In this context, the Adaptive Security Framework (ASF) becomes an important reference, with four main, mutually integrated domains, namely predict, prevent, detect, and respond (Lopes & Oliveira, 2015). These four elements form the foundation for evaluating institutional readiness in confronting continuously and dynamically developing cyber risk, and ensure that the applied security system is able to adapt to changes in the threat landscape in real time and on an intelligence basis.

Analysis Using the Adaptive Security Framework — Predict, Prevent, Detect, Respond

In order to evaluate the banking sector's readiness to confront the escalation of cyber attacks, this research analyzes the intensity of implementation of the four main stages within the Adaptive Security Framework (ASF), namely predict, prevent, detect, and respond. Each stage is examined based on the number of references and the strength of the coefficient emerging from the triangulation of primary and secondary data. The classification results show that dominance still lies with the detection and response stages, while the prediction and prevention portion shows a relatively low intensity level. This reflects that the security approach applied by banking institutions tends to be reactive, with limitations on the anticipatory and preventive aspects. The details of this analysis are presented in the following table.

Table 4.3. Analysis of the Implementation Intensity of the Adaptive Security Framework Stages in the Banking Sector

Source: Researcher's Data

ASF Stage	Total References	ASF Intensity Coefficient	Stage Intensity Level	Interpretation
Predict	8	0.67	Moderate	The ability to predict cyber attacks is still not optimal. The limited references at this stage reflect the not-yet-built intelligence system capable of recognizing patterns, trends, and indications of attack early on. The low prediction index shows that CTI- and foresight-based preparedness has not yet fully become part of security governance.
Prevent	6	0.5	Low	The preventive stage is at the lowest position, indicating weak systemic prevention measures in confronting cyber attacks. This shows that policy, early-detection technology, and system-strengthening mechanisms (e.g. Zero Trust) have not yet become a main focus, so the system has not yet reached the ideal adaptive-security stage.
Detect	12	1	Dominant	The detection stage is the most dominant stage in the analyzed security system, showing that banking institutions tend to focus

ASF Stage	Total References	ASF Intensity Coefficient	Stage Intensity Level	Interpretation
				on identification and incident-reporting aspects after an attack occurs. This dominance reflects a reactive approach, in which the system responds more to symptoms than it anticipates causes, so preventive effectiveness becomes limited.
Respond	10	0.83	High	The response stage occupies second position in the display of reactivity, indicating that incident handling — such as system recovery and post-event reporting — remains a priority. This indicates a strong emergency-handling pattern, but one not yet fully supported by a proactive, long-term mitigation strategy.

First Stage — Predict

The predict phase serves as the initial stage in forming adaptive cyber resilience. This stage focuses on an institution's effort to carry out early anticipation of potential attacks through risk modeling, reading threat trends, and the use of Cyber Threat Intelligence (CTI) data. At the technical level, this approach covers the threat-modeling process, integration of vulnerability intelligence, and analysis of indicators of compromise aimed at identifying gaps before they are exploited by malicious actors.

Nevertheless, in practice within the Indonesian banking sector, the predict stage is often not carried out to its fullest extent. Julio, a white hat hacker, reveals that "many institutions haven't made threat modeling a strategic culture. They usually only look for a pattern after there's already been an incident, not before." This statement indicates that the predictive function has not thoroughly become part of a systematic cyber-security framework, but remains post-event driven.

A similar view is also conveyed by Dr. Pratama Persadha, who highlights the weak role of strategic intelligence at many institutions. He explains that "often the cyber-security personnel at an institution only act as system operators, not as risk analysts or scenario designers. Whereas predict is about the ability to read the direction of an attack based on technical and non-technical indicators." This emphasis shows that the not-yet-integrated function of strategic analysis within the digital-security architecture causes the prediction process to fail to have an impact on the design of long-term security policy.

Hilal Akbar of PT Mitra Integrasi Informatika also highlights that although a number of large banks have begun to use tools such as Recorded Future or ThreatConnect, the main obstacle is not the availability of technology, but the imbalance in human resources and the weak attention of policymakers to the use of threat intelligence. He states that "the biggest obstacle isn't the tool, it's the HR gap and the awareness at the managerial level to really follow through on that intelligence information into preventive policy." This confirms that the results of threat analysis are not always translated into strategic decisions, so it risks losing its anticipatory function.

In general, the weakness of the national banking sector at the predict phase stems from the minimal evidence-based-forecasting capability, the limited internal intelligence analysts, and the not-yet-optimal transformation of horizon scanning into digital-protection policy. Thus, for the predict pillar to function effectively, harmonization is needed between technological tools, HR analytical capability, and strategic leadership able to place early detection as the main foundation within an adaptive cyber-defense architecture.

Second Stage — Prevent

The prevent phase within the Adaptive Security Framework serves as the initial protective layer designed to stop an attack before it reaches the active-execution stage. This pillar emphasizes the importance of strengthening the system through closing technical vulnerabilities, strengthening security configuration, and applying protective policy such as zero trust architecture, periodic system updates, the least-privilege principle in access management, and improving authentication and encryption. In the banking ecosystem, these steps should be a foundation for dampening the potential success of an attack before the cyber attacker is able to implant or run a malicious tool.

Nevertheless, the reality of implementation at national banking institutions shows that this stage still faces various obstacles. Julio, a white hat hacker, reveals that a number of banks still use old protocols and have not carried out regular patching practices. He adds that "there's even a system that still relies on single-layer authentication without multi-factor authentication (MFA)," which significantly increases the chance of a successful attack at the initial entry point. This technological gap is mainly found in the use of legacy systems that are difficult to integrate with modern protection mechanisms.

Dr. Pratama Persadha also highlights that the prevention approach applied at many institutions is still procedural and merely administrative in nature, not based on in-depth technical evaluation. He asserts that "security-audit procedures are often only administrative, not technical. So gaps like default configuration, an open port, or weak authentication often escape attention." This account depicts that prevention effectiveness heavily depends on the depth of technical governance, not merely formal compliance.

Meanwhile, Hilal Akbar of PT Mitra Integrasi Informatika emphasizes the presence of structural obstacles in preventive decision-making. He explains that "often the IT department already has a technical recommendation for risk mitigation, but it takes a long time for approval from management because it's considered not a priority." This shows an imbalance between technical capacity and managerial sensitivity in understanding the urgency of protecting the cyber system.

In general, the weakness at the prevent stage can be traced to three main aspects: first, limited budget allocation for infrastructure updates and the elimination of vulnerable systems; second, weak application of granular, adaptive access control; and third, a lack of strategic awareness at the management level regarding the importance of strengthening

the prevention architecture as the main foundation of digital resilience. Without addressing these aspects, the prevent phase will remain a weak point that can be systematically exploited by threat actors.

Thus, the success of this pillar is not solely determined by the availability of security technology or formal procedure, but is very much determined by an institution's readiness to build a strong security culture, consistent application of the least-privilege principle, and the implementation of continuous technical audits able to proactively uncover vulnerability.

Third Stage — Detect

The detect stage within the Adaptive Security Framework functions as a strategic bridge between prevention efforts and response steps against a cyber attack. The main goal of this phase is to identify abnormal activity, system anomaly, or indicators of compromise (IoC) in real time, through a continuous monitoring system for the entirety of the digital infrastructure. The mechanism used covers the use of a Security Information and Event Management (SIEM) system, network-traffic classification, and artificial-intelligence (AI)-based behavior modeling. Frameworks such as the Cyber Kill Chain are used to systematically map each attack phase, from reconnaissance through to actions on objectives, to improve detection precision and the likelihood of early intervention (Hutchins, Cloppert, & Amin, 2011).

Even so, the effectiveness of this stage at a number of national banking institutions still faces various limitations. Julio, a white hat hacker, explains that "detection mostly still uses signature-based detection, whereas now attackers have widely adopted fileless techniques or operate in memory." He also highlights that some banks have indeed adopted SIEM such as Splunk or QRadar, but have not maximized its capability due to the minimal human resources with threat-hunting competence and the ability to compile contextual correlation rules.

A similar opinion is conveyed by Dr. Pratama Persadha, who assesses that most institutions still apply detection reactively. In his view, "the SOC team's role is often limited to logging and delivering notifications, when ideally they should be an active detection center able to initiate an initial investigation." This condition shows that the

detection challenge is not only technological in nature, but also related to organizational structure and operational authority that have not been optimally integrated.

Hilal Akbar of PT Mitra Integrasi Informatika adds that detection-system accuracy is highly determined by the availability of accurate historical data and behavior baseline. He reveals that "if the user's or system's behavior baseline is inconsistent or incomplete, an AI-based detection system will also struggle to distinguish what's an anomaly." In other words, detection-system effectiveness is heavily dependent on the quality of log management and the ability to map normal behavior as a reference.

In general, the main challenges at the detect stage in the Indonesian banking context include a lack of HR trained in threat analysis, not-yet-optimal use of SIEM and threat intelligence, and low intensity of simulative testing such as red teaming. If not immediately addressed, this condition will lengthen the detection lag for an incident and increase the risk of adaptive-response failure.

In closing, success at the detect stage requires an institutional transformation from a passive-reporting orientation toward a context- and behavior-based detection approach. The Cyber Kill Chain framework offers a systematic scheme for mapping the stages of an attack that enables early intervention, thereby increasing an institution's cyber resilience and strengthening the effectiveness of an integrated and adaptive cyber-intelligence system (Hutchins, Cloppert, & Amin, 2011).

Fourth Stage — Respond

The respond phase within the Adaptive Security Framework represents a crucial stage that determines an institution's level of effectiveness in handling and mitigating the impact of a previously detected cyber incident. This pillar's focus covers a series of fast, systematic actions including incident management, comprehensive reporting, isolation of the affected system, and a sustainable, measurable recovery strategy. This approach does not focus solely on the technical response, but also emphasizes the importance of cross-divisional coordination, crisis communication, and precisely targeted, intelligence-data-based decision-making (Lopes & Oliveira, 2015).

Nevertheless, in practice, this stage often becomes a weak point at various national banking institutions. According to Julio, a white hat hacker, the main obstacle lies in the slow internal-escalation process. He states that "the response is often late because internal approval or escalation is slow. Sometimes it's not the SOC that finds the incident but an outside party, only then do they start investigating." This indicates that dependence on a lengthy bureaucratic procedure has weakened response speed, while the internal security unit is often not empowered to take mitigative action independently.

In Dr. Pratama Persadha's view, this weakness is rooted in the absence of a playbook or standard operating procedures (SOPs) that are tested and updated regularly. He explains that "many institutions don't yet have a playbook or SOP that's tested regularly. When a big incident happens, they're confused about who should decide what, especially when the attack impacts reputation." This situation depicts a deficit in the incident-governance structure that should be able to provide clear, coordinated action guidance, including in handling external communication.

In line with this, Hilal Akbar adds that a weak reporting culture and slow decision-making also worsen response effectiveness. He reveals that "there was once an incident where the IT team already knew about the exfiltration, but because they were afraid of being seen as exaggerating, they waited until there was really a leak. In the end, the response was late." This statement reflects a psychological and cultural resistance in acknowledging and responding to an incident, which actually opens room for a wider impact.

In addition, the structural issue hindering response effectiveness also lies in the lack of system-recovery capacity and the low frequency of post-incident review. Many banking institutions do not routinely carry out incident-response simulations or tabletop exercises needed to test a team's readiness in confronting a real scenario. The absence of a mechanism for learning from prior incidents causes the potential for the same mistake to be repeated without procedural improvement.

Considering these various challenges, the success of executing the respond phase is greatly determined by three main components: speed and precision in technical execution, clarity of command structure and the decision-making process, and the ability to carry out reflection and continuous improvement based on the evaluation of prior incidents. If

these three elements are not met, an institution's response will tend to be reactive and fragmented, rather than adaptive and integrated.

Overall, the Adaptive Security Framework positions the respond stage as a key element in creating a layered, resilient cyber-security system. In the context of rising cyber risk marked by ransomware, phishing, and data-breach attacks in the banking sector, the speed and precision of the response become a crucial factor determining the success of recovery and minimizing loss on the reputational, operational, and financial side.

Analysis of Adaptive Security Framework Implementation

The application of an adaptive security approach becomes a necessity in confronting the increasingly complex and persistent dynamics of cyber-threat escalation in the Indonesian banking sector. This adaptivity concept is closely tied to the Adaptive Security Framework (ASF), which emphasizes the importance of integration between threat detection, rapid response, risk prediction, and continuous system recovery (Lopes & Oliveira, 2015). Security is no longer viewed as a fixed condition, but as a dynamic process that continues to evolve alongside changes in attack patterns.

Julio also explains that Indonesian banking institutions' readiness to adopt an adaptive security approach still faces various fundamental challenges. When asked whether current banking cyber protection is already optimal, he answers firmly:

"Not optimal. Even for the big banks. They're generally still very dependent on systems from foreign vendors and not developed modularly. Whereas modularity is important for scalable response and targeted mitigation. Many financial institutions also haven't applied the zero-trust principle. What exists is only basic network segmentation, and even that's often not updated regularly. Patch management is also often manual."

This statement indicates that the level of adaptation of the national banking system has not met the characteristics of an adaptive-security system, which ideally has the ability to learn from an incident, update the system automatically, and evolve against the

attacker's tactics. In addition, the high dependence on closed foreign vendor systems limits the flexibility to adjust security modules quickly and contextually. Within the ASF framework, this shows a weakness in the ability to predict and respond to changes in the attack landscape (Lopes & Oliveira, 2015).

When asked to explain Indonesia's cyber-security position based on the four available categories, Julio states explicitly:

"If I had to choose, I'd pick: high threat, low resilience. Because there are so many bank systems in Indonesia, especially the regional ones, that aren't technically inspected regularly. Even basic system hardening is often not maximized. Let alone vulnerability audits, log review, or SIEM. Many don't even have a SIEM. So even if there's an attack, they only realize it after the damage has been done."

This context indicates that weaknesses in technical defense do not only occur at small or regional banks, but are also still found at several large banks that structurally already have IT infrastructure. This shows that the adaptation problem is not solely due to limited institutional capacity, but also due to uneven or inconsistent implementation of security systems across all layers of the organization. This evaluation becomes an important argument supporting the need for adopting a systemic and thorough adaptive security approach.

Further, when asked about a suitable short-, medium-, and long-term strategy for banking institutions, Julio states:

"Short term: training and cyber-incident simulation for all levels of employees. Don't let it be just the IT team that understands there's an attack, because a lot of entry points come from an ordinary user. Medium term: build an AI-based detection system and data correlation. A lot of anomalies can be detected before they become an incident if we have correlative analysis. Long term: form an internal SOC, not just rent a vendor. This SOC has to be independent and connected to the global threat-sharing community."

The short-term strategy conveyed emphasizes strengthening the human-resource aspect through regular training and cyber-incident simulation. An adaptive approach in the short term needs to be realized through a security-literacy program targeting every level of the organization, from top management through to the most basic operational staff. Every individual within the institution needs to have the awareness that cyber threats are real, not merely technical, and can start from even the smallest human error. Understanding of psychological-manipulation techniques such as phishing and social engineering is a fundamental first step for building cyber hygiene and forming an organizational culture resilient to potential internal exploitation.

Meanwhile, the medium- and long-term strategy shows a direction of more structural system transformation. The application of an AI- and data-correlation-based detection system supports the implementation of predictive analytics in the ASF context. Meanwhile, the establishment of an internal Security Operation Center (SOC) independent of an external vendor reflects an effort to build sustainable system independence and resilience. An internal SOC enables real-time threat-intelligence integration and supports a faster detection-response process, as recommended in the Cyber Threat Intelligence model (Tounsi & Rais, 2018).

When asked whether new policy needs to be made or whether it is enough to strengthen existing regulation, Julio explains:

"In my view, strengthening implementation is enough. POJK 11/2022 is already good. But there are still many banks that don't understand its technical details. OJK shouldn't just check documents, but should also have a technical team that can review implementation. For example, directly checking the audit-trail system, automatic patching, or the use of encryption. If this is carried out, it'd be very good already."

This statement reinforces that the problem does not lie in a regulatory vacuum, but in weak execution and technical oversight. This asserts the need to apply an adaptive-governance approach, namely policy flexibility to respond to real conditions rather than merely following an administrative structure. In the context of Strategic Intelligence theory (Walsh, 2017), this condition shows that decision-making at the regulator level has not yet been fully based on real-time operational analysis, whereas banking technical

data could be used as an intelligence source for setting a more responsive and adaptive policy strategy.

Overall, Julio's answers reflect that cyber-security adaptation in the banking sector has not occurred systemically, whether from the technical, institutional, or policy side. To answer the escalation of increasingly sophisticated cyber attacks, more than mere security procedure is needed; a new paradigm is needed that makes the security system a self-evolving process driven by contextual intelligence and continuous monitoring.

An adaptive security approach is a strategic necessity for the Indonesian banking sector in responding to the increasingly dynamic escalation of cyber threats. The Adaptive Security Framework model emphasizes the importance of a security mechanism able not only to detect and respond, but also to continuously predict threats and carry out intelligent system recovery (Lopes & Oliveira, 2015). Security is no longer static, but becomes a process that continues to evolve following new attack patterns and vulnerabilities that emerge.

In this context, an adaptive approach cannot rely on only one technology or protocol, but must cover thorough organizational transformation. This is reflected in the following statement:

"Cyber security can't just stop at a firewall or antivirus. Banking needs a system that can respond automatically and learn from prior incidents. The system also has to be able to detect anomalies in real time and give strategic insight to leadership."

An effective response cannot be produced without an in-depth understanding of attack patterns and characteristics. Therefore, integration between an AI-based monitoring system, correlative analytics, and policy feedback becomes a key element of adaptive security. In this regard, a detection system must be able to identify a threat vector from the earliest phase and automatically trigger mitigation action based on predetermined policy.

Nevertheless, the implementation of this approach still faces structural obstacles. It is stated that:

"The problem is that a lot of banks just buy the tool, but don't use it to the max. Sometimes because they don't understand it, sometimes because there really isn't a clear SOP for response. A lot are also systems that stand alone, when they should really be interconnected so decisions can be made quickly."

This issue indicates that the obstacle does not lie in a shortage of tools or budget, but in the lack of integration and cross-unit operational capability. An unconnected (siloe) system causes delays in detection and decision-making, and does not support the execution of a context-based detection process. This shows the importance of the principle of modularity, system orchestration, and cross-functional training to strengthen organizational readiness.

Further, it is added that:

"Sometimes the problem isn't the tools or the technology, it's the mindset. Many still think cyber security is just the IT team's business. When in fact, many attacks actually come in through an ordinary user, for example through a phishing email or social engineering."

This statement highlights the cognitive and organizational-culture dimension as a crucial element in the adaptive-security model. As explained in the Strategic Intelligence approach, an organizational decision responsive to threats requires situational awareness from every level, not only technical but also managerial (Walsh, 2017). Therefore, an adaptive approach requires a security-literacy program that reaches every layer, including training on recognizing psychological manipulation and incident simulation.

Regarding a short-, medium-, and long-term strategy, it is conveyed that:

"Short term is audit and simulation. Medium term, build a correlative analytic system using AI. Long term, form a national threat-intelligence center that can collaborate between banks and government agencies."

This statement clarifies the adaptive framework at the operational stage. Audits and simulations carried out routinely become the basis for closing existing gaps. Meanwhile, a correlative analytic system enables early detection based on pattern, not merely on a

sign of crime that has already occurred. The formation of a national threat-intelligence center is also a strategic step for building a resilient ecosystem, strengthening inter-institutional coordination, and increasing national predictive capacity.

An evaluation of banking readiness in confronting the escalation of threats is also explicitly stated:

"If I were to assess it now, our position is at high threat, but medium resilience. There's a SOC, there's SIEM, but not everything can auto-respond yet. Threat hunting also isn't widely running yet. So the awareness is there, but it's still halfway there."

This assessment indicates that although infrastructure such as a Security Operations Center (SOC) and SIEM is already available, it has not all been supported by autonomous-response capability and active threat-intelligence-based practice. In the context of the Adaptive Security Framework, this shows a weakness in the respond and learn dimensions, which should be the basis for forming an evolving security system.

When asked about the effectiveness of existing regulation, it is explained that:

"OJK's regulation is already sufficient. What's lacking is the implementation. Sometimes the supervisors also don't understand the technical details, so they can't assess implementation. Whereas if a real technical audit was done, it'd be clear which is just a document and which actually runs."

This shows that the imbalance is not in a vacuum of rules, but in weak adaptive governance, namely implementation not accompanied by competent technical oversight. As explained in cyber-security-policy literature, good policy must be accompanied by technical-audit tools and regulator capacity to conduct evidence-based assessment (Gentry, 2023).

Finally, in answering how the system must continue to be developed so as not to stagnate, it is conveyed:

"Security isn't a one-time project. It has to keep being evaluated. There has to be a feedback loop, for example after an incident there's an immediate evaluation, it goes into the SOP, training is given, the system is updated. So the system learns."

This statement explicitly depicts the importance of the learning cycle within the adaptive-security framework. The feedback loop becomes a vital element enabling the system to continue being adjusted to new conditions and threats. This is a manifestation of the resilience capability built through an iterative process, not a one-time procedure (Linkov & Kott, 2019).

The application of an adaptive-security approach becomes an urgent need in confronting the dynamics of cyber-threat escalation in the national banking sector. The Adaptive Security Framework (ASF) emphasizes that information security is not a static condition, but a dynamic process involving the integration of early detection, rapid response, system recovery, and the ability to learn from every incident (Lopes & Oliveira, 2015).

Dani explains that an ideal short-term strategy for banking institutions to increase security adaptivity is to thoroughly update the system and cyber-defense tools, and to apply periodic backup on a separate network. He states that:

"Short term — the bank updates its digital-infrastructure security systems and tools and carries out a periodic backup process on a separate network."

This step reflects a commitment to operational system resilience, in line with the detect and respond pillars in ASF. By applying a segmentation and independent-backup strategy, financial institutions are able to maintain service continuity even if part of the system is compromised.

At the medium-term scale, Dani emphasizes the importance of carrying out a technology-transfer process from developer to financial institution gradually and continuously. The government is also urged to issue more comprehensive customer-data-

protection regulation, to strengthen the legal framework against digital exploitation. Dani states:

"Medium term — the bank carries out technology transfer periodically and the government issues laws related to customer-data protection as well as improvement of the procurement process for banking digital-infrastructure construction."

This approach asserts that adaptive security does not only depend on technology, but also on governance and progressive regulation. In this context, adaptive governance becomes an important concept enabling policy formulation based on actual operational needs and the continuously developing dynamics of threat (Walsh, 2017).

On the long-term horizon, Dani suggests strengthening technological independence by learning from other countries that have succeeded in developing their digital-security systems. He states that this process covers the stages of watching, imitating, and developing technology to achieve national independence.

"Long term — watch, imitate and develop technology so it becomes independent."

This strategy reflects the effort to build cyber sovereignty, namely a country's independence in developing and managing a digital-security system without dependence on a foreign technology provider. This aligns with the predict phase in ASF, which requires predictive and evolving capability against future threats (Lopes & Oliveira, 2015).

When asked about the objective condition of Indonesia's cyber security today, Dani states that Indonesia's position is still at the level of "high threat, low resilience." He explains that:

"This is because Indonesia is a major consumer country so consumer data has the potential to be stolen and used in an economic-war situation, and conversely, technology dependence is still high so it's judged far from technologically independent (vulnerable to being influenced by the development of foreign technology)."

This statement shows that the biggest challenge does not only come from the technical side, but also from structural dependence on foreign technology and weak domestic detection infrastructure. This limits national capability in building an adaptive and resilient cyber-security system.

Next, when asked about the character of current national cyber-security policy, whether it is already adaptive and proactive, or still reactive, Dani explains that:

"Most policy in Indonesia is still reactive in nature, tending to wait for an incident to happen before setting concrete steps. It should be able to be more anticipatory, for example through monitoring threat indicators and involving technical parties in designing policy from the outset."

This statement shows that institutionally, the policy-formulation system has not yet been fully integrated with cyber-intelligence technical capacity. In Strategic Intelligence theory, an effective decision-making process must rest on actual information and early indicators provided by the operational unit (Walsh, 2017). The absence of this synergy can hinder the state's ability to confront dynamic and structured threats such as large-scale ransomware attacks or data breaches.

Overall, Dani's explanation reflects that the application of an adaptive security approach in the Indonesian banking sector still faces various challenges. Although there are initiatives at the operational level and clear strategic recommendations, national policy in general has not yet shown a systemic response based on prediction and long-term mitigation. Therefore, integration between the principles of the Adaptive Security Framework and Strategic Intelligence becomes very important to ensure that Indonesia's digital-security policy is no longer reactive but more adaptive and proactive.

4.1.1.3 Strengthening the Implementation of Information-Protection Policy in the Banking Sector's Cyber-Security System

In an effort to answer the three questions and research objectives that have been formulated, a review of national regulatory documents is essential to assess the extent to which an adaptive cyber-security approach, a proactive cyber-intelligence system, and an integrated information-protection policy have been designed and enacted by the state.

The following analysis presents five main regulations and one Policy Guideline of strategic relevance to the banking sector's cyber-security system in Indonesia.

Table 4.4 Evaluation of the Adoption and Implementation of Cyber-Security Regulation by Banking Institutions

Source: Researcher's Data

Type of Regulation & Policy Guideline	Adoption Status by the Banking Sector	Weaknesses in Implementation
POJK No. 11/POJK.03/2022	Already adopted structurally, but not yet optimal in technical practice	Early detection is not yet based on active intelligence, system testing is formalistic, the IT-oversight unit is not yet autonomous
POJK No. 38/POJK.03/2016	Already adopted by most banks, but not yet thorough in adaptive technology	Network monitoring not yet integrated with threat intelligence, firewall and encryption not yet adaptive
Presidential Regulation No. 82 of 2022	Applied in the form of CSIRT formation and cross-sector coordination	Inter-agency coordination effectiveness still varies, CSIRT readiness not yet even
2024 National Cyber Security Landscape	Serves as a strategic reference for national roadmap and policy	Implementation is still macro in nature and not yet directly tied to a bank's operational system
Ministry of Communication and Digital Affairs Regulation No. 5 of 2025	Applied at public institutions, some banks are adjusting to the data-classification provisions	There are still obstacles in thorough encryption and data classification in the banking sector
Ministry of Communication and Digital Affairs	Some banks are beginning to adjust application	Software-testing standardization has not yet been fully implemented across all

Type of Regulation & Policy Guideline	Adoption Status by the Banking Sector	Weaknesses in Implementation
Regulation No. 6 of 2025	development to the SPBE standard	banking applications

4.1.1.3.1. Financial Services Authority Regulation No. 11/POJK.03/2022

One of the key policies serving as a strategic foothold for strengthening the information-security system in the banking sector is Financial Services Authority Regulation (POJK) No. 11/POJK.03/2022 on the implementation of information technology by commercial banks. This regulation requires every bank to implement a holistic information-technology risk-management system, with an approach based on formal documentation and internal-oversight mechanisms. This regulation has six main pillars, namely: first, the formulation of an information-technology strategy and data-security policy based on risk analysis; second, the establishment of an organizational structure accommodating a technology-oversight and control function; third, the establishment of an obligation to report an incident to the authority in a tiered, timely manner; fourth, the periodic implementation of vulnerability assessment and penetration testing; fifth, the development of human-resource capacity through cyber-security certification; and sixth, the establishment of an independent unit responsible for internal oversight of the information-technology system (OJK, 2022).

In addition, this regulation emphasizes the application of a three-lines-of-defense approach, which divides the control function into three layers: the business unit as the first line, compliance as the second line, and internal auditors as the third line. Banking institutions are required to compile an incident response plan and to hold periodic incident-handling simulations as part of the organization's readiness for cyber risk.

Nevertheless, the results of policy observation and interviews with a number of informants show that the implementation of this regulation at the operational level still faces a number of obstacles. First, the information-security strategy applied by many banks — particularly regional development banks and non-bank financial institutions — tends to be administrative in nature and has not been fully supported by an adaptive, integrated risk-monitoring system. Some incident response plans that have been designed

have not gone through a periodic testing process or are not directly linked to real-time risk indicators. This has an impact on delayed detection of ransomware-as-a-service-based attacks and AI-driven phishing, which are becoming ever more common in the domestic context (BSSN, 2024; Julio, 2025).

Second, although the regulation has required the implementation of vulnerability assessment and penetration testing, many banking institutions have not systematically applied continuous vulnerability assessment or adversarial testing. The testing carried out is generally formalistic, meant merely to satisfy the annual audit obligation, without touching the rapidly developing technical dimension of threat, such as zero-day vulnerabilities. This condition opens a gap for threat actors to exploit systems that have not been updated, as occurred in the LockBit 3.0 attack case against the national banking information system (Hilal, 2025).

Third, the independent IT-oversight units established at various institutions are still merely structural, without adequate functional-authority support. In many cases, this unit remains under the authority of the information-technology directorate and lacks direct access to threat intelligence or the authority to make a mitigation decision quickly and autonomously.

Overall, although POJK No. 11/POJK.03/2022 has formulated a comprehensive regulatory framework consistent with adaptive-security principles, the gap in implementation at the institutional level is the main factor explaining why the Indonesian banking sector still faces significant vulnerability to cyber attacks. The real challenge lies in the application of intelligence-based early detection, continuous system testing, and the systemic and collaborative strengthening of incident-response readiness.

4.1.1.3.2. POJK No. 38/POJK.03/2016

One regulatory framework playing a fundamental role in shaping the resilience of the national banking sector's information-technology system is Financial Services Authority Regulation No. 38/POJK.03/2016 on the Implementation of Risk Management in the Use of Information Technology by Commercial Banks. This regulation contains technical provisions designed to guarantee the confidentiality, integrity, and availability aspects of

the information system through a defense-in-depth approach, namely a layered defense model covering physical, systemic, and operational-policy protection (OJK, 2016).

In detail, the regulation requires every bank to apply various main protection mechanisms. Among these is the implementation of a firewall with strict configuration to filter risky data traffic, the application of an encryption system for data at rest and data in transit, periodic implementation of vulnerability assessment to detect security gaps, and real-time network monitoring based on technical threat indicators. In addition, banks are also required to compile an access-rights-management policy using the least-privilege principle and to adopt a role-based authorization system, or role-based access control, to limit the potential for misuse of internal access.

From a conceptual perspective, this regulation supports the construction of a responsive cyber-intelligence system through early detection and rapid response to a potential threat, in line with the first research objective and the second research question. However, in practical implementation, a number of structural and technological weaknesses still limit the effectiveness of applying that principle.

Many banking institutions have not dynamically updated their firewall configuration. The filtering parameters used are often not updated to keep pace with the evolution of the latest digital threats, including zero-day attacks and advanced persistent threats (APT). The absence of this adaptive mechanism opens a gap for threat actors to infiltrate through protocol-obfuscation or tunneling techniques, which are difficult to detect by traditional systems.

Although encryption is required, not all institutions have adopted the end-to-end-encryption standard in their communication systems, particularly for internal data transmission and inter-application integration. Many banks still rely on limited encryption such as the Transport Layer Security (TLS) protocol for website access alone, without applying thorough protection to their IT architecture.

The implementation of vulnerability audits at several financial institutions is still administrative in nature and does not yet reflect a scenario-based testing approach or adversarial simulation, such as the red-teaming method. Non-involvement in a simulation of attack designed to mimic the strategy of a real threat actor causes a lack of readiness

to confront exploitation methods tailored to the target's specific infrastructure, as reflected in the LockBit ransomware incident against a national banking institution (Hilal, 2025).

The network monitoring mandated in this regulation has not been entirely connected to the national cyber-threat-intelligence system. Most banks have not yet implemented Security Information and Event Management (SIEM) as an integrative system for incident-data analysis. As a result, detection of a security anomaly is often reactive and based on manual notification, rather than a prediction-based or predictive-analytics-based system.

Thus, although POJK No. 38/POJK.03/2016 has provided a comprehensive technical-security structure, there is a significant gap between the normative provisions and their execution capability in the field. The lack of integration with adaptive technology, the minimal dynamic system testing, and the limited access to up-to-date threat intelligence are the dominant factors explaining why the Indonesian banking sector remains vulnerable to the escalation of sophisticated, machine-enabled-threat-vector-based cyber attacks.

4.1.1.3.3. Presidential Regulation No. 82 of 2022

on the Protection of National Vital Information Infrastructure affirms that the financial sector is part of strategic infrastructure that must be systematically protected. This regulation mandates the formation of a Computer Security Incident Response Team (CSIRT), the periodic implementation of technical audit, and inter-agency coordination for responding to digital threats. Thus, this regulation answers all of the research questions and objectives because it encourages an adaptive, proactive, and cross-sector-integrated cyber-security approach (BSSN, 2022).

4.1.1.3.4. The 2024 National Cyber Security Landscape

released by BSSN describes the challenges, priorities, and national action plan in confronting the escalation of digital threats. Several main focuses of this document include improving human-resource competence, applying the principle of cyber hygiene, and the use of a threat-intelligence system to detect and analyze attack patterns such as phishing, ransomware, and data leaks. Therefore, this document is highly relevant to the

second question and first objective, because it emphasizes the importance of strengthening a structured, proactive cyber-intelligence system (BSSN, 2024).

4.1.1.3.5. Ministry of Communication and Digital Affairs Regulation No. 5 of 2025

on Public-Scope Electronic-System Operators sets the classification of digital data into three risk categories, namely open, restricted, and closed data. It regulates the obligation to apply encryption at storage (at rest) and transmission (in transit), and prohibits the use of an online network for managing closed data. This regulation answers the third question and third objective because it provides a clear technical and normative framework for protecting the strategic information of the public sector, including designated financial institutions (Ministry of Communication and Informatics, 2025a).

4.1.1.3.6. Ministry of Communication and Digital Affairs Regulation No. 6 of 2025

on Technical Standards and Procedures for Building Electronic-Based Government-System (SPBE) Applications requires application development to follow national and international standards, such as SNI ISO/IEC/IEEE 12207 regarding the software life cycle, ISO/IEC 25000 for software-quality evaluation, and ISO/IEC/IEEE 29119 for system testing. This regulation answers the first question and second objective because it requires digital-system governance based on quality testing and documented development, while also encouraging the adoption of an adaptive security framework (Ministry of Communication and Informatics, 2025b).

Overall, these six regulations reflect a national cyber-security policy architecture designed to be able to adaptively respond to the dynamics of cyber threats, encourage proactive threat detection, and strengthen cross-sector integration within the strategic-information-protection system. Nevertheless, the effectiveness of this regulatory framework's implementation is highly dependent on institutional readiness, the availability of technical resources, and the continuity of coordination between stakeholders in the national financial sector.

The application of a comprehensive information-protection policy is a central element in strengthening the banking sector's cyber-defense system. In Indonesia, the existence of

regulation such as Financial Services Authority Regulation (POJK) No. 11/POJK.03/2022 has provided a clear normative framework regarding information-technology management, cyber-risk management, and the protection of banking data and digital infrastructure. Nevertheless, that regulation's effectiveness is highly determined by the quality of its implementation in the field. Within the Strategic Intelligence framework, policymaking does not only stop at the formulation of a norm, but must be directed at responding strategically to developing technical and institutional reality (Walsh, 2017).

The results of an in-depth interview conducted with an informant using the alias Julio assert that the main challenge does not lie in the weakness of the policy, but in the weak understanding and technical implementation by banking institutions. Julio states:

"In my view, strengthening implementation is enough. POJK 11/2022 is already good. But there are still many banks that don't understand its technical details. OJK shouldn't just check documents, but should also have a technical team that can review implementation. For example, directly checking the audit-trail system, automatic patching, or the use of encryption. If this is carried out, it'd be very good already."

This statement reflects that most institutions have not been able to translate normative provisions into operational steps consistent with the latest cyber-security standards. Technical-management weaknesses such as non-optimal patch-management implementation, the absence of a thorough audit-trail system, or the minimal use of encryption on critical data, become main indicators of a weak strengthening of policy implementation. In this context, the Adaptive Security Framework approach requires a security system able to adapt, learn from incidents, and continuously update its strategy and protection infrastructure (Lopes & Oliveira, 2015).

Julio also highlights that oversight by the regulator tends to be administrative and not yet supported by adequate technical verification. He further explains:

"OJK shouldn't just check documents, but should also have a technical team that can review implementation. For example, directly checking the audit-trail system, automatic patching, or the use of encryption."

This statement depicts the imbalance between formal compliance and the actual condition of the security system run by a bank. In the author's view, an oversight approach of this kind is not yet in line with the adaptive-governance principle, which should provide flexibility in responding to the dynamics of risk with an evidence-based approach. Direct technical evaluation of a bank's security system becomes important, given that policy effectiveness heavily depends on the connection between the strategic and operational dimension (Walsh, 2017).

This condition also reflects the need for synergy between the regulator and banking institutions in building a cyber-resilience culture that does not rely solely on administrative reporting. In the context of Cyber Resilience, an institution's ability to recover and adapt from a cyber incident heavily depends on a policy structure that can be applied technically and developed based on lessons from prior events (Linkov & Kott, 2019). Therefore, strengthening policy implementation means not only enforcing compliance, but also ensuring that the policy can be translated into a resilient, adaptive system.

In addition, Julio's statement indicates that many banks still lack an internal team with adequate technical understanding to bridge regulation with cyber-security practice. The lack of a cyber-policy-translation unit at many institutions causes a policy such as POJK No. 11/POJK.03/2022 not to be optimally absorbed. This reinforces the importance of improving institutional capacity in the banking sector so that it can carry out regulation substantively, not merely as administrative formality.

Overall, it can be concluded that information-protection-policy effectiveness is heavily determined by the integration between the regulatory dimension and a bank's internal technical capability. Strengthening implementation cannot be sufficiently achieved through brief training or regulation socialization, but requires a fundamental change in oversight design, technical-resource allocation, and strategic partnership between the regulator and financial-industry actors. If this step is not immediately taken, normatively compiled policy will fail to become the main foundation in confronting the increasingly complex and systemic escalation of cyber threats.

In confronting the increasingly complex dynamics of cyber-threat escalation, banking-sector information-protection policy is not only required to be reactive, but must also foreground preventive, proactive, and adaptive principles. According to Hilal Akbar, policy effectiveness is highly dependent on its consistent application across all organizational units. He asserts that:

"Policy that can strengthen the information-protection system, in my view, needs to be preventive, proactive, and adaptive. It's not enough to just regulate, it has to be able to be carried out. So it's not just having an SOP, but also simulating it regularly. And everyone in the organization knows their responsibility."

This statement shows that a normative policy is not yet effective enough if not accompanied by a periodic-testing mechanism and cross-functional training. This is in line with the adaptive-security approach developed by Lopes and Oliveira (2015), in which a security system must be able to dynamically adjust to changes in threat, both technically and structurally.

Further, Hilal also emphasizes the importance of governance oriented toward risk assessment and institutional context. He explains that:

"Every bank has a different ecosystem. So the protection and information-security approach has to be based on each institution's own risk assessment. Don't make it uniform for everyone. But OJK could make a kind of baseline guideline and maturity model so a bank knows where its level stands."

This view reinforces the risk-based-governance concept, which underscores the importance of policy differentiation based on the level of vulnerability, resource capability, and digital exposure of each institution. With a maturity-level assessment framework, protection policy can be adjusted contextually without losing the minimum standard set by the regulator.

Beyond that, Hilal highlights the importance of cross-unit integration in the implementation of information protection. He states:

"Security isn't just the IT team's business. Other units like legal, audit, up to customer service also have to understand and be involved. Because

sometimes an attack doesn't go straight to the system, but through email, phone, or a fake document. If the other units don't understand, it'll get breached too."

This statement asserts the importance of applying the security-by-design principle involving the entire business unit, not just the information-technology team. This integration aligns with the Cyber Resilience framework as explained by Linkov and Kott (2019), which places people and organizational process as an inseparable part of a resilient digital-security system.

To ensure the sustainability of policy execution, Hilal also suggests an audit-and-evaluation mechanism that is not only administrative, but also covers the technical aspect. He adds that:

"Audit shouldn't just look at the document. It has to look at the system log, patch management, and the encryption mechanism. Sometimes everything looks good on paper, but there's no implementation. So there has to be a team that understands the technical side too."

This recommendation aligns with the evidence-based-governance principle, which requires empirical validation of policy-implementation effectiveness. Within the Strategic Intelligence framework (Walsh, 2017), this shows the need to use operational data as a basis for regulatory decision-making and compliance evaluation.

Finally, Hilal underscores the need to improve cyber-security literacy among management and decision-makers to prevent a gap between policy and technical reality. He states that:

"Sometimes the problem is with the decision-makers. They make policy without understanding the technical side. In the end, the policy is good, but can't be carried out. So training and regular discussion between the technical team and management is also needed."

This statement asserts the need for an effective communication bridge between the policy and operational domain, to ensure that every policy compiled is not only normatively ideal, but also applicable in real conditions.

Dani states that a regulatory framework such as the Financial Services Authority Regulation (POJK) has actually provided an adequate normative basis for supporting an information-protection system in the banking sector. However, he highlights that implementation in the field still faces various technical and institutional obstacles. He states that:

"Regulation like POJK is already good enough as a basic framework, but the challenge is in technical implementation. Many banks don't yet have an adequate early-detection system and incident-reporting system, and there's still weak technical literacy at the policymaker level in the financial sector."

This statement indicates that there is a significant gap between policy construction and an institution's technical capability to carry it out effectively. Within the adaptive-governance framework, this gap reflects a lack of synergy between regulation and implementation practice, which ideally could mutually adjust to the continuously changing dynamics of the cyber-risk landscape (Walsh, 2017).

Further, Dani asserts that the challenge in implementing policy does not only originate from within banking institutions, but is also due to the absence of an independent institution specifically mandated to carry out a thorough technical audit of the financial sector's digital-security system. He explains:

"Ideally there'd be an independent technical body, not just from the financial regulator, that could carry out a thorough audit of a bank's security system. Not just an administrative audit, but one that really goes into the technical side like a source-code audit, penetration testing, and evaluation of the encryption system."

This statement reflects the urgency of forming an oversight body with high technical competence to ensure that system protection is carried out comprehensively, not merely administratively. That technical audit is part of the cyber-hygiene cycle, which is crucial in the context of Cyber Threat Intelligence (Tounsi & Rais, 2018). The CTI principle requires that the entire system be able to respond adaptively based on technical data, indicators of compromise, and relevant threat-vector mapping.

Within the framework of long-term strengthening, Dani emphasizes the importance of building an integrative policy and ensuring active coordination among stakeholders. In his view:

"There has to be a policy that guarantees rapid threat-data exchange between banks, regulators, intelligence, and law enforcement. Without this coordination, the response will always be slow. Even a big attack can be detected late because the systems aren't connected."

This context reflects the collective-defense principle, namely a response pattern built through collaboration between entities within a national security ecosystem. This approach is also in line with the strategic-foresight framework, which foregrounds the formulation of policy based on multidimensional and cross-sector threat scenarios (Walsh, 2017).

Dani also adds that the direction of future policy should place greater emphasis on the preventive, adaptive, and proactive aspect, not merely be reactive when an incident occurs. He states:

"Future policy has to drive readiness, not just response. For example by mandating threat simulation, incident response drills, and the formation of an internal rapid-reaction team at every bank."

This proposal reflects the Cyber Resilience principle, which underscores the importance of a system's resilience in confronting disruption and its ability to continuously adapt (Linkov & Kott, 2019). This approach also strengthens the need to transition from a reactive risk-management paradigm toward an adaptive and proactive cyber-defense system.

4.2. Foresight Horizon Scanning Analysis Method

This section describes the foresight approach used as a strategic analysis method for navigating the dynamics of cyber threats against national banking institutions. The foresight approach was chosen to complement the intelligence-analysis framework by systematically and structurally providing a medium- to long-term perspective. This method consists of five main stages — event, trend, driver, scenario planning, and

roadmap — all of which are integrated to map and anticipate the increasingly complex, coordinated, and disruptive change in cyber-attack patterns.

4.2.1. Event

The event-identification stage in horizon scanning is used to trace empirical occurrences reflecting the rising intensity and complexity of threat to the national banking sector's information-security system. A number of crucial incidents have been successfully mapped and show an increasingly coordinated, developing, and systematic attack pattern, particularly in the form of ransomware, phishing, and data breach.

Table 4.5. Cyber-Attack Events Against Banking Institutions in Indonesia

Year	Institution	Type of Attack	Attack Description
2022	Bank Rakyat Indonesia (BRI)	Phishing	Phishing hit thousands of BRI customers due to a fake message resembling a mobile-banking service notification.
2023	Bank Syariah Indonesia (BSI)	Ransomware	A LockBit 3.0 ransomware attack paralyzed BSI's service for >5 days and leaked 1.5TB of customer data.
2024	Bank Rakyat Indonesia (BRI)	Ransomware (AI-powered)	A suspected AI-powered attack caused a critical disruption to BRI's digital services.

The event-identification stage in horizon scanning is used to trace empirical occurrences reflecting the rising intensity and complexity of threat to the national banking sector's information-security system. A number of crucial incidents have been successfully mapped and show an increasingly coordinated, developing, and systematic attack pattern, particularly in the form of ransomware, phishing, and data breach.

One major incident occurred in May 2023, when Bank Syariah Indonesia's (BSI) information-technology system was successfully infiltrated by the LockBit 3.0 ransomware variant. This attack not only caused a digital-service halt for more than five days, but also caused a leak of around 1.5 terabytes of sensitive customer data. The attack modus showed complex stages, starting from initial access, takeover of the internal system, through to the data-exfiltration stage, showing the not-yet-optimal early-detection and incident-response capability held by the institution (Chakravarti, 2023). Given that BSI is the result of a merger of several national sharia entities, this incident raised public concern about the institution's level of digital readiness in confronting strategic cyber risk.

Another important event involved a suspected attack against Bank Rakyat Indonesia (BRI) at the end of 2024. Although there has not yet been an official statement regarding a data leak, several sources state that a number of digital services experienced a critical disruption. The reported attack pattern reflects the use of artificial-intelligence-based technology to penetrate a system through a pathway undetectable by signature-based-detection defense methods, indicating an escalation both technically and on the threat-actor side (Kompas, 2024).

The phenomenon of escalating phishing attacks is also worth noting, particularly throughout 2021 to 2022. The Financial Services Authority's annual report shows an increase in social-engineering practices targeting banking customers through fake text messages and websites resembling an official application. An incident in 2022 hit thousands of BRI customers who lost funds due to clicking a phishing link disguised as a mobile-banking service notification. This event highlights a large gap in user protection, both from the minimal digital-security education and the absence of an effective multi-layer authentication system (Financial Services Authority, 2023).

In addition, data-breach incidents have also risen significantly. One event in 2022 indicated that the data of hundreds of thousands of customers, including account and transaction-history information, had been traded on a dark digital forum. Although there has not yet been an official acknowledgment from the authorities, an independent investigation concluded that the data breach was made possible through a security gap at a third-party service provider that had not been fully protected by the parent bank's security policy (Cyberthreat.id, 2022).

Data from the 2024 National Cyber Security Landscape document published by the National Cyber and Crypto Agency shows that throughout 2024, 514,508 ransomware activities were recorded, along with more than one million phishing-attack attempts and illegal data-harvesting attempts. Of that total, the LockBit variant is the most dominant, with more than 102,000 detected activities. This fact indicates that the character of attack has changed from opportunistic to more targeted, using vulnerability-mapping techniques, digital psychological manipulation (social engineering), and machine-learning-based attack automation (National Cyber and Crypto Agency, 2024).

In the context of threat actors, the involvement of organized cyber-criminal groups is very prominent, particularly a group running ransomware-as-a-service (RaaS) such as LockBit and Conti. The attacker's main motive is economic gain, but their strategy has transformed to become more systematic, resembling the tactics of an advanced-persistent-threat (APT) group that uses vulnerability-mapping techniques to target weak points in digital infrastructure. Several data breaches, such as those alleged to have occurred at BCA and Bank Jatim, even involve a possible failure of third-party oversight or an insider threat, which becomes an entry point for external infiltration.

In terms of incident classification, the resulting impact covers a broad spectrum ranging from technical damage, operational disruption, reputational loss, through to potential legal consequence. The LockBit case at BSI not only paralyzed basic services such as ATM and mobile banking, but also gave rise to a public-trust crisis and the risk of violating personal-data-protection regulation. The BRI phishing incident directly impacted customer finances, while the case at Bank Indonesia raised concern over systemic vulnerability at the national level.

Analysis of the systemic point of failure indicates that the weak point often originates from the people and process factor. Inadequate cyber-security training for staff, an unresponsive incident-handling procedure, and the absence of a real-time monitoring system contributed to the attack's success. Although several banks have implemented a firewall and static-scanning system, not all are equipped with endpoint detection and response (EDR) or extended detection and response (XDR) technology adaptive to zero-day attacks.

A review of early indications shows that there were a number of anomaly signals or logs that escaped detection or were not followed up quickly. In the BSI incident, for example, suspicious activity from an internal account was found before the mass-encryption attack occurred. However, the absence of a behavioral-based-alert mechanism caused a delayed response, which ultimately worsened the attack's impact.

The above series of events shows that the banking sector is now not only the main target of large-scale cyber attacks, but also faces escalation both technically and tactically. Therefore, within the framework of strategic-intelligence-based foresight, identification of these events becomes the initial foothold for compiling a mitigation scenario for the 2025–2029 period. This stage also strengthens the urgency of needing a defense strategy designed based on a proactive threat-detection system, an adaptive information-security structure, and an information-protection policy integrated across institutions and sectors.

4.2.2. Trend

To identify the pattern of increase and shift in cyber threat, the trend stage in the foresight analysis is used to trace the quantitative dynamics of various types of attack and data exposure in cyberspace. Table 4.6 presents the trend of ransomware, phishing, data-breach attacks, and data exposure on the dark net during 2023 and 2024. This data is adapted from a cyber-security-landscape report released by BSSN, and provides a picture of the significant escalation and shift in threat intensity, particularly on the aspect of data leakage and the expansion of the attack surface in the hidden domain.

Table 4.2. Table 4.6. Cyber-Attack Trends and Dark Net Exposure

Source: Adapted from the BSSN cyber-security landscape

Year	Ransomware	Phishing	Data Breach	Dark Net Exposure
2021	218,043	6,686,616	20	83,991
2022	441,320	11,812,336	36	459,879
2023	1,011,209	26,619,476	103	1,674,185

2024	514,508	26,771,610	241	56,128,160
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Based on the data in Table 4.2, the dynamics of cyber attacks against the national digital sector during the 2021–2024 period show a complex, non-linear escalatory trend. Ransomware cases rose from 218,043 in 2021 to a peak of 1,011,209 in 2023, then declined drastically to 514,508 in 2024. This decrease of $\pm 49\%$ does not automatically indicate reduced risk, given that up-to-date ransomware variants generally operate covertly through a fileless attack, hybrid payload, or are embedded within another attack such as phishing and the misuse of an open API.

Meanwhile, phishing attacks show a consistently high pattern. From 6,686,616 activities in 2021, this number rose significantly to over 26 million since 2023, with a total of 26,771,610 activities in 2024. This high consistency confirms that social engineering remains the main attack vector, particularly due to the low cyber-security literacy among end users and the weak multi-layer authentication control across various digital systems.

The data-breach category recorded a significant surge, from 20 cases in 2021 to 241 cases in 2024. This increase indicates a structural weakness in the data-management system, particularly at organizations relying on external integration without adequate security audit. Weaknesses in access control, activity logging, and monitoring of data use also increased the risk of a large-scale leak.

The accumulation of ransomware, phishing, and data-breach attacks directly impacted the escalation of dark-net exposure. The amount of Indonesian data exposed rose from 83,991 in 2021 to 56,128,160 in 2024 — a surge of more than 660 times in four years. This trend shows that data leakage has become a main commodity in dark forums and opens opportunity for cyber actors to carry out repeated attacks, such as identity fraud, distribution of further malware, and manipulation of financial systems based on stolen data.

This development reflects a shift in attack pattern from probabilistic — that is, random and mass-scale — toward a targeted pattern, in which the attacker already has detailed data about the target, such as ID number, transaction history, device pattern, and digital-

communication preference. That data is then used to design an attack scenario with precision, one that is difficult to detect by conventional protection systems.

Further, the transformation of the threat landscape is also marked by the emergence of an attack model that is fully automated and driven by artificial intelligence (AI-powered automated attack). With this approach, an attacker is able to carry out automated reconnaissance, user-pattern learning, and the compilation of a payload that is adaptive and real-time. Technology such as machine learning and generative AI has begun to be used to design an attack that no longer depends on manual skill, but on an algorithm able to evolve independently. Therefore, future defense systems must adopt an AI-driven-defense approach that combines predictive detection and automatic, digital-environment-context-based response.

The attack trend over the past four years shows a substantial change from opportunistic attack toward a more aggressive, intelligent-driven pattern, supported by automated-attack tools and the use of artificial intelligence to exploit the human-factor weak point. This complexity strengthens the urgency of building a cyber-security system that is not only reactive, but also able to adapt to the evolution of tactics and technology used by attackers. Therefore, a deep understanding of this trend becomes a prerequisite in compiling a 2025–2029 mitigation scenario based on an adaptive, proactive approach integrated across systems and institutions.

The development of cyber-attack patterns against the banking sector in Indonesia over the past four years shows a significant shift from a conventional attack toward a more complex, artificial-intelligence-based threat form. This transformation is marked by the increasing intensity of a mass attack that exploits an automation algorithm. Technology such as botnet and automated credential-stuffing tools is used to attack a bank's digital system simultaneously with high efficiency, particularly against institutions that have not applied a machine-learning-based anomaly-detection system (BSSN, 2024).

The change in attack strategy is also reflected in a tactic that is increasingly personalized and intelligent-driven. Phishing techniques, which previously relied on the random distribution of fake links, have now developed into spear phishing targeting individuals with a message tailored based on public information. Even the vishing method, using deepfake based on text-to-speech AI, has begun to be used to impersonate

a bank official to deceive internal staff. In addition, the emergence of the ransomware-as-a-service (RaaS) model such as LockBit and Conti gives a non-technical actor the opportunity to launch an attack merely by renting the malware, making the attack easier to carry out and more widespread (Chakravarti, 2023; Kompas, 2024).

In the spatial and temporal aspect, the cyber-attack trend shows a significant increase in the end-of-fiscal-year period, namely October to December. The increase in digital-transaction volume and system load during that period increases exploitation opportunity. On the other hand, infrastructure that has moved to a cloud-based environment also becomes a main target due to the complexity of the authorization system and weak protection on the application programming interfaces (APIs) used in system integration (OJK, 2023; BSSN, 2024).

The policy response developing in this sector tends to be reactive. Most new regulation is designed as a response to an incident that has already occurred, rather than based on anticipation based on medium-term threat projection. Although OJK Regulation No. 11/POJK.03/2022 and Presidential Regulation No. 82 of 2022 are beginning to show a more adaptive risk-management approach, their application at the technical level does not yet fully reflect the character of predictive policy. The lack of integration of the cyber-threat-intelligence approach in the policy-formulation process remains a main obstacle in responding to the attacker's increasingly complex tactics (OJK, 2023; BSSN, 2024).

The tendency for recurrence and increasing scale of impact is also an important indicator in reading the trend. For example, the ransomware attack pattern experienced by BSI in 2023 has a similar structure and impact to the incident that hit Bank Indonesia in 2022, both in terms of service disruption and the initial-access method used. This shows that the post-incident-evaluation (post-incident learning) process within the digital-defense system has not been carried out systematically. Whereas, the experience from a prior incident should be able to be used to strengthen layered defense against a similar threat in the future (Chakravarti, 2023; Kompas, 2024; BSSN, 2024).

4.2.3. Driver

The development of the character of cyber attacks against the banking sector in Indonesia shows a systematic and layered escalation pattern, which begins with

opportunistic attack — that is, random, massive-scale attack against a common vulnerability — then shifts to targeted attack that specifically targets high-value institutions, and now enters a more complex phase in the form of automated adaptive attack supported by artificial intelligence (AI-driven). This shift reflects a significant increase in the technical capability, tactical flexibility, and resources used by threat actors. Within the horizon-scanning analytical framework, mapping the driving forces (drivers) of these dynamics becomes an important element in compiling an anticipatory scenario and a medium-term mitigative policy. Three main drivers that can be identified from this trend include: (1) the increasing use of artificial-intelligence technology and automation by cyber-crime perpetrators, which enables the attack to become increasingly adaptive and hard to detect; (2) the security-readiness and security-system-implementation gap between banking institutions, particularly between a large bank with complete resources and a financial institution still limited in early-detection and incident-response capacity; and (3) increasingly open access to cybercrime-as-a-service-based crime services through a decentralized digital ecosystem that is difficult to regulate (Simorangkir, Purba, & Hartati, 2025; SOCRadar, 2024; Gentry, 2023).

4.2.3.1. Increasing Use of AI Technology and Automation by Threat Actors

The use of artificial intelligence (AI) technology and automation systems by cyber actors has become one of the main drivers in the shift of attack pattern against the banking sector, from what was originally opportunistic and mass-scale, shifting to a more precise targeted attack, and now developing into an automated adaptive attack powered by adaptive, complex algorithms. This evolution is made possible by the integration of large language models (LLMs), which allow attackers to design a spear-phishing scheme with a high level of personalization, along with the use of generative AI to create deepfake content in the form of voice and video to support a sophisticated social-engineering scheme (Hahnagy & Fincher, 2021; Olavsrud, 2023).

This phenomenon is reinforced through the statement of a white hat hacker informant using the alias "Julio," who explains that "the use of an automated toolkit based on open-source AI is increasingly widespread on digital-crime forums, allowing an attacker to mimic internal bank communication with a high level of precision." He also adds that a deepfake voice has been used in a voice-phishing technique against helpdesk staff, and in

his view "this approach cannot be countered merely through basic digital-security-awareness training."

The sophistication of the attack is also visible from the presence of polymorphic malware able to automatically adapt to a defense system, and the use of an AI-based botnet in a distributed-denial-of-service (DDoS) attack that adjusts its strategy based on the target's response. Validation of this escalation is given by an informant from the State Intelligence Agency (BIN) using the alias "Dani," who states that "we have detected a transition from generic phishing attacks toward a more specific, AI-based pattern, and there's a signal that a foreign actor has been testing that technique against national digital-financial infrastructure."

In line with that, Hilal Akbar, who serves as Division Head at PT Mitra Integrasi Informatika, reveals that "the main challenge actually lies in human readiness, not just the technology, because an AI-powered spear-phishing attack can now fake an internal official's identity with high accuracy." He also highlights the phenomenon of the commercialization of technology through a ransomware-as-a-service (RaaS) service that now includes an AI-automation feature, thereby increasing the opportunity for a non-state actor to launch a sophisticated attack without needing high technical capability (SOCRadar, 2024).

The impact of this development spreads through to a bank's internal defense system. Malware developed with a machine-learning algorithm is now able to learn from the target's defense pattern repeatedly (feedback loop), and then adjust the payload and exfiltration method based on a system structure detected in real time. Some attackers have even used AI to map a detection gap in the endpoint-detection-and-response (EDR), security-information-and-event-management (SIEM), and firewall systems commonly used within the banking environment (Asmar & Tuqan, 2024; Alaeifar, Shakarian, & Nunes, 2024).

Taking these dynamics into account, it is clear that the integration of AI and automation by threat actors not only increases the efficiency and success of an attack, but also accelerates the tactical-adaptation cycle and broadens technology access across a wider spectrum of actors. Therefore, this factor can be identified as one of the main

structural drivers within the horizon-scanning framework for the escalation of cyber threat faced by the Indonesian banking sector in the 2025–2029 period.

4.2.3.2. The Security-Readiness and Security-System-Implementation Gap Between Banking Institutions

The gap in cyber-security readiness and implementation between banking institutions shows a significant correlation with the increasing complexity of the cyber-attack-threat landscape in recent years. The inhomogeneity in the application of global security standards, such as ISO/IEC 27001, ISO/IEC 27035, ISO/IEC 27002:2022, ISO/IEC 27005:2022, and the National Institute of Standards and Technology Cybersecurity Framework (NIST CSF), has created a structural imbalance in institutions' capability to confront various attack vectors. Even large institutions such as Bank Syariah Indonesia (BSI), Bank Rakyat Indonesia (BRI), Bank Indonesia (BI), and BCA — which have all along been assumed to already have a solid foundation on the people, process, and technology aspect — have proven to have experienced an incident showing a penetration of their defense systems. This phenomenon shows that the level of security maturity does not necessarily align with the effectiveness of the protection system against continuously evolving cyber threat. Further, the condition of institutions still structurally at a developmental stage, such as a regional development bank (BPD) and other smaller-scale banks, further widens the readiness and defense-resilience gap.

This imbalance is visible not only in the technological aspect, but also in the risk-management framework and mitigation strategy used. Many mid-to-lower-tier banks do not yet have internal security infrastructure such as a Security Operations Center (SOC), or specific personnel at the level of a Chief Information Security Officer (CISO) responsible for compiling threat-intelligence-based adaptive policy. In an interview, "Hilal Akbar," who serves as Division Head at PT Mitra Integrasi Informatika, asserts that "there are still many banking institutions that see security as a cost burden, not as a strategic investment. This causes a low budget for IT-personnel training and the ongoing strengthening of the threat-monitoring system."

In addition, the gap is also reflected in the variation in the system architecture used. Large institutions tend to adopt a cloud-native or hybrid-system approach with an integrated control layer, while small institutions still rely on legacy systems that are less

adequate in detecting and responding to contemporary cyber attacks. "Julio," a white hat hacker, states that "attackers often target a bank whose infrastructure doesn't yet support thorough encryption and network segmentation, so lateral penetration is very easy to carry out."

Vulnerability also arises from the dependence on a third party in providing information technology (IT), where not every external vendor holds appropriate security certification or explicitly includes a protection clause in the service agreement. "Dani," an informant from the State Intelligence Agency (BIN), reveals that "even a large institution has now become a main target, not just because of its high value, but also because it's often considered already secure, so it's lax in early detection."

From a human-resources perspective, many banks still face a shortage of certified experts in the field of incident response, digital forensic analysis, and penetration testing. This causes many incidents to be handled only administratively, without a comprehensive investigation or post-incident learning. At the managerial level, limited understanding of social-engineering techniques and advanced-persistent-threat (APT) threat also increases the potential for exploitation.

This disparity is also reinforced by differences in budget allocation. Many institutions view the security aspect as a cost center, so they only allocate minimal funds for regulatory compliance, without developing a long-term strategy for an early-detection system, penetration testing, or a disaster-recovery plan. Conversely, a bank that places security as a strategic element tends to have a more mature, proactive mitigation roadmap.

On the regulatory side, the implementation of Financial Services Authority Regulation (POJK) No. 11/POJK.03/2022 is still not evenly applied across all financial institutions. Many institutions apply it only documentatively to fulfill the annual-audit requirement, without internalizing that framework into daily operations. This causes access control, the detection system, and mitigation procedures to run partially and be less responsive to persistent, repeated attacks.

International standards such as ISO/IEC 27005 concerning risk management, ISO/IEC 27017 and 27018 governing cloud-service security and personal-data protection, and ISO 22301 concerning business-continuity management, have not yet been fully adopted by most banking institutions. Whereas layered integration of these standards

could strengthen the defense system to become more resilient against various forms of cyber attack.

The difference in audit implementation is also an important aspect forming vulnerability. Large banks generally run internal and external audits quarterly, while small banks only do so once a year or when a major incident occurs. This imbalance lengthens the exposure period to system weaknesses that are not immediately detected.

This situation forms an asymmetric battlefield, in which the attacker tends to target the weakest point in the national financial ecosystem. "Julio" states that "hackers now don't need to attack a strong system, they just need to find an institution that doesn't yet have a SOC or that hasn't even enabled 2FA to log into their internal dashboard."

Based on all these factors, it can be concluded that the gap in readiness and security-system implementation between banking institutions not only functions as a risk indicator, but also strengthens an adaptive attack pattern increasingly selective toward a target with structural weakness. Thus, this factor holds strategic value in horizon scanning, particularly in compiling a medium-term scenario responsive to changes in the threat landscape and the risk-mitigation policy of Indonesia's financial sector.

4.2.3.3. Opening Access to Cybercrime-as-a-Service-Based Cyber-Crime Services

Increasingly open access to cybercrime-as-a-service (CaaS)-based crime services represents a fundamental shift in the modus operandi of digital attacks against financial institutions. This phenomenon not only marks the technological evolution of threat, but also reflects the formation of a structured, cross-border digital criminal-economy ecosystem. Within this ecosystem, a criminal no longer needs complex technical capability to launch an attack. With just a cryptocurrency wallet and a virtual private network such as Tor, anyone can now access a digital dark market selling various cybercrime tools, from phishing kits and malware builders, through to initial-access credentials offered by an initial access broker (Trend Micro, 2023).

The cybercrime business model is now developing modularly and in fragmented form, allowing an actor to adopt a service-oriented approach. One of the most-developed forms is ransomware-as-a-service (RaaS), namely an affiliate model that allows a local actor to run an attack using infrastructure provided by a global actor. Thus, an attack against an

Indonesian banking institution is not always carried out by the main actor, but by a domestic affiliate with an understanding of local context and the country's system weak points. The National Cyber and Crypto Agency (BSSN, 2024) records an increase in this type of attack as a form of a threat actor's adaptation to the progress of conventional cyber defense.

In an interview, informant "Julio," a white hat hacker, confirms that "the majority of the attack tools we've found originate from ready-to-use attack packages sold freely on Telegram and dark forums, complete with technical documentation in Indonesian." This indicates that not only the dark web, but also popular messaging applications such as Telegram, have shifted into an open market for cyber crime, where a bank customer's data and account credentials are sold via a subscription system or a buyer-reputation-based auction.

Further, this criminal structure is supported by weaknesses in cross-border law enforcement. Many digital-crime service providers operate from a jurisdiction lacking a mutual legal assistance (MLA) agreement with Indonesia, or are even protected by the local legal system. When the Indonesian authority wants to take enforcement action, the legal process against the actor is hampered by minimal international cooperation. This covers difficulty in extradition, blocking digital assets, and seizure of cryptocurrency-based crime infrastructure (INTERPOL, 2024).

As a result, the barrier to entry into the digital-crime world becomes very low. Where an attack against a banking system could once only be carried out by an actor with high technical expertise, now even an ordinary criminal can carry out an attack relying on a CaaS-based automated tool. This condition explains why large institutions such as Bank Syariah Indonesia (BSI), Bank Rakyat Indonesia (BRI), Bank Indonesia (BI), and BCA remain targets, not because of a weak defense system, but because the number of actors with access to sophisticated tools has increased significantly with a low legal risk.

An informant from the State Intelligence Agency (BIN) using the alias "Dani" emphasizes that "attack structure is no longer linear. Attackers now assemble an attack from a modular exploit toolkit, bought from various sources and assembled into a complex attack scheme that isn't detected by an old signature-detection-based system."

He adds that there has been an increase in the use of international third-party services to distribute a malware loader aimed at the national financial system.

This condition is worsened by the low awareness of several banking institutions toward the dynamics of the digital criminal market. Hilal Akbar, Division Head at PT Mitra Integrasi Informatika, explains that "many banks don't yet have a cyber-intelligence team that actively monitors the movement of threat actors on the dark market, when information like this is essential for threat-hunting activity and early-detection-based prevention." This lack of readiness in monitoring and analyzing the movement of the cyber-crime market lengthens the exposure period and heightens the likelihood of a successful attack.

The CaaS phenomenon also systemically broadens the attack surface. The flexible service model, affordable and available 24/7, makes CaaS a "cheap, fast, and adaptive" service, meeting a criminal actor's need to adjust an attack tool to the target's condition. In the horizon-scanning context, this trend represents a structural threat that is not only technical, but also economic and geopolitical. Without a cross-disciplinary approach and effective international collaboration, CaaS will continue to develop as global criminal infrastructure threatening the stability of Indonesia's digital financial sector through 2029 and beyond.

4.2.4 Scenario Planning

To anticipate the escalation of cyber threat in the national banking sector, a predictive approach is needed able to depict various possible future conditions. Through the scenario-planning method, this research constructs four scenarios based on two main axes, namely the resilience capability of banking institutions and the attack capability of hackers. Each scenario contains a different spectrum of weakness, potential risk, and threat complexity, and becomes the basis for formulating an adaptive strategy for increasing cyber resilience going forward.



Figure 4.1 Scenario Planning for Banking-Institution Cyber Resilience

Source: Researcher's Data

4.2.4.1. Scenario 1: Low Banking-Institution Resilience Capability, High Hacker Attack Capability

The systemic vulnerability faced by banking institutions in this scenario is caused not only by technical limitations, but also by weak overall information-security governance. Many institutions still rely on outdated legacy systems that do not receive a regular security-patch update. The mismatch between this old system and modern cyber-defense technology, such as artificial-intelligence (AI)-based endpoint detection and response (EDR), creates a significant security gap. The absence of a zero-trust architecture implementation, which emphasizes strict access restriction based on identity and context, further increases the likelihood of a lateral, hard-to-detect cyber attack.

On the other hand, delay in implementing information-security policy is also a factor weakening the system's resilience. Regulation such as POJK No. 11/POJK.03/2022 is often only translated into documentary form for audit purposes, without concrete technical steps being supported in the field. Whereas international standards such as ISO/IEC 27001:2022 concerning an information-security-management system and ISO/IEC 27005:2022 focused on information-security-risk management should be adopted comprehensively. Many institutions have not yet implemented ISO/IEC 27035 as the main framework for incident handling, and do not yet have a Standard Operating Procedure (SOP) ready to be activated quickly when a digital crisis occurs.

Immature digital leadership further worsens this situation. Many institutions have not yet formed a Computer Security Incident Response Team (CSIRT) with adequate authority and budget support. Coordination between units, particularly between the information-technology department and the risk-management unit, is often not synergistic. As a result, the process of escalating and responding to a cyber incident runs slowly and without integration. This shows that the principles of business-continuity management as set out in ISO 22301:2019 have not yet been fully applied operationally, although they are already available as formal guidance.

Meanwhile, on the opposing side, the cyber attacker has undergone a drastic increase in capability. Malware used by threat actors is now adaptive and able to evade a conventional, signature-based detection system. The use of an automated vulnerability scanner allows them to efficiently identify an institution still using unupdated software or not aligned with the latest information-security standard. This finding is reinforced by informant "Dani" from the State Intelligence Agency (BIN), who reveals that many cyber actors have used a modular exploit kit freely sold on dark forums and can be reassembled to precisely attack a bank's system according to its respective weakness.

The expansion of the digital-crime market facilitated by the ransomware-as-a-service (RaaS) ecosystem is also a main driver of the increasing volume and reach of the attack. This affiliate-based model allows a domestic actor to access global attack technology and infrastructure with just a crypto-wallet stake and a connection to a dark-web forum or public application such as Telegram. This activity makes an attack against a banking institution cheaper, faster, and harder to trace, while significantly expanding the attack surface. According to a Trend Micro report (2023), the dynamics of this digital criminal economy have developed into an organized commercial system offering various products and services, from phishing kits and malware builders, through to initial access into a bank's internal network sold by initial access brokers.

Thus, the spectrum of vulnerability in this scenario reflects a structural failure in building thorough cyber resilience. Technological unpreparedness, weak regulatory execution, minimal leadership capacity, and aggressive development on the threat-actor side, all combine to form serious systemic risk against the continuity of national digital financial services. Without strategic transformation based on international standards such

as ISO/IEC 27001:2022, ISO/IEC 27002:2022, ISO/IEC 27005:2022, ISO/IEC 27017:2015, and strengthening software-life-cycle management through SNI ISO/IEC/IEEE 12207, and system-quality evaluation with SNI ISO/IEC 25000 and ISO/IEC/IEEE 29119, institutions will remain in an uncompetitive defensive position. This also requires the real implementation of the Ministry of Communication and Informatics Regulation No. 5 and 6 of 2025, which emphasizes personal-data protection, risk management, periodic audit, and safe use of cloud technology.

SWOT Analysis — Scenario 1

Strengths

A portion of national banking institutions have shown an initial initiative in applying a global information-security framework, including the ISO/IEC 27001:2022 standard emphasizing risk-approach-based security management, along with ISO/IEC 27035, which specifically governs cyber-incident-handling governance. Beyond the application of these international standards, domestic policy such as POJK No. 11/POJK.03/2022 also becomes a main reference in information-technology risk management in the banking sector. Institutional support from regulator entities such as OJK and BSSN reinforces this position, through the publication of technical guidance encouraging policy harmonization with global best practice. In this context, there exists a strong foundation for integrating an adaptive-security-framework (ASF) approach, enabling the development of a cyber-defense system adaptive to the dynamics of threat and responsive to the escalation of systemic risk.

Weaknesses

Although progress exists at the regulatory level, the defense system used at various banking institutions is still dominated by a signature-based detection model, which has fundamental limitations in detecting a dynamic cyber threat, such as a zero-day attack, polymorphic malware, and an artificial-intelligence-based infiltration technique (BSSN, 2024). This dependence reflects the not-yet-optimal integration of the cyber-intelligence function into the strategic-decision-making process. In addition, the weak commitment from the leadership level in making digital security a long-term priority is also a significant obstacle. This is reflected in the tendency toward merely administrative policy

implementation, with a focus on documentary compliance, but not yet reaching technical effectiveness in day-to-day operational management.

Opportunities

The opening of space for policy reform and security-system modernization can emerge through leveraging the momentum of a cyber incident as a catalyst for transformation. Within a crisis-driven-transformation framework, banking institutions can carry out a thorough assessment of system resilience and begin adopting an adaptive security approach relying on the integration of cyber threat intelligence, behavioral-based analytics, and an AI-based defense system. In addition, there is great potential for strengthening cross-institutional collaboration — particularly between regulators such as OJK, BSSN, BI, and state-intelligence institutions — in building a more progressive and sustainable information-protection policy framework. This opportunity can be used to build digital resilience that is not only reactive, but also based on a predictive, integrative system responsive to the dynamics of contemporary threat.

Threats

The risk faced by the banking sector originates not only from an advanced actor group, but also from the ease of access to cybercrime-as-a-service, which enables an individual with low technical capability to systematically launch a cyber attack (BSSN, 2024). In addition, the surge in the use of AI- and deepfake-based attacks further increases the potential for data exploitation and social manipulation difficult to recognize by a conventional detection system. One other critical threat is the occurrence of fragmentation between defense systems in the banking-institution environment. Lack of cohesion in coordination and response can create a gap causing a domino effect — namely, vulnerability at one institution can have a systemic impact on other institutions interconnected within the national digital financial ecosystem. Therefore, a collective, cross-sector-integrated defense system needs to be strengthened to address the increasingly complex challenge.

Financial Impact

Banking institutions risk experiencing significant loss due to data theft, system encryption, and post-incident recovery costs, which can reach billions of Rupiah per case. In addition, the potential decline in share value, cash-flow disruption, and declining investor confidence will directly impact short-term liquidity stability.

Reputational Impact

A data-leak incident or service disruption can erode public and customer trust. Media coverage of an institution's weak response in handling a cyber crisis can also worsen the managerial and governance image in the eyes of the broader public.

Operational Impact

Disruption to a critical service such as mobile banking, ATM, through to the core banking system will cause disruption to a customer's daily transaction process. Procedural unpreparedness and the absence of a structured incident-handling guide will slow the recovery process and increase the risk of a cascading operational disruption.

This scenario marks the strategic need to carry out a thorough reform of the banking sector's cyber-defense readiness before entering an era dominated by automated, modular, high-precision AI-based attacks. Without an adaptive, cross-institutional-collaborative response on the technical, policy, and oversight aspect, Indonesia risks facing serious systemic vulnerability in its national financial infrastructure. Moreover, the existence of policy such as POJK No. 11/POJK.03/2022 and Presidential Regulation No. 82 of 2022 will not have a real impact if their implementation is not carried out substantively, but merely becomes administrative formality. Therefore, a comprehensive foresight approach is needed to ensure that the information-protection system is able to dynamically adapt to the evolution of increasingly complex threat.

Solution

In the context of the increasing systemic vulnerability of national banking institutions, a transformative approach is needed emphasizing the thorough strengthening of cyber-security governance. The most urgent strategic effort is thorough application of an Information Security Management System (ISMS) based on the ISO/IEC 27001:2022 framework, which functions as the main pillar for designing a standardized, adaptive

information-security system that can be continuously evaluated. This standard must be implemented not merely as administrative formality, but as an operational instrument embedded in sensitive-data management, user-access-rights arrangement, and the digital-incident-detection-and-response mechanism (ISO/IEC, 2022a).

In addition, the implementation of ISO/IEC 27002:2022, which provides technical guidance in security control such as encryption, network segmentation, and access-rights management, needs to be fully adopted into the daily protocol of the banking digital-service system (ISO/IEC, 2022b). To strengthen the preventive approach against cyber threat, an institution needs to use the ISO/IEC 27005:2022 framework, which focuses on identifying, evaluating, and controlling risk arising from information-system vulnerability. This integration should be complemented by the implementation of ISO/IEC 27035, which provides a structural methodology for cyber-incident management, including the compilation of a realistic, actual-threat-based cyber drill (ISO/IEC, 2022c; ISO/IEC, 2016).

At the national policy level, Regulation of the Minister of Communication and Informatics of the Republic of Indonesia Number 5 of 2025 on Public-Scope Electronic-System Operators (PSE) provides an important framework for electronic-data classification and protection. This regulation sets three levels of data classification — namely open, restricted, and closed — with an obligation to apply encryption when data is stored (at-rest) or sent (in-transit) (Ministry of Communication and Informatics of the Republic of Indonesia, 2025a). This policy is highly relevant to the banking sector, which inherently stores highly sensitive customer data and is vulnerable to exploitation by threat actors as a credential-based attack entry point.

Next, Regulation of the Minister of Communication and Informatics of the Republic of Indonesia Number 6 of 2025 on technical standards for developing SPBE (Electronic-Based Government-System) applications requires the application of national software-engineering standards. This covers SNI ISO/IEC/IEEE 12207, which sets the software life cycle based on secure-software-development-life-cycle (SSDLC) principles, SNI ISO/IEC 25000 for system-quality assurance, and SNI ISO/IEC/IEEE 29119, which governs software- and information-technology-system-testing standards (Ministry of Communication and Informatics of the Republic of Indonesia, 2025b). Implementation of these three standards is crucial to prevent the entry of vulnerability from internal

applications or a third-party vendor that has not gone through a security-validation process.

The sustainability strategy must also be strengthened through the adoption of the principle in ISO 22301:2019, which focuses on a business-continuity-management system (BCMS). In this context, a bank must have a clear contingency scenario, including priority-service recovery and an emergency-response simulation integrated into organizational policy (ISO/IEC, 2019).

In addition, the growth in the use of cloud-computing services in the banking sector requires the application of ISO/IEC 27017:2015 and ISO/IEC 27018:2019. Both standards emphasize shared-responsibility risk management between the banking institution and the cloud-service provider, particularly in data protection, access control, and audit of the data center used (ISO/IEC, 2015; ISO/IEC, 2019b).

Overall, these solutions show that success in confronting an increasingly complex cyber-threat landscape is determined not only by technological sophistication, but also by the effectiveness of regulation, organizational structure, and progressive digital leadership. Strengthening the role of the Chief Information Security Officer (CISO) becomes very important, not only as a policy executor, but also as a strategic actor in crisis decision-making. Without thorough integration and implementation across the technical, regulatory, and institutional dimension, the entire effort to meet the security standard will only produce symbolic compliance, without giving a real impact on improving the digital resilience of national banking institutions.

4.2.4.2. Scenario 2: High Banking-Institution Resilience Capability, High Hacker Attack Capability

In a condition where national banking institutions have built a more comprehensive cyber-defense system, a number of strategic initiatives have been applied to confront the continuously developing threat landscape. An adaptive detection-and-response system is now beginning to be supported by machine-learning technology, enabling more precise and rapid anomaly identification. The integration of cyber threat intelligence (CTI) with a security-information-and-event-management (SIEM) system and the existence of a security operation center (SOC) have strengthened monitoring capability and real-time incident response. A threat-hunting approach is actively used to detect a hidden attack

pattern unrecognizable by a conventional signature-based-detection approach (Mell, Scarfone, & Romanosky, 2020). The implementation of endpoint detection and response (EDR) is also assessed as able to provide protection against a zero-day attack and fileless malware (Nguyen et al., 2021). In addition, a system relying on network-behavior anomaly detection and user-and-entity-behavior analytics (UEBA) has broadened detection coverage through observation of network traffic and user behavior deviating from a normal pattern (Almseidin et al., 2020).

Although this technical foundation has become established, its effectiveness is tested by the increasingly complex dynamics of cyber attack. Attackers now exploit a ransomware-as-a-service (RaaS) scheme combined with a deepfake-based social-engineering technique, and a living-off-the-land (LotL) method that exploits a system's built-in process and application to infiltrate without triggering a conventional detection system (Hassan et al., 2020). This AI-based multi-vector attack approach has surpassed the effectiveness boundary of security technology not yet strengthened by a holistically predictive and adaptive model.

Furthermore, institutional resilience against a repeated attack is highly determined by the mean time to detect (MTTD) and mean time to response (MTTR), two main indicators in cyber-incident management. An institution that consistently tests and updates its business-continuity plan (BCP) and disaster-recovery plan (DRP) generally shows a better recovery speed. Nevertheless, an advanced tactic such as lateral movement and command and control (C2), which targets a backup system and secondary infrastructure, remains a serious gap in the digital-defense structure (Goutam et al., 2021).

At the institutional level, coordination between banks, the regulator, and oversight bodies has increased. Incident-reporting procedure to OJK and the involvement of BSSN in a cyber-incident response are part of a protocol beginning to run systematically. Nevertheless, cross-institutional response effectiveness still faces obstacles in terms of communication protocol, clarity of technical authority, and collective response time. In this context, cyber wargaming as a crisis-simulation method has proven useful in strengthening systemic resilience against a large-scale attack, although its implementation is still limited within Indonesia's financial-industry environment (Culnan & Williams, 2022).

Further, an attacker's ability to adjust their attack technique to the evolution of a defense system becomes its own challenge. Several tactics used include breaching a sandboxing mechanism, evading a deception system such as a honeypot, and exploiting a system configuration not updated in time (Alasmary et al., 2021). In another case, the presence of an advanced persistent threat (APT) able to remain within a network for months becomes increasingly difficult to recognize because the attacker uses normal network traffic as camouflage to hide malicious activity.

This condition depicts the complexity of a modern, asymmetric cyber conflict, in which an institution's high defense capacity does not automatically eliminate risk if not accompanied by continuous innovation, increased human-resource capacity, and solid cross-institutional coordination. Therefore, an open-protocol-based threat-sharing mechanism, strengthening of national-policy interoperability, and digital leadership able to respond quickly to threat escalation, become an important foundation for guaranteeing the banking sector's cyber resilience going forward.

SWOT Analysis — Scenario 2

Strengths

Banking institutions in Indonesia have succeeded in building a relatively solid cyber-defense framework through the adoption of a threat-hunting system and machine-learning-based detection technology, as well as the real-time integration of cyber threat intelligence (CTI) into the security-information-and-event-management (SIEM) system and the operation of a security-control center (security operation center/SOC) (Mell, Scarfone, & Romanosky, 2020). Endpoint-detection-and-response (EDR) technology makes a major contribution in detecting an attack type such as zero-day and fileless malware generally undetected by a conventional perimeter mechanism (Nguyen et al., 2021). Further, oversight capability over network behavior through network-behavior anomaly detection and user activity through user-and-entity-behavior analytics (UEBA) becomes the backbone of a layered security system in confronting continuously developing threat (Almseidin et al., 2020). On the operational-readiness side, most banks have compiled and periodically tested a business-continuity-plan (BCP) and disaster-recovery-plan (DRP) document, ensuring service continuity if a large-scale digital disruption occurs (Goutam et al., 2021).

Weaknesses

Although the technology approach used is fairly up to date, the main challenge still lies in human-resource capacity, particularly in the limited number of cyber-security analysts skilled in tactical analysis based on real-time threat intelligence (Culnan & Williams, 2022). Dependence on a foreign software and security-system provider also creates strategic vulnerability, both from the possibility of an undetected backdoor and potential technical obstacle due to a technology restriction from the vendor's country of origin (Alasmary et al., 2021). In addition, although the threat's potential continues to develop, the practice of emergency-response simulation such as cyber wargaming has not yet become part of the routine activity of most financial institutions, so the validity of an institution's response capability against a layered attack is not yet fully tested (Culnan & Williams, 2022).

Opportunities

The current condition opens strategic space for strengthening synergy between authorities such as OJK, BSSN, and Kominfo, in order to design an open national threat-sharing platform accessible across institutions as a national cyber-intelligence-coordination tool (Culnan & Williams, 2022). A technological breakthrough in the field of deception technology, honeynet, and a zero-trust-architecture approach presents an innovative opportunity for strengthening an adaptive cyber-defense system not dependent on a single perimeter (Alasmary et al., 2021). The government is also accelerating the digitalization of the financial system through SPBE policy and PSE Public-Scope regulation, which can be used as a regulatory framework for strengthening cross-sector cyber-security governance in an integrated and accountable manner (Hassan et al., 2020).

Threats

Threat actors now use artificial-intelligence technology to develop a multi-vector-based attack strategy, including the use of ransomware-as-a-service (RaaS) and deepfake-based phishing, which significantly broadens the attack surface and increases the probability of successful intrusion (Hassan et al., 2020). An advanced-persistent-threat (APT) tactic allows an actor to infiltrate covertly and remain within a network system for a long duration before then carrying out a highly precise, coordinated attack (Culnan & Williams, 2022). In addition, a manipulative technique such as sandboxing breach and

evasion of deception-based defense such as a honeypot has been used to deceive a protection system not optimally updated (Alasmary et al., 2021). This tactic is worsened by the patch-delay condition still occurring in many old infrastructures or a system not yet fully integrated with an automatic update system (Hassan et al., 2020). Threat also increasingly targets a backup system and an institution's internal pathway using a lateral-movement technique, which shows that the attacker has an in-depth understanding of the design and weaknesses of banking digital-security architecture (Alasmary et al., 2021).

Financial Impact

Although banking institutions' cyber-defense systems have been developed comprehensively, penetration by a highly adaptive threat actor can still cause substantial financial loss. One of the most directly felt consequences is disruption to a digital-transaction service, both for the needs of individual customers and cross-sector business transactions. When technology infrastructure has to be temporarily halted for forensic-analysis and system-recovery purposes, the bank loses service-fee revenue while also bearing a surge in operational burden caused by the need for additional technical resources and a significant increase in IT-staff working hours.

Beyond that direct loss, a complex attack such as artificial-intelligence-based ransomware causes further financial burden in the form of recovery financing. That cost covers hardware re-procurement, rebuilding the system from scratch, and carrying out a thorough security audit. Although a digital ransom is often avoided, the cost of post-attack system strengthening often reaches billions of Rupiah. This situation is worsened by an increase in cyber-insurance premium and potential fines from the regulator if there is a violation of provisions set by OJK or the Ministry of Kominfo Regulation related to the management of high-risk electronic data.

Further, financial impact can also be systemic. For an institution already listed on the capital market, a cyber incident has the potential to trigger a share-price decline due to a decrease in investor confidence toward the internal security system. This uncertainty creates market fluctuation harming the institution's strategic position in the eyes of shareholders and business partners. In addition, the capital-expenditure burden that must be increased to fix and strengthen the cyber-defense system will reduce financial flexibility in developing digital services or medium-term business expansion. Thus, the

threat to a bank's financial stability occurs not only directly, but also has implications for institutional competitiveness in the long term.

Reputational Impact

A cyber attack that successfully penetrates a banking security system also has a very significant reputational impact. In this era of information openness and mass dissemination via social media, public perception of an institution's credibility can change drastically in a short time. When a digital-service disruption or data leak occurs, even on a limited scale, the public tends to doubt the institution's ability to protect their financial security. This situation creates psychological pressure impacting customer loyalty and the overall use of digital service.

This reputational impact becomes more severe if the incident involves the exploitation of customer personal data, such as identity use in phishing or another social-engineering action. In a condition like this, public expectation of transparency, accountability, and a form of compensation from the bank will sharply increase. Institutional failure to respond quickly and openly will worsen the trust crisis. Pressure also comes from the regulator and oversight bodies, which demand accountability and a thorough audit of data-protection practice and cyber-emergency-response readiness.

Strategically, a reputational crisis does not only impact a bank's relationship with individual customers, but also with corporate partners, other financial institutions, and the government. Partner trust can erode if they assess that the institution has failed to maintain system integrity and information confidentiality. In some cases, a third party can choose to halt cooperation or transfer its portfolio to an institution considered safer. This situation lowers the bank's competitive position within the digital financial ecosystem, and disrupts business-growth potential dependent on long-term reputation and stability.

Operational Impact

The impact on the operational aspect in this scenario is complex and multidimensional. Although most banks already have a business-continuity plan (BCP) and disaster-recovery plan (DRP), the success of implementation depends on the actual level of readiness when confronting a real attack. In practice, the execution process of the BCP and DRP often faces a coordination challenge, particularly in ensuring the continuity of a main function such as the payment system, liquidity management, and real-time

financial-transaction reporting. If a threat actor succeeds in penetrating not only the main system but also the backup, operational disruption can last for a significant duration and impact business activity across divisions.

In addition, a technical response to a cyber incident requires system-isolation action as a prevention effort against the spread of further malware. This process causes access restriction both internally (by employees) and externally (by customers), which impacts productivity and causes an accumulation of unfinished transaction processes. In this phase, a banking institution must involve various work units — including IT, legal, internal audit, and risk management — to carry out investigation and remedy, which often slows overall daily operational activity.

At the medium-term recovery stage, the bank faces pressure to renew information-technology policy, rearrange the work flow, and accelerate the adoption of a new security solution such as zero-trust architecture and a software-defined perimeter. This transformation requires a human-resource retraining process, system update, and a not-insignificant budget allocation. In addition, cross-unit coordination in executing an emergency response is often hindered by the absence of a standardized incident-response framework, thereby lengthening the time needed for a critical decision. All of these challenges show that the operational impact of a cyber attack is not only technical, but also touches the broader institutional-governance foundation.

Solution

In responding to the increasingly complex and dynamic escalation of cyber threat, national banking institutions need to build an integrated cyber-threat-intelligence (CTI)-based detection-and-response system. This approach is realized through real-time integration of a security-information-and-event-management (SIEM) system, which functions to centralize incident monitoring through a security operation center (SOC) as the cyber operational-control center (Mell, 2020). Further, the application of a machine-learning-based threat-hunting method becomes a main element in developing an adaptive-security-framework (ASF), able to identify a new attack pattern and run an automatic response mechanism (Scarfone, 2020). On the other hand, an anomaly-detection algorithm built based on a machine-learning pattern also increases early-detection capability against deviant network behavior and a system showing an indication of compromise (Romanosky, 2020).

Strengthening on the technical aspect requires support from a strategic-intelligence (SI) approach, which focuses on decision-making at the managerial and regulator level for strategic cyber-crisis management. One recommended instrument is the implementation of cyber wargaming, namely a comprehensive simulation of a cyber incident involving cross-strategic units as part of the process of testing response readiness and institutional-command-chain efficiency (Culnan, 2022). This method functions as a testing tool for the detection-speed (mean time to detect — MTTD) and response (mean time to respond — MTTR) indicator, while also strengthening decision-making capability in an emergency condition that is layered and simultaneous (Williams, 2022).

At the national-regulation level, Presidential Regulation Number 82 of 2022 has established the strengthening of a cyber-security system to protect vital information infrastructure through a cross-sector-collaboration scheme, which makes the banking sector one of the protection priorities. This regulation requires synergy between government bodies and industry actors in building a coordinated national digital-defense system (Kominfo, 2022). On the other hand, the Financial Services Authority (OJK) through POJK No. 11/POJK.03/2022 sets an information-technology-risk-governance principle covering an incident-reporting mechanism, periodic system audit, and strengthening of customer-data protection based on a risk-based approach (OJK, 2022).

As part of the commitment to digital transformation of government and the financial sector, banking institutions must also fully implement Kominfo Ministerial Regulation No. 6 of 2025, which governs technical standards and procedures for building SPBE applications. This regulation requires the use of an international standard such as SNI ISO/IEC/IEEE 12207 concerning the software life cycle (Kominfo, 2025a), SNI ISO/IEC 25000 related to software-quality evaluation (Kominfo, 2025a), and SNI ISO/IEC/IEEE 29119 concerning software-testing methodology (Kominfo, 2025a). These three standards provide a technical foundation for building an application system that is secure from the design stage, in line with the cyber-resilience (CR) principle that emphasizes long-term, layered resilience.

Next, implementation of Kominfo Ministerial Regulation No. 5 of 2025 concerning Public-Scope Electronic-System Operators also holds a central role in the protection of

strategic data. Financial institutions are required to classify electronic data based on sensitivity level — open, restricted, and closed — and apply encryption in the storage condition (encryption at-rest) and during transmission (encryption in-transit) for restricted and closed data (Kominfo, 2025b). To support this, zero-trust-security architecture must be applied, by removing implicit trust and replacing it with risk-based authentication and authorization for every access to data (Kominfo, 2025b).

On the other hand, optimization of the ISO/IEC 27035 standard related to information-security-incident management needs to be improved. This standard covers the reporting, root-cause-analysis, incident-escalation, and complete documentation stage as part of the incident-management cycle (Alasmary, 2021). Although most institutions have partially adopted it, practice such as systematic documentation and the compilation of lessons learned from an incident still need strengthening to guarantee compliance with the audit standard required by BSSN and OJK (Alasmary, 2021).

From a collaborative perspective, connectivity between a bank's internal risk-management system and a national threat-sharing platform must be expanded and standardized. The exchange of indicator of compromise (IOC), carried out in real time and securely, enables collective detection and early mitigation against an attack pattern emerging across sectors (Hassan, 2020). This approach becomes a concrete manifestation of CTI and SI integration in building a collaborative, sustainable national cyber-security ecosystem (Hassan, 2020).

Overall, the solution strategy in this scenario does not require the formation of new regulation, but focuses on strengthening the implementation and integration of existing regulation. The effectiveness of SPBE, POJK, and ISO-standard implementation must be assembled into a single harmonious system based on CTI, ASF, SI, and CR as the main pillars of a resilient, responsive, and adaptive banking-sector cyber-defense system for confronting the evolution of digital threat going forward.

4.2.4.3. Scenario 3: Low Banking-Institution Resilience Capability, Low Hacker Attack Capability

In a scenario where the level of adaptive readiness and threat intensity are both at a low level, there is significant latent risk in the form of a false sense of security. This

condition arises when an institution experiences a period without a cyber incident, so a mistaken perception emerges that the existing system is already adequate in guaranteeing security. This perception drives management to delay or even ignore investment in information technology, both from the security hardware and software side. Whereas the nature of cyber threat is very dynamic and often exploits a moment of vulnerability that is not visibly detected. When an early-warning system or threat monitoring based on cyber threat intelligence (CTI) is not implemented, risk will develop without detection, and an attack that seems to appear suddenly can cause serious damage due to weak readiness — structurally, mentally, and technologically.

Further, this situation is often accompanied by negligence in updating and strengthening digital-security infrastructure. The absence of a visible incident causes the institution not to actively evaluate whether the system has already met the required security baseline. Many banks still rely on old architecture without a network-segmentation protocol in line with the zero-trust-architecture principle. The internal-security-audit process is also often run as a mere formality, without a resilience-testing method against a real attack. The absence of an adaptive-security-framework (ASF)-based assessment makes IT-risk management reactive, not anticipatory. A gap like this is highly vulnerable to a lateral-movement-type attack or a fileless attack, in which the attacker exploits a configuration weakness previously considered insignificant.

The next impact is visible in the low level of cyber-security awareness among internal human resources. When digital-security training is not carried out consistently and not adjusted to the latest threat dynamics, employees become the weak point in the defense system. The minimal execution of testing such as phishing simulation, crisis simulation, or scenario-based training makes individuals within the organization lack the capacity to recognize an early sign of compromise such as anomalous activity or an unauthorized access request. This condition becomes an obstacle to building a cyber-resilience (CR) system, which should emphasize readiness and resilience — both individual and institutional — in confronting a cyber incident quickly and in a structured manner.

Although the level of attack in this scenario is relatively low, the risk of a sudden escalation remains unable to be ignored. An attacker can exploit this situation to launch a sudden attack based on a zero-day exploit, credential stuffing, or exploitation of an unupdated system. Without a proactive monitoring mechanism, the potential presence

of a backdoor or hidden gap that has thus far not been noticed can be easily exploited. The absence of situational awareness shows that the system has failed to integrate CTI into the intrusion-detection-system (IDS) mechanism or holistic incident management. When an attack truly occurs, its impact can be very large because the system has not been prepared to adapt to a sudden change in threat status.

In addition, this scenario also highlights the issue of dependence on old technology that has passed the technical-support period from the provider or vendor. An important system such as a teller, middleware, or ATM network still uses legacy infrastructure not designed to confront a modern cyber threat. Dependence on a third-party vendor also increases the risk of a supply-chain attack and reduces the institution's control over system security. Often, the decision to delay system migration or technology update is not based on an objective risk assessment, but on cost or operational-comfort considerations. The absence of pressure from the regulator or minimal supervision from oversight bodies lengthens this stagnant condition. In the context of strategic intelligence (SI), the detection and mitigation of vulnerability points originating from the technology chain and third party should be a main focus in a sustainable long-term defense strategy.

SWOT Analysis — Scenario 3

Strength

In a condition where cyber-attack intensity is still relatively low, many banking institutions in Indonesia remain able to consistently maintain operational stability. The continuity of daily financial service without major disruption creates the impression that the running system is fairly reliable. This stability, although not necessarily reflecting true cyber resilience, still provides room for operations to run without significant technical pressure. In addition, the absence of threat escalation in the short term is often interpreted as a reason to reduce information-security budget allocation. Cost efficiency in the form of delaying system update or human-resource training is considered a rational policy in a situation that appears calm. On the other hand, an old technology system or legacy system still used together with an external partner continues to run its basic function functionally. Although this infrastructure is not fully designed to confront a modern cyber-world challenge, in reality that system remains able to serve conventional banking activity without significant obstacle.

Weakness

Nevertheless, this surface-level stability hides a number of structural vulnerabilities that can weaken an institution's resilience to a sudden attack. One of the most fundamental weak points is the emergence of a security illusion or false sense of security, in which management judges the absence of an incident as proof that the system is already adequate. This assumption drives an attitude of delaying investment in information technology and strengthening human-resource capacity. In a situation like this, the absence of an early-detection system such as cyber threat intelligence (CTI) or an early-warning-system mechanism creates a highly potential gap for exploitation by a cyber actor. The technology infrastructure used, mostly still a legacy system not receiving a periodic update and very dependent on an external vendor, becomes an additional weak point. This condition is worsened by the absence of a thorough security audit of that system. Another weakness is the low level of literacy and cyber awareness among employees. The irregularity of training, the absence of an attack simulation, and the lack of real-case-based education cause employees to become the weak point in an institution's defense chain. Without the ability to recognize an early indicator of compromise, the institution loses one important component in the effort to build thorough cyber resilience.

Opportunities

Although in a condition of low adaptation and threat level, financial institutions still have significant opportunity to proactively strengthen the security system. One main opportunity available is more optimal implementation of already-enacted regulation, such as POJK No. 11/POJK.03/2022 and Presidential Regulation No. 82 of 2022. That regulation has provided a fairly complete normative framework for building an integrated information-technology-protection system. In a situation not yet filled with the pressure of an incident, a banking institution has the leeway of time to carry out a structured digital transformation, including in the aspect of migration from an old system to a new one without needing to be reactive. This momentum also enables the building of synergy between the strategic-intelligence (SI), CTI, and cyber-resilience (CR) approach into a standard procedure and institutional policy. Early implementation of these three approaches allows the institution's readiness to increase alongside the changing threat landscape, thereby creating a system that is not only responsive, but also anticipatory.

Threats

Although it appears calm, this latent condition holds threat potential that can appear suddenly and have a large impact. One of the most significant risks is the possibility of system exploitation by an opportunistic cyber actor exploiting an old vulnerability such as a zero-day vulnerability, a credential-stuffing technique, or an undetected backdoor pathway because it has never been used before. The absence of a real-time threat-monitoring system causes the institution to lose the crucial situational awareness needed to recognize the early escalation of threat. Dependence on an external technology vendor and the use of an outdated system also open a gap for a supply-chain-attack-based threat, in which third-party protection is not fully under internal control. Another threat no less serious is a delay in detection and response when an incident finally occurs. Without a security-information-and-event-management (SIEM) system and reliable intelligence-data integration, the incident-handling process becomes slow, fragmented, and inefficient. This condition is real evidence of the weak integration between CTI, SI, and an adaptive approach, which should be the backbone of the national banking cyber-defense system in an increasingly complex digital era.

Financial Impact

In a condition that appears stable — where there is no major cyber attack over a certain time period — a banking institution's financial vulnerability actually increases latently. One of its implications is the absence of budget allocation for preventive-technology investment such as endpoint detection and response (EDR), user and entity behavior analytics (UEBA), and a threat intelligence platform (TIP), which should function as an early-anomaly-detection system (Exabeam, 2025). When an attack occurs suddenly, the institution is forced to spend a significant recovery cost, ranging from technical handling, digital-forensic effort, customer compensation, through to the possibility of a fine or legal claim. This cost burden generally far exceeds the preventive-investment value that should be periodically allocated.

Loss of revenue also arises as a direct consequence of unanticipated operational disruption. Without an adequate business-continuity plan (BCP) and disaster-recovery plan (DRP), the system-recovery process requires a long time, causing a temporary halt

to a digital-transaction service, fee-based services, and e-banking channels. In the tight competitive climate of digital banking, every downtime period risks losing customers and lowering trust in the institution.

A long-term impact is also unavoidable, such as an increase in cyber-insurance premium due to a worsening risk profile, and the possibility of a declining score from a risk-rating body. An insurance company and rating authority generally set compliance with a standard such as ISO/IEC 27001:2022 and POJK No. 11/POJK.03/2022 as a protection prerequisite. Failure to meet this standard can cause an institution to lose an external financial safety net when a major incident occurs.

Reputational Impact

Reputational risk in the context of the absence of a major incident over a certain period actually becomes more significant. When a cyber incident finally occurs, the public tends to judge that the institution has been negligent in building an adequate detection-and-protection system. A slow crisis reaction and lack of transparency in public communication worsen the negative perception, creating the assumption that the institution is unprofessional in safeguarding the digital security of its customers' data.

Reputational loss also impacts the trust of the regulator and strategic partners. In a worsening incident situation, the oversight authority can apply an administrative sanction, establish special supervision, or even halt part of the digital service. Meanwhile, a partner such as a global financial-infrastructure provider and an international financing institution can re-evaluate cooperation with an institution considered unfit in terms of digital security. A decline in reputation within the international financial network will impact declining transaction volume and exclusion from a cross-country collaborative initiative.

Mass media and social media also amplify the reputational impact. When public narrative is formed from media coverage highlighting the institution's failure to detect and respond to an attack, negative perception spreads quickly. This does not only affect retail customers, but also weakens the institution's position in various strategic partnerships and reduces its appeal as a main actor in the national digital-transformation effort.

Operational Impact

From the operational aspect, the non-occurrence of a major attack over the long term often creates an illusion of stability, when there is actually a large gap in readiness to confront a real threat. Dependence on a legacy technology system not being updated causes a number of crucial processes — such as user authentication, customer-database management, and internal communication — to become vulnerable. The absence of a periodic security audit causes an accumulation of unidentified risk, which in a crisis situation will produce a layered system failure.

Response to an incident also tends to be unstructured. Without an incident-automation system and a trained emergency-response team, the escalation and recovery process is manual and slow. The absence of integration between security-information-and-event-management (SIEM), an ISO/IEC-27035-based incident-reporting system, and the ASF framework slows the recovery of a main service function such as mobile banking, ATM, and other electronic transactions.

In addition, an operational structure still dependent on a manual system or technology incompatible with modern cloud infrastructure makes it difficult for the institution to adapt to the demands of digitalization and policy change. Without an architectural update and strengthening of interoperability, the institution not only loses operational efficiency, but also risks falling behind in competitiveness within the increasingly digitalized national financial ecosystem.

Solution

In a latent situation where external pressure appears low but internal system readiness is not yet adequate, the main priority is not to compile new regulation, but to optimize the implementation of already comprehensively available regulation and information-security standards. The initial effort that must be carried out is strengthening the early detection-and-warning system through thorough integration of cyber threat intelligence (CTI) into the security operation center (SOC) held by the banking institution. This step is in line with the ISO/IEC 27001:2022 framework, which emphasizes the importance of managing information security systemically and based on risk management. In addition, to support fast, documented incident management, the SOC needs to adopt a security-

information-and-event-management (SIEM) system referencing ISO/IEC 27035, so that incident reporting, escalation, and response can be carried out proactively. In the context of adaptation, the adaptive-security-framework (ASF) concept becomes important to apply, particularly through a revision of security policy emphasizing the principles of network segmentation, tiered authentication, and the periodic, continuous implementation of a vulnerability test.

A structural solution must also be directed at reducing dependence on an old technology system (legacy system) and an external provider that has not met the periodic-security-audit qualification. Ministry of Communication and Informatics Regulation Number 6 of 2025 has mandated the use of a national standard such as SNI ISO/IEC/IEEE 12207 for software-life-cycle management, and SNI ISO/IEC/IEEE 29119 for thorough software-testing processes. The application of this standard will ensure that the system used has gone through a software-engineering process attentive to integrity, confidentiality, and availability from the earliest design stage. In addition, system-quality evaluation needs to refer to SNI ISO/IEC 25000, which emphasizes the importance of functional quality, interoperability, performance efficiency, and the security aspect. This strategy aligns with the strategic-intelligence (SI) approach, which emphasizes the importance of planning and risk mitigation from the conceptual stage, not merely reacting to an incident.

The next step no less important is rearranging the information-system architecture and data management based on sensitivity and risk classification. Kominfo Ministerial Regulation Number 5 of 2025 has set the classification of electronic data into the open, restricted, and closed category, each with its own security treatment. For restricted and closed data, an institution is required to apply encryption at-rest and in-transit, and to restrict online access for data classified as closed. In addition, the application of zero-trust architecture becomes crucial, in which all access to an internal resource must go through context-based authentication and minimum access-right restriction. This strategy aligns with the ISO/IEC 27002:2022 guidance, which focuses on risk-based operational control, and ISO/IEC 27018:2019, which emphasizes personal-data protection in a cloud-computing service. Thorough execution of this control will strengthen an organization's resilience against a potential sudden attack that is often undetected in an ostensibly stable situation.

To address the institution's weakness in the human-resource aspect, cyber-security training based on real-incident simulation must become an integral part of the HR-strengthening strategy. Practices such as cyber wargaming and attack simulation need to be carried out periodically, in accordance with POJK No. 11/POJK.03/2022's mandate concerning information-technology-risk management. Through this training, an institution can measure the effectiveness of an internal response through parameters such as mean time to detect (MTTD) and mean time to respond (MTTR). Within the cyber-resilience (CR) framework, strengthening cyber literacy and emergency-response capability across every employee line is a main foundation that must be built at the stage before an escalation occurs. This process connects the cyber-intelligence (CTI) function with institutional readiness to carry out structured incident mitigation and recovery.

At the institutional and inter-agency level, the development of a national platform for safe, standardized intelligence-information exchange is needed. This platform will facilitate the real-time exchange of indicator of compromise (IOC) between banking institutions and security authorities, thereby increasing collective-detection effectiveness and accelerating cross-sector response. This step is consistent with the direction of Presidential Regulation No. 82 of 2022 on the protection of vital information infrastructure (IIV), which emphasizes the importance of inter-institutional coordination in confronting a national-scale cyber attack. By integrating the CTI, ASF, SI, and CR approach into an internal-policy framework, operational procedure, and strategic planning referencing the aforementioned national and international standards, a banking institution will be able to strengthen its resilience in confronting today's digital-security challenge while also building long-term resilience adaptive to the complexity of future cyber threat.

4.2.4.4. Scenario 4: High Banking-Institution Resilience Capability, Low Hacker Attack Capability

In a strategic configuration where a banking institution's cyber-defense capability has reached an optimal level, while the intensity and sophistication of an external attack such as phishing, ransomware, and data breach tend to decline, a new issue arises in the form of overengineering potential. This situation is reflected when an institution allocates large resources for developing an advanced security system such as zero-trust architecture,

artificial-intelligence (AI)-based segmentation, and machine-learning-based threat detection without being accompanied by adjustment to the actual risk spectrum (Mell, Scarfone, & Romanosky, 2020). In this context, the adaptive-security-framework (ASF) approach becomes relevant as a control mechanism so that the security-system design remains aligned with a realistic threat profile and does not become an excessive cost burden without justification from an actual threat.

This relatively safe situation should not become a reason for stagnation, but rather momentum for carrying out systemic consolidation and building sectoral cooperation. One strategy that needs to be prioritized is the strengthening of a cyber-threat-intelligence (CTI)-based information-sharing ecosystem, through the formation of a cross-institutional platform such as an Information Sharing and Analysis Center (ISAC) or a sectoral incident-response team (CSIRT). Strategic connection with regulatory bodies such as BSSN, OJK, and the police becomes vital so that an attack indicator (indicators of compromise) such as a malicious domain address, a malware signature, or a social-engineering pattern can be shared in real time. This initiative will strengthen early-detection capability and broaden the coverage of collective response to a cyber attack, including a mass phishing attack or a covert ransomware attempt appearing sporadically.

Security stability must also be optimized through strengthening human-resource capacity and improving layered defense. Real-attack-simulation training, red teaming, and cyber war-gaming need to become part of a structured periodic training agenda. This is in line with the cyber-resilience (CR) principle, which emphasizes that institutional resilience is not only built through a technology system, but also through human readiness in detecting, responding to, and recovering the post-incident condition. POJK No. 11/POJK.03/2022 becomes the main reference in requiring periodic incident-response testing as part of information-technology-risk management in the financial sector.

On the other hand, the role of digital leadership becomes an aspect that cannot be ignored. A leadership commitment to make cyber security a corporate strategic element will determine the direction of thorough digital-security transformation. Through the strategic-intelligence (SI) approach, an institution can compile a long-term security roadmap covering a success indicator such as incident-response time, mean time to detect (MTTD), and mean time to recover (MTTR). The information-protection policy formed

is not only reactive, but proactive and visionary in responding to the escalation of a stealth or distributed attack such as a staged data breach or a targeted phishing attack.

Nevertheless, the hidden challenge in this scenario is the emergence of a security-fatigue symptom, namely a false-satisfaction condition due to the absence of a major incident within a certain time period. This risks weakening evaluative spirit and giving rise to complacency toward a latent weak point not routinely identified. Therefore, a preventive, periodic security-audit mechanism is needed, referencing the ISO/IEC 27001 and ISO/IEC 27035 standard, covering penetration testing, incident-response testing, and thorough vulnerability analysis as part of a standard operational process, not merely an incidental response.

Through the integration of CTI, ASF, SI, and CR across every stage of cyber-security management, banking institutions in Indonesia can maintain adaptive readiness on an ongoing basis and remain strategically superior even though the level of external threat is low. A regulatory framework such as POJK No. 11/POJK.03/2022 and Presidential Regulation No. 82 of 2022 provides a strong legal and technical basis for developing an adaptive, collaborative, actual-risk-based digital-defense system, in order to confront the complexity of phishing, ransomware, and data-leak threat that continues to evolve within the national digital-financial landscape.

Financial Impact

In a situation where a banking institution's cyber-security readiness is already at an optimal level, potential financial loss from an incident can be significantly suppressed. The implementation of advanced security technology such as zero-trust architecture, a machine-learning-based detection system, and the application of ISO/IEC 27001:2022 and ISO/IEC 27035:2022 allow an institution to detect anomaly and mitigate a cyber attack before it reaches a crucial phase. This strategy reflects the effective application of the adaptive security framework (ASF) and answers the first and second research questions, namely about how an adaptive approach and cyber intelligence support the prevention of, and response to, phishing, ransomware, and data breach.

Even so, there is potential for a challenge in the form of expenditure inefficiency. In a low-threat condition, an institution that continues massive investment in the security system risks experiencing overengineering, in which the cost incurred is no longer proportional to the actual level of threat. This mismatch can cause non-optimal budget

allocation and reduce investment capacity in other fields such as digital-service development or the strengthening of inclusive access. This phenomenon underscores the urgency of evaluating security-expenditure policy so it aligns with the current risk profile, as relevant to the third research question related to the effectiveness of a progressive, risk-based information-protection policy.

In addition, external pressure from a certification body and global auditor to maintain a high standard can cause budget allocation to be focused solely on achieving symbolic compliance. If spending is not adjusted to real performance and protection effectiveness, an imbalance will arise between administrative compliance and operational technical need. Therefore, an institution needs to apply an indicator such as return on security investment (ROSI) and cost-benefit analysis to ensure that the security system remains adaptive to the dynamics of risk and supports the sustainable strengthening of digital economic resilience.

Reputational Impact

The high-readiness condition held by a banking institution creates a positive perception of credibility, governance, and institutional commitment in protecting customer data and digital transactions. This reputation is reinforced through evidence of participation in cross-institutional threat-intelligence sharing, compliance with international standards, and active involvement in a forum such as a sectoral ISAC or CSIRT. In this context, the long-term-oriented strategic-intelligence (SI) approach has proven effective in building the trust of strategic partners and the regulator (Gentry, 2023), and is relevant in answering the third research question related to information-protection policy.

Nevertheless, digital reputation remains vulnerable to decline if an institution does not continuously maintain an evaluative culture. The security-fatigue phenomenon — namely a false sense of security due to the absence of a major attack — has the potential to weaken the commitment to system updating and periodic testing. In this situation, even a small-scale incident such as a customer-data leak or a phishing scam can trigger a reputational crisis if the response is not carried out openly and quickly. Therefore, a banking institution's reputation is determined not only by the absence of an incident, but also by transparency, consistency, and the courage to continuously evaluate its institutional readiness.

Further, external perception from a risk-rating body or international partner will heavily depend on the institution's openness in showing evidence of adaptive-security strengthening. When the public sees alignment between the technology system, policy, and human capacity in handling digital risk, the institution's reputation will become increasingly solid. Conversely, the gap between the claim of readiness and the reality in the field can create a systemic doubt that is hard to recover from. Therefore, the formation of digital reputation must be based on performance-based compliance, not merely policy-based assertion.

Operational Impact

From the operational side, high readiness allows a banking institution to optimally maintain service continuity. A system such as a Security Operations Center (SOC) equipped with cyber threat intelligence (CTI) and security information and event management (SIEM) is able to monitor anomaly in real time and respond to it automatically. This guarantees that an important service such as digital transaction, ATM, mobile banking, and retail payment can run stably even in an abnormal technical condition. Thus, the second question in the research — regarding cyber intelligence's contribution to early detection and response — can be answered empirically through maintained operational performance.

However, this stability situation also holds structural-inertia risk. When an institution feels already "safe," the drive to update the system, carry out further testing, or expand digital transformation can decline. This can create a gradual technological lag and cause a system that was originally adaptive to become stagnant. Therefore, operational readiness is not an end point, but a foundation for accelerating a more sustainable and innovative process of institutional modernization.

In addition, the sustainability of operational readiness depends on the consistency of training, attack simulation, and SOP updates. Without cross-divisional training, incident-response testing such as red teaming, and cyber war-gaming, organizational capability will decline over time even though its technology system is sophisticated. Therefore, adaptive security must be accompanied by human- and business-process-based resilience-capability building. Thorough integration between policy, operations, and strategic decision-making becomes the key in systemically and future-orientedly answering all of the research questions.

SWOT Analysis — Scenario 4

Strengths

A banking institution that has built a cyber-defense system at a high readiness level obtains a strategic advantage in dampening the impact of a sporadic, non-escalating cyber attack. Successful implementation of zero-trust architecture, the application of machine-learning-based threat-detection technology, and conformity with an international standard such as ISO/IEC 27001:2022 and ISO/IEC 27035:2022 reflect that the adaptive-security-framework (ASF) approach has been systemically internalized into the institution's digital-security structure. The existence of a Security Operations Center (SOC) integrated with a cyber-threat-intelligence (CTI) platform and Security Information and Event Management (SIEM), along with active involvement in sectoral collaboration such as the Information Sharing and Analysis Center (ISAC) and Computer Security Incident Response Team (CSIRT), becomes evidence of organizational readiness in identifying and countering an attack in real time. This capability directly answers the first and second research questions, which relate to how an adaptive security system and proactive cyber intelligence are able to effectively maintain service continuity and digital resilience.

Weaknesses

Although high technological readiness has been achieved, there is a potential latent weakness that needs to be watched, particularly related to the overengineering phenomenon. An institution risks over-allocating budget to build an advanced defense system disproportionate to the actual threat. This can result in an imbalance in resource allocation, in which funds that should be used for digital-service development, security literacy, or improving inclusivity are instead absorbed to meet a symbolically administrative compliance demand. In addition, the security-fatigue symptom — namely a false-satisfaction condition due to the absence of a major incident within a certain time period — can weaken the intensity of evaluation and innovation of the security system. The accumulation of this condition has the potential to lower organizational readiness against a latent or low-frequency–high-impact attack. Therefore, this weakness indirectly answers the third research question, which highlights the urgency of forming a progressive, adaptive information-protection policy based on periodic risk evaluation.

Opportunities

The relatively stable external condition can be used as strategic momentum for institutional consolidation and strengthening the cross-sector information-sharing ecosystem. A banking institution has great opportunity to shift from a policy-based-compliance approach toward performance-based compliance, with emphasis on an effectiveness indicator such as mean time to detect (MTTD), mean time to recover (MTTR), and a resilience-maturity index. In addition, national regulation such as the establishment of the Indonesian National Work Competency Standard (SKKNI) for Cyber Incident Response in 2025 opens opportunity for an institution to develop HR capacity in a structured manner, through a training program, simulation, and long-term-resilience-oriented incident management. This opportunity contributes to answering the third research question, namely about how an institution can strengthen its information-protection policy and system to adaptively and anticipatorily address the dynamics of cyber-risk escalation.

Threats

Although the intensity of attack from an external cyber actor declines, potential threat still exists in the form of a systemic blind spot arising from excessive dependence on an automation system without periodic validation and testing. When system strengthening is not balanced with an increase in human capacity and business process, an institution becomes vulnerable to a low-frequency–high-impact attack, such as targeted spear phishing or staged data theft (slow data exfiltration). In addition, fragmentation of risk understanding between work units or functional divisions can hinder an organization's response to an incident appearing suddenly and unexpectedly. Therefore, thorough integration of the four strategic approaches — CTI, ASF, strategic intelligence (SI), and cyber resilience (CR) — is needed as a systemic foundation for maintaining technical and organizational adaptive readiness. Addressing threat in this scenario simultaneously strengthens the research narrative that a dynamic, cross-disciplinary security approach is a necessity in confronting the continuously developing cyber-threat landscape.

Solution

The application of an adaptive security approach in confronting a high-readiness–low-threat configuration can begin from thorough integration between the technology system, managerial process, and improvement of human-resource capability supported by the latest national regulation. The establishment of the Indonesian National Work

Competency Standard (SKKNI) in the field of Cyber Incident Response in May 2025 allows a banking institution to build incident-response capacity holistically and to a national standard. With 13 competency units covering the entire incident-handling cycle from planning through to reporting, the SKKNI TIS becomes a foundation for continuously strengthening the cyber-resilience (CR) aspect. This implementation answers the first research question regarding how the adaptive security framework (ASF) is measured not only from sophisticated technology, but also from an institution's ability to adapt humanly and procedurally to the dynamics of cyber attack.

On the other hand, to support a proactive cyber-intelligence system, a data-based approach and inter-institutional collaboration becomes crucial. Kominfo Ministerial Regulation No. 5 of 2025 on PSE Public Scope provides an electronic-data-classification structure into three categories: open, restricted, and closed. This policy provides a technical basis for developing a threat-prioritization and risk-informed-response system, which is very important in developing a cyber-threat-intelligence (CTI) system. With different treatment based on data sensitivity — such as at-rest and in-transit encryption — a banking institution can focus its monitoring-and-response system on the most vulnerable area. This approach strengthens CTI's role as an early-warning system while also becoming a pillar in answering the second research question regarding the effectiveness of the intelligence system in responding to phishing, ransomware, and data leak.

Strengthening the software-based security system also becomes a crucial element that needs attention. Kominfo Ministerial Regulation No. 6 of 2025 on SPBE Application Technical Standards requires that every digital application follow a software-development life cycle based on SNI ISO/IEC 12207 and SNI ISO/IEC 25000. In this context, a banking digital-service system such as mobile banking, e-payment, and digital onboarding can be ensured to be not only secure by default, but also resilient by design. This standard enables periodic application-security-quality testing and evaluation — including penetration testing and code review — which directly contributes to improving strategic intelligence (SI). With SI, defense policy is not merely reactive, but based on long-term analysis and a developing threat pattern, as asked in the third research question.

If these three regulations are integrated simultaneously, they can form a multi-layered defense ecosystem in the banking sector, combining technical readiness, policy precision,

and human-resource preparedness. This model supports the development of a security system that does not depend on the current intensity of attack, but is able to remain proactive and dynamic in responding to the changing tactics and attack tools used by cyber criminals. Within an adaptive framework, a banking institution can use an indicator such as mean time to detect (MTTD) and mean time to recover (MTTR) as a security-system performance metric, strengthening a performance-based-compliance evaluation system rather than mere formal compliance.

Finally, this solution asserts that high readiness does not mean stopping innovation. In fact, in a condition of low attack, the biggest challenge is maintaining the urgency and speed of adaptation without losing efficiency. Therefore, the use of the SKKNI TIS as a training reference, the application of PSE data classification for a risk-based protection system, and strengthening of the SPBE standard in the digital-application-development cycle become a policy synergy directly answering the three research questions. An integrative approach based on CTI, ASF, SI, and CR enables the Indonesian banking sector to strategically and sustainably maintain its digital resilience amid a threat landscape that continues to develop and become increasingly hidden.

4.2.5. Roadmap

As a follow-up to the constructed scenarios, compiling a roadmap becomes a crucial step in designing a short-, medium-, and long-term implementation strategy to strengthen the banking sector's cyber resilience. The following figure maps a roadmap recommendation based on a two-axis foresight matrix, namely an institution's resilience-capability level and the sophistication level of a cyber attack. Each quadrant contains a policy direction, capacity-building priority, and reform agenda tailored to the readiness condition and threat potential faced.

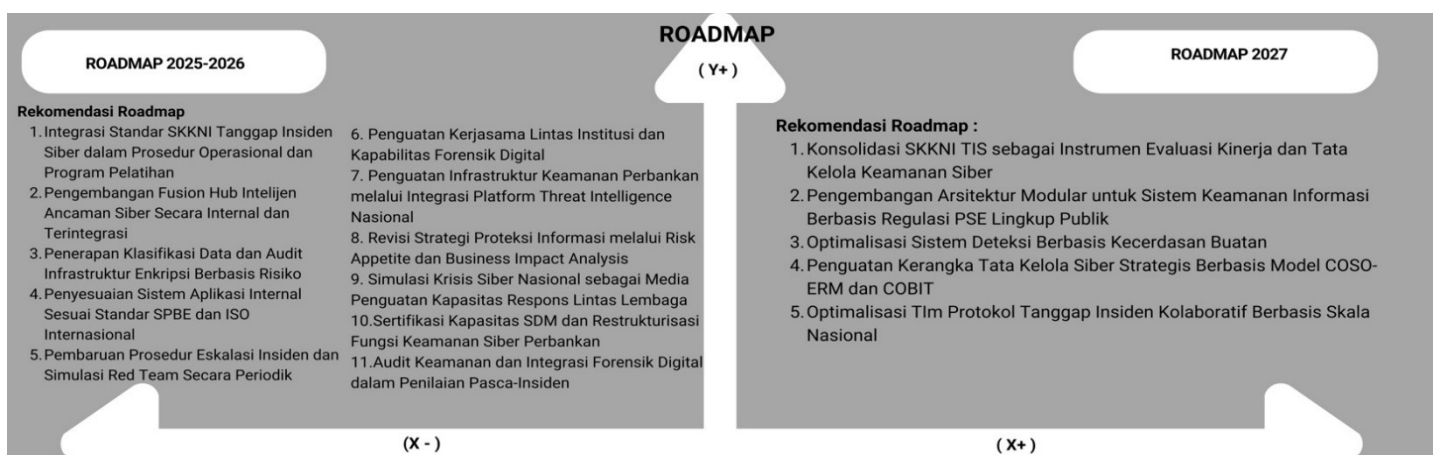


Figure 4.2 Strategic Roadmap for Indonesian Banking Cyber Resilience 2025–2029

Source: Researcher's Data

4.2.5.1. Cyber-Resilience Roadmap for 2025

The year 2025 becomes the initial foundation for building an integrated national cyber-resilience system oriented toward a rapid response to an incident. The main focus is directed at integrating the SKKNI Cyber Incident Response Standard into operational procedure and a national training program, in order to shape human-resource readiness for confronting the escalation of digital attack systematically. At the same time, the development of a Cyber Threat Intelligence Fusion Hub, internally and across institutions, is carried out as part of an effort to unify the flow of threat data and information that has thus far been scattered across various institutional silos. The effort to improve security posture is also realized through the application of data classification and a risk-based encryption-infrastructure audit, as a mitigation step against potential leakage and exploitation of vital information systems. Adjusting internal application systems to the SPBE standard and international ISO reference asserts the importance of interoperability and strengthening technology governance. Updating the incident-escalation scenario through periodic Red Team simulation is carried out to test organizational resilience against a complex, dynamic attack. This entire agenda is closed with strengthening cross-institutional cooperation and improving digital-forensic capability, which not only plays a role in the post-incident investigation process, but also

in building a sustainable learning system as the basis for an adaptive, collaborative national cyber resilience.

4.2.5.1.1. Integration of the SKKNI Cyber Incident Response Standard into Operational Procedure and Training Program

The initial step that can concretely be taken is adapting every element of the 13 competency units of the SKKNI Cyber Incident Response (TIS) into a banking institution's operational policy. This covers the establishment of a standardized cyber-incident-handling procedure, from the early-detection stage, incident reporting, mitigation, system recovery, through to the compilation of post-incident documentation. In addition, intensive training is needed for Security Operation Center (SOC) personnel and the incident-response team (CSIRT), held with a simulation approach based on a real-attack case study such as the LockBit ransomware. Assessment of the mastery of this competency can be used as an indicator in the internal-audit process to measure human-resource readiness for the cyber-resilience principle.

4.2.5.1.2. Internal and Integrated Development of a Cyber-Threat-Intelligence Fusion Hub

To strengthen the ability to detect and respond early to an incident such as phishing, ransomware, or data breach, an institution needs to form an intelligence-integration center (fusion hub) consolidating the CTI function, Security Information and Event Management (SIEM), and an endpoint-log system in real time. In the short term, automatic integration with a national CTI source such as OJK, BSSN, and the ISAC forum can be realized through an integrated-monitoring dashboard. In addition, threat-intelligence enrichment from various open or commercial feeds needs to be carried out to proactively recognize an indicator of compromise (IoC). An alert-threshold-based early-warning system can then be configured to produce automatic action at the SOC with a focus on the efficiency of mean time to detect (MTTD) and mean time to recovery (MTTR).

4.2.5.1.3. Application of Risk-Based Data Classification and Encryption-Infrastructure Audit

Based on the provisions in Kominfo Ministerial Regulation No. 5 of 2025, an institution must classify electronic data into three sensitivity levels, namely open, restricted, and closed. To ensure appropriate protection, the data-storage and -transmission system needs to be audited to meet an encryption standard, both in a static (at rest) and sent (in transit) condition. The application of an access-control model based on the zero-trust principle is also important, particularly for data with a closed classification such as strategic customer identity and transaction information. This step is part of a risk-based protection approach and supports the application of an adaptive security framework within the context of a more contextual, non-uniform information-protection policy.

4.2.5.1.4. Adjusting the Internal Application System to the SPBE Standard and International ISO

On the application-technology side, a banking institution needs to align its software development with the SNI ISO/IEC 12207 and ISO/IEC 25000 standard as set out in Kominfo Ministerial Regulation No. 6 of 2025. This adjustment covers systematic documentation of the application life cycle, execution of functional testing, and technical-vulnerability mitigation through periodic regression testing. A module related to financial transaction, user authorization, and the mobile-banking channel must become the main priority. The formation of a cross-divisional application-audit team (application security/AppSec) is also recommended to conduct a quarterly examination based on the SPBE development stage.

4.2.5.1.5. Updating the Incident-Escalation Procedure and Periodic Red Team Simulation

To ensure cross-divisional response readiness, an institution needs to establish an automatic cyber-incident-escalation procedure based on severity-level classification. The

maximum time limit for reporting and recovery must be firmly set. In addition, the execution of red-teaming simulation at least twice a year is highly recommended. This activity must involve the IT-security unit, legal, through to corporate communication to test tactical and coordinative readiness when confronting a complex attack. The logging-and-audit-trail system also needs to be strengthened, including to detect an attack pattern such as brute-force login, privilege escalation, or social engineering.

4.2.5.1.6. Strengthening Cross-Institutional Cooperation and Digital-Forensic Capability

The next step is strengthening external collaboration through the formation of a memorandum of understanding (MoU) for cyber-intelligence-information exchange with an institution joined in the ISAC and CSIRT. In addition, forming an internal digital-forensic team is very crucial for carrying out a post-incident investigation with a fast response time. The investigation results must be systematically compiled as lessons-learned material evaluated in a periodic forum. Finally, the process of aligning internal security-audit results with a report or recommendation from an authority such as BSSN and OJK will ensure that the direction of the adaptive roadmap remains relevant to the developing national risk context.

4.2.5.2. Indonesian Banking Cyber-Resilience Roadmap for 2026

Entering 2026, the national cyber-resilience strategy begins to focus on strengthening an early-warning system and improving proactive, intelligence-based detection capability. The main agenda is directed at developing an integrated Cyber Threat Intelligence Platform system supported by an artificial-intelligence (AI)- and machine-learning-based analytical architecture, in order to improve the capacity for automatic and contextual attack-indicator detection. In an effort to build data-based resilience, refinement of the cyber-risk-classification framework is also carried out through a layered approach integrating technical data, behavioral indicators, and infrastructure-vulnerability evaluation.

This year's initiative also covers the acceleration of organizational-capacity strengthening through the establishment of a rapid-response unit in the banking sector and other strategic institutions, equipped with a cross-sector coordination mechanism and a real-time incident-reporting dashboard. This is carried out to encourage a more integrated response ecosystem between financial institutions, the regulatory authority, and the national cyber-security institution. In line with this, 2026 also becomes an important moment for the implementation of a dynamic-policy-engine system — an

adaptive framework enabling cyber-security policy to be automatically reconfigured based on the threat level and actual incident context. In addition, this year is marked by the start of the development of a collective inter-institutional learning system through the integration of an incident-learning repository, which allows an institution to share an attack pattern and handling mechanism based on real cases. Through this approach, cyber resilience is formed not only from the technology and regulation side, but also through the strengthening of an intelligence culture and sustainable collaboration among stakeholders.

4.2.5.2.1. Strengthening Banking-Security Infrastructure Through the Integration of a National Threat-Intelligence Platform

In 2026, national banking institutions are encouraged to strengthen their cyber resilience through direct integration with a national threat-intelligence platform developed by the National Cyber and Crypto Agency (BSSN) together with cross-sector partners such as the Information Sharing and Analysis Center (ISAC) for the financial sector. This process covers aligning an internal system such as Security Information and Event Management (SIEM), Endpoint Detection and Response (EDR), and Network Detection and Response (NDR) with an external intelligence-data source, using an already-standardized Application Programming Interface (API). As a follow-up step, a threat-classification scheme needs to be applied based on tactics, techniques, and procedures, or Tactics, Techniques, and Procedures (TTP), as set out in the MITRE ATT&CK framework. In addition, the implementation of artificial-intelligence-based threat hunting becomes important to identify anomaly and a latent attack undetected by a signature-based system. This strategy represents the application of an adaptive security approach and the use of a proactive cyber-intelligence system in detecting and responding to increasingly complex threat.

4.2.5.2.2. Revising the Information-Protection Strategy Through Risk Appetite and Business Impact Analysis

Another important step is reformulating data-protection policy based on risk appetite and the results of a Business Impact Analysis (BIA). This approach emphasizes a thorough evaluation of the value and sensitivity of digital assets through a standard BIA methodology, as set out in ISO/IEC 27005:2022. Based on those results, a bank can set an information-protection baseline in accordance with the criticality level and business impact of each information category. Each baseline then becomes the reference for

applying an appropriate protection system, such as strengthening Data Loss Prevention (DLP) control, applying the zero-trust principle, and using a layered encryption scheme. This approach supports the integration of a more contextual, adaptive, and progressive information-protection policy, thereby answering the need for strengthening information resilience in confronting continuously developing cyber risk.

4.2.5.2.3. A National Cyber-Crisis Simulation as a Medium for Strengthening Cross-Institutional Response Capacity

In responding to the rising risk of a large-scale cyber attack, a banking institution needs to actively participate in the execution of a national-scale cyber-crisis simulation. This simulation can be developed through collaboration between BSSN, OJK, the banking sector, and law-enforcement bodies. The designed scenario covers a ransomware-incident simulation involving many actors and requiring a standardized national response protocol. That protocol contains a division of tasks, an escalation scheme, and a post-incident public and internal communication strategy. Execution of this simulation is expected to evaluate an institution's capability in incident containment, service recovery, and the effectiveness of intelligence-based coordination. This effort not only reflects the application of the intelligence-driven-security principle, but also shows the importance of systemic, cross-institutionally integrated adaptive readiness.

4.2.5.2.4. HR-Capacity Certification and Restructuring of the Banking Cyber-Security Function

To optimize the role of human resources in confronting the dynamics of digital threat, a banking institution must ensure that all personnel joined in the Security Operation Center (SOC) and Computer Security Incident Response Team (CSIRT) already hold a technical-competency certification, such as CEH (Certified Ethical Hacker), CompTIA CySA+ (Cybersecurity Analyst), or a standard equivalent to SKKNI TIS 2025 ratified by the Ministry of Kominfo (2025a). In addition, the cyber-security organizational structure also needs to be recalibrated by forming specific units handling threat analysis, vulnerability research, and digital-forensic investigation. An institution is also recommended to develop an internal learning system based on a Learning Management System (LMS), equipped with a real-attack case study in Indonesia's financial sector such as LockBit 3.0. This approach strengthens organizational readiness in building an

adaptive, resilient security system, and answers the need to integrate a human-resource strategy with information-security policy.

4.2.5.2.5. Security Audit and Digital-Forensic Integration in Post-Incident Assessment

Routinely conducted security evaluation must undergo a transformation toward a scenario-based approach to replace a rigid checklist-based audit system. In this approach, an audit is carried out with a realistic incident simulation in order to assess procedural and technical resilience contextually. In addition, post-incident-analysis activity must be expanded to cover digital-forensics investigation, with the aim of identifying the root cause of the incident and the accompanying process weakness. To support data-based reporting and decision-making, a bank is also recommended to develop an automated audit dashboard able to visualize an important indicator such as Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), and a vulnerability score based on the Common Vulnerability Scoring System (CVSS). This strategy answers all of the research questions with a systematic, in-depth, and measured approach, creating a defense system that is not only reactive, but also predictive and precise.

4.2.5.3. Indonesian Banking Cyber-Resilience Roadmap for 2027

The year 2027 becomes a consolidation phase in the journey of building strategic-sector cyber resilience, with an emphasis on expanding security-system interoperability and cross-institutional-platform integration. The strategic focus is directed at the full implementation of Security Orchestration, Automation, and Response (SOAR), which combines automatic response, intelligence analysis, and incident correlation into a single integrated system. This approach is aimed at minimizing the time lag between threat detection and mitigation through automation of the technical decision-making process previously carried out manually. The effort to improve policy effectiveness is also accelerated through the integration of a dynamic-compliance-evaluation system (dynamic compliance engine) directly linked to national regulation such as POJK No. 11/POJK.03/2022 and Presidential Regulation No. 82/2022, as well as an international standard reference such as ISO/IEC 27002 and ISO/IEC 27035. With this approach, every institution is able to adjust internal policy in real time based on an evaluation of detected risk and potential attack escalation.

At the same time, strengthening a collaborative cyber drill between institutions, both domestically and regionally, is carried out to test a multi-layer incident scenario and response capability in confronting a coordinated large-scale attack. This exercise is also accompanied by the compilation of a banking-sector resilience index based on a technical, organizational, and policy indicator, used as a benchmark for national-system readiness in responding to systemic threat. Through this stage, the strengthening of cyber resilience begins to shift from a detection- and response-based approach toward a strategic recovery capability that is measured, metrics-based, and strengthened by an institutional learning system adaptive to the increasingly dynamic and unpredictable threat landscape.

4.2.5.3.1. Consolidation of the SKKNI TIS as an Instrument for Evaluating Cyber-Security Performance and Governance

Entering 2027, the strengthening of institutional capacity in the banking sector is directed at integrating the national competency framework into the cyber-security-governance system. The Indonesian National Work Competency Standard (SKKNI) in the field of Cyber Incident Response (TIS), which began to be enacted since 2025, needs to be converted into an applicable performance-measurement indicator within the internal scope. A competency unit such as incident-recovery capability (recovery time objective), accuracy in analyzing a cyber attack, and the quality of incident reporting can be used as an evaluative matrix for assessing the effectiveness of the Security Operation Center (SOC) and Computer Security Incident Response Team (CSIRT) team. This consolidation not only strengthens the adaptive-security approach, but also answers a research question related to improving adaptive-response readiness and the implementation of a proactive intelligence system in confronting the dynamics of threat.

4.2.5.3.2. Developing a Modular Architecture for an Information-Security System Based on Public-Scope PSE Regulation

Along with the enactment of Ministry of Communication and Informatics Regulation Number 5 of 2025 on Public-Scope Electronic-System Operators (PSE), 2027 becomes an important momentum for the banking sector to rearrange its information-security-system architecture modularly. This system must have the ability to automatically identify and differentiate the protection of electronic data based on classification — open, restricted, and closed — as set out in that regulation. Its implementation can be carried

out through the application of an adaptive encryption engine and an attribute-based access control (ABAC) system. This modular design functions as the technical foundation of a more progressive, risk-responsive information-protection policy, answering the research focus regarding the integration of information policy within the context of banking digital resilience.

4.2.5.3.3. Optimizing an Artificial-Intelligence-Based Detection System and Adjusting the Security-Infrastructure Baseline

The increasingly rising trend of artificial-intelligence-based cyber attacks, such as ransomware-as-a-service and the deepfake-spear-phishing technique, demands a transformation in the detection system used by financial institutions. Therefore, the 2027 roadmap recommends optimizing the detection system through integrating machine-learning anomaly-detection technology and behavioral analytics. Updating the security-infrastructure baseline can be carried out by combining User and Entity Behavior Analytics (UEBA) capability into the SIEM (Security Information and Event Management) system, and implementing periodic adversarial simulation to test system resilience against a new attack pattern unrecognizable by a conventional system. This strategy is a real manifestation of a predictive intelligence system, while also answering two crucial aspects in the research: early-detection capability and an adaptive security approach.

4.2.5.3.4. Strengthening the Strategic Cyber-Governance Framework Based on the COSO-ERM and COBIT Model

In confronting a systemic threat requiring cross-institutional coordination, the 2027 roadmap sets the urgency of strengthening cybersecurity governance by adapting the COSO-ERM (Enterprise Risk Management) and COBIT (Control Objectives for Information and Related Technologies) framework. This governance model places the strategic role of the board of directors, the risk-management committee, and the information-security unit within a coordinated decision-making ecosystem. A periodic agenda such as a cyber risk committee quarterly review and a governance-maturity-level audit becomes an indicator of an institution's maturity in aligning cyber-security policy with business objectives and oversight regulation. This configuration supports the achievement of the third research objective by asserting the importance of an integrated,

accountable, strategic-risk-management-based information-protection policy (COSO, 2023).

4.2.5.3.5. Optimizing a National-Scale Collaborative Incident-Response Protocol

Response to a wide-impact cyber incident requires a coordination framework going beyond institutional boundaries. Therefore, 2027 is marked by the development of an industry-scale incident-response protocol compiled collaboratively by the banking sector, a regulator such as OJK, the BSSN cyber-security authority, and a national CSIRT community network. This protocol covers the development of an incident-severity-scoring method, activation of a cross-institutional emergency-communication pathway, and the provision of a real-time collaborative dashboard displaying response-status information, action recommendation, and current threat distribution. This initiative demonstrates the application of the strategic-foresight principle and strengthens collective cyber resilience, while also answering the second question and the research objective regarding a collaborative, national, intelligence-based response system.

4.2.5.4. Indonesian Banking Cyber-Resilience Roadmap for 2028

Entering 2028, the orientation of national cyber-resilience policy is directed toward strengthening predictive capacity and long-term strategic planning to anticipate a disruptive threat. The main priority lies in developing a National Cyber Risk Register containing a cross-sector strategic-risk classification, based on the integration of big data, historical-incident-analysis results, and artificial-intelligence-based threat-scenario modeling (AI-based threat simulation). This initiative aims to form a prospective national database usable in the cross-institutional decision-making process quickly and precisely.

Within a policy-governance framework, a policy-foresight-framework approach is applied that supports regulation design based on the dynamics of future threat. This instrument integrates a strategic-assessment process, technology-trend analysis, and institutional-resilience projection to reduce the gap between threat reality and regulatory readiness. Through this approach, the cyber-security-policy system is not only reactive, but also able to adapt to the non-linear complexity of digital threat.

In addition, strengthening digital literacy and leadership becomes a crucial element in this year's roadmap, with the national application of an executive cybersecurity leadership program for the leadership of strategic institutions. Its purpose is to create a policymaker ecosystem not only understanding the technical security aspect, but also possessing the competency to navigate strategic risk and build an information-protection culture at the institutional level. This year's initiative reflects the transition toward an anticipatory, intelligence-based cyber-security system supported by strong cross-sector collaboration.

4.2.5.4.1. Integrating the SKKNI TIS Competency Standard into a Professional Certification and Evaluation Mechanism

As a response to the need to improve response capacity for an increasingly complex cyber incident, banking institutions in 2028 are directed to adopt the Indonesian National Work Competency Standard (SKKNI) in the field of Cyber Incident Response into the certification and professional-competency-recognition system for information security (BNSP, 2025). This process covers measuring the technical performance of a unit such as the Security Operation Center (SOC) and Computer Security Incident Response Team (CSIRT), referencing the 13 work units within the SKKNI, covering procedure compilation, incident-scenario testing, through to post-incident reporting. Evaluation is carried out based on an operational-performance indicator such as Mean Time to Detect (MTTD) and Mean Time to Recover (MTTR). In addition, partnership with a BNSP-certified training institution will strengthen the quality of human-resource-capacity development. This approach also answers the need to implement a system that is adaptive and based on intelligence-driven detection.

4.2.5.4.2. Optimizing the Implementation of an Information-Security Architecture Based on Electronic-Data-Risk Classification

A follow-up to Kominfo Ministerial Regulation No. 5 of 2025 drives the strengthening of a data-classification-based security system, differentiated into the open, restricted, and closed category (Ministry of Communication and Informatics, 2025). In 2028, every bank is required to apply a data-centric cybersecurity architecture adjusting its protection mechanism to the sensitivity level of the information. Protection of restricted and closed data must use encryption-at-rest and encryption-in-transit, along

with attribute- and context-based access management (attribute-based access control and context-aware access management). Online processing activity for closed data is entirely prohibited, so a system design with physical and virtual security segmentation is needed. This approach answers the progressive information-protection-policy aspect needed within the context of strategic risk and the critical value of financial-sector digital data.

4.2.5.4.3. Developing a Threat Fusion Center for Financial-Sector Collaborative Intelligence

As part of the strategy for strengthening cross-institutional-collaboration-based early detection, the 2028 roadmap includes the formation of a Threat Fusion Center functioning as a technical and tactical intelligence-coordination center between the regulator (OJK, BSSN) and banking-industry actors (BSSN, 2024). This center will manage the integration of threat-telemetry data, honeypot-system observation results, and an indicator from the SIEM and EDR system through a secure-API-sharing scheme. Every analytical model will be built referencing the MITRE ATT&CK framework and periodically updated through a sectoral coordination forum. This initiative enables a standardized, predictive detection-and-response approach able to accommodate the dynamics of sophisticated threat. This roadmap answers the need to form a structured collective-detection system within the framework of national cyber resilience.

4.2.5.4.4. Development and Implementation of Artificial-Intelligence-Based Detection Technology and Adversarial Simulation

The increasing prevalence of artificial-intelligence-based cyber attacks such as ransomware-as-a-service, deepfake phishing, and large-language-model exploitation requires an adaptive, highly precise security-technology approach (Deloitte, 2024; BSSN, 2024). For that, financial institutions are expected to implement the integration of the User and Entity Behavior Analytics (UEBA) system with an adversarial-simulation environment to test defense strength against a non-conventional attack. The simulation covers a voice-cloning technique, automated scripting, through to an AI-based social-engineering scenario. This detection model is designed as the backbone of a big-data-based prediction system functioning to detect anomaly in real time before threat escalation occurs. This approach strengthens a cyber-security pillar responsive to the dynamics of contemporary cyber technology.

4.2.5.4.5. Implementation of a Service-Recovery System and Operationalization of Financial-Sector Sustainability

As an anticipation of systemic-disruption risk due to a large-scale cyber attack, the 2028 roadmap drives the application of a Business Continuity and Recovery Integration Plan (BCRIP) based on the international standard ISO/TS 22317:2021 and ISO/IEC 27035-3:2020 (ISO, 2020; ISO, 2021). This strategy covers the integration of an inter-bank Recovery Time Objective (RTO) target, updating a failover scenario to a disaster-recovery site, and strengthening the interoperability of a system such as BI-FAST, ATM Bersama, and QRIS. Execution of the recovery simulation is carried out periodically through industry-wide simulation and a table-top exercise directly supervised by the financial authority. This step ensures that operational recovery is not only reactive, but also able to maintain business continuity in confronting a measured, systemic crisis scenario.

4.2.5.5. Indonesian Banking Cyber-Resilience Roadmap for 2029

The year 2029 marks the maturation phase of a national cyber-resilience system oriented toward sustainability and long-term cyber resilience. At this stage, policy is focused on the consolidation of a strategic information-protection ecosystem based on institutional interoperability, learning continuity, and the strengthening of an audit-and-public-accountability mechanism against a cyber incident. The main initiative is directed at applying an adaptive-cyber-governance model that is dynamic and reflective in nature, allowing for the periodic renewal of regulation and organizational structure based on the evaluation results of national digital-resilience performance.

The development of long-term cyber-intelligence infrastructure is also improved through the full integration of a national cyber-early-warning-grid system, able to monitor a cross-sector anomaly signal and present a strategic indicator based on predictive modeling. This system does not only function as an early-detection tool, but also as a basis for building a more accurate, real-time cyber situational awareness for policy decision-making.

In addition, this year's agenda covers the establishment of a Cyber Resilience Knowledge Center serving as a cross-sector collaboration center and an evidence-based-policy-lab innovation center. This institution's function is to manage the evaluation results

of prior years' roadmap, compile a national-data-based resilience index, and provide an advocacy, dissemination, and public-education platform. With the formation of this center, national cyber-resilience strategy is no longer ad hoc or incidental, but is systematically built through reflection, institutional learning, and measured long-term collaboration.

4.2.5.5.1. Implementation of National-Competency-Based Incident-Response Automation and AI Predictive Response

The year 2029 marks the start of the implementation of an adaptive-automation-based incident-response system integrated with the SKKNI TIS national competency framework. All banking institutions are required to adopt a Security Orchestration, Automation, and Response (SOAR) platform configured based on the 13 SKKNI work units, such as incident containment, system recovery, and real-time digital reporting (BNSP, 2025). This platform is connected to national and local threat-intelligence data, and supports predictive detection with the strengthening of machine reasoning to detect a not-yet-classified digital-behavior anomaly. This initiative functions as a strategic answer to the escalation of AI-based attack and answers the research question about early detection and long-term adaptive resilience.

4.2.5.5.2. Development and Optimization of a Resilient Data Architecture Based on Zero Trust and Risk Segmentation

As a continuation of the implementation of Kominfo Ministerial Regulation No. 5 of 2025 on Public-Scope PSE, the 2029 roadmap requires the national banking sector to thoroughly adopt the zero-trust-architecture (ZTA) principle, including biometric- and operational-context-based layered authentication, and network microsegmentation based on electronic-data classification (open, restricted, closed) (Ministry of Communication and Informatics, 2025). Treatment of closed data involves an offline-processing system, double encryption, and dynamic-role-based access control. This step aligns with the need for progressive protection policy, answering the research question regarding the integration of security policy with the critical value of information.

4.2.5.5.3. Transforming Threat Intelligence into a Collaborative Predictive System Based on Agentic AI

The year 2029 marks the transition from a traditional threat-sharing model toward collaborative predictive intelligence, using agentic artificial intelligence to accelerate the intelligence cycle from observation to mitigation. This system allows a banking entity to model a possible attack based on threat telemetry, MITRE ATT&CK tactics, and an anomaly pattern based on adversarial machine learning (Kandaswamy et al., 2023). Every bank is required to send a threat signal to a national fusion center jointly controlled by BSSN and OJK, and to hold an inter-bank evaluation scenario every quarter. This development answers the second research question regarding the integration of strategic intelligence in the early response to complex threat.

4.2.5.5.4. Application of Continuous Cybersecurity Assurance Through Adaptive Audit and Dynamic Assessment

As a strengthening of institutional resilience, this roadmap drives the implementation of a COBIT- and COSO-based cybersecurity-assurance model formulated in the form of a dynamic audit (continuous adaptive control assessment) and residual-risk-based reporting (COSO, 2023). The banking sector is required to compile a digital-resilience-evaluation dashboard measuring protection capability against five main vectors: system integrity, recovery readiness, access control, monitoring effectiveness, and policy-adaptation flexibility. Assessment is no longer annual, but ongoing through the integration of an automatic indicator from the SIEM and endpoint system. This is a direct answer to the urgency of an evaluative policy able to adapt to the complexity of cyber threat.

4.2.5.5.5. National Simulation and Replication of Attack on Critical Infrastructure

Answering the need to strengthen systemic resilience, the 2029 roadmap requires a banking institution to take part in a strategic cyber warfare simulation held twice a year. The simulation is carried out using a digital twin of the banking-service system — including core banking, the BI-FAST payment system, and QRIS infrastructure — tested against a ransomware-based attack, AI-driven phishing, and deepfake manipulation. This training is combined with cross-institutional red-blue teaming, under the supervision of BIN, OJK, and BSSN. Its purpose is to test response speed, mitigation effectiveness, and the continuity of public banking service, while also answering the research objective regarding a resilient, collaborative national resilience system.

Chapter 5 Conclusion and Recommendations

5.1. Conclusion

This research examines the cyber-attack pattern targeting the banking sector in Indonesia using a proactive-cyber-intelligence-based approach and an adaptive security framework to understand the level of escalation of continuously developing digital threat. Through the application of the Cyber Kill Chain model, it was found that attacks against financial institutions occur in structured, layered stages, with a combination of social-engineering techniques and technical exploitation of digital-infrastructure weakness. This series of cyber activity begins from the reconnaissance stage through to system takeover, which ultimately impacts service disruption or the leakage of sensitive information. This finding shows that every weakness in an organization's system and internal process can become a strategic gap if not anticipated through an integrated early-detection and adaptive-response system.

Within that framework, the role of Cyber Threat Intelligence (CTI) becomes fundamental in detecting potential threat before an attack reaches a critical stage. However, CTI implementation at various banking institutions is still not optimal. Many financial institutions have not made CTI part of the decision-making process or an early-warning system connected to a digital-risk-management strategy. Several identified obstacles include limited HR technical competency, weak internal data structure, and the not-yet-formed interoperability and cross-institutional-collaboration system that could strengthen threat-analysis accuracy (Goel et al., 2022).

Further, the use of the Adaptive Security Framework (ASF) indicates that a flexible, layered cyber-defense system is an urgent need. The four main elements in this framework — predict, prevent, detect, and respond — have not yet fully run at most institutions. The lack of cohesion between CTI analysis results and technical policy causes weak anticipation of new attack patterns, while detection and response capability is often hampered by a bureaucratic process and limited technical infrastructure.

In line with this, the Strategic Intelligence (SI) approach becomes a complementary pillar in creating an anticipatory, scenario-based decision-making system. By considering external dynamics and long-term trends, SI allows banking institutions to assess the

strategic impact of every cyber incident and integrate intelligence results into a more future-oriented protection policy (Walsh, 2017). SI also plays a role in bridging understanding between the technical side and strategic decision-makers, thereby supporting a thorough risk-governance process.

Within a long-term-resilience framework, the Cyber Resilience (CR) principle needs to be integrated to ensure that an institution is not only able to withstand a cyber incident, but can also recover quickly, learn from every incident, and strengthen its adaptive capacity going forward. CR does not merely focus on prevention and protection, but also covers business continuity, crisis management, and the formation of a sustainable, risk-aware cyber culture (Linkov & Kott, 2019).

On the policy side, the national regulatory structure governing cyber resilience in the banking sector has been formed fairly comprehensively. Financial Services Authority Regulation No. 11/POJK.03/2022 and POJK No. 38/POJK.03/2016 become the foundation in information-technology-risk management, while Presidential Regulation No. 82 of 2022 categorizes the financial sector as vital information infrastructure that must be protected. This regulation drives the formation of a Computer Security Incident Response Team (CSIRT) and periodic technical audit as an oversight mechanism. In addition, Ministry of Communication and Informatics Regulation No. 5 and No. 6 of 2025 strengthen the responsibility of private institutions in handling digital incidents. A national strategic document such as BSSN's 2024 National Cyber Security Landscape also becomes a reference for setting a threat map and mitigative policy at the macro level.

Nevertheless, the existence of this regulatory framework does not yet fully guarantee effectiveness in the field. Many banking institutions, particularly those outside the group of large banks, do not yet have the technical and managerial readiness to translate regulatory provisions into a concrete protection system. The main challenge lies in the HR-quality gap, limited investment in security technology, and a lack of strategic attention from institutional leadership toward digital-security issues as part of core governance. As a result, security audit and incident reporting tend to be administrative in nature, and do not reflect organizational readiness in managing real risk. On the other hand, weak sanction enforcement and minimal audit-result transparency also hinder the emergence of a strong accountability culture in this sector.

The foresight approach through horizon scanning applied in this research identifies three main drivers pushing the escalation of cyber risk over the next five years. First, the increasing use of automation and artificial intelligence by cyber attackers allows them to dynamically adjust tactics. Second, the security-readiness gap between banking institutions — including even among large institutions — causes the emergence of easily exploitable weak points. Third, an increasingly accessible cybercrime-as-a-service service opens opportunity for non-expert actors to carry out an attack with sophisticated tools available online (Interpol, 2022). These three factors create a threat landscape that is scattered, adaptive, and difficult to predict with a traditional approach.

Taking all of these findings into account, it can be concluded that the Indonesian banking sector's cyber-security system is at a crucial phase for undertaking thorough transformation. A security approach still reactive and administrative in nature needs to be replaced with a system grounded in cyber intelligence, adaptive integration, and visionary strategic leadership. Digital security must be positioned as a dynamic, evolving process, based on organizational learning, and unified within institutional governance. Therefore, thorough integration between the CTI approach, the ASF framework, the SI and CR principles, and a consistently enforced policy system, becomes the main foundation for strengthening the digital resilience of the Indonesian banking sector in confronting a complex, adaptive, and multidimensional cyber threat.

5.2. Recommendations

Recommendations for Banking Top-Level Management

Top management at banking institutions in Indonesia needs to place cyber-security issues as an integral part of the institution's overall governance strategy, not merely treated as an operational technical aspect. Given the continuously developing character of a cyber attack in terms of complexity and adaptability, an approach that relies solely on post-incident response (a reactive approach) is no longer adequate for confronting a challenge that is dynamic and often unpredictable. This condition requires a paradigm shift toward a digital-defense system that is predictive, intelligence-based, and integrated with the institutional risk-management framework.

One strategic policy that can be adopted is the formation of a functional unit specifically handling cyber threat intelligence (CTI), placed within an organizational structure directly connected to the highest managerial ranks. This unit functions to facilitate the process of identifying threat indicators (indicators of compromise), mapping attack tactics and techniques, and assessing systemic vulnerability that can be used to support strategic decision-making in a timely manner. CTI's role becomes increasingly important when integrated with the Cyber Kill Chain approach, which parses the stages of a cyber attack from the reconnaissance phase, weaponization, through to action on objectives. Through this approach, financial institutions can recognize the pattern and exploitation gap used by an attacker, and formulate more accurate, evidence-based mitigative steps.

Applying Cyber-Kill-Chain-based analysis within the context of CTI provides added value in providing information that is not only technical, but also strategic. This information covers the type of social engineering used by the attacker, the infiltration pathway into the internal information system, and the evasion technique used to deceive the defense system. This information is highly crucial in compiling prevention-and-early-detection policy, and in handling a cyber incident in real time. The integration of CTI and Cyber-Kill-Chain analysis can form the foundation of a proactive, adaptive cyber-defense system, supported by continuous learning and institutional-capacity building. This becomes increasingly important in the context of the banking industry, which is very dependent on public trust, service continuity, and information protection as a key element of institutional resilience.

Further, the commitment to fully adopt the principles contained in the Adaptive Security Framework (ASF) needs to become an institutional priority. The four main pillars in this framework — predict, prevent, detect, and respond — must be implemented in an integrated, sustainable manner. This covers the development of threat-modeling-based technology, optimization of an adaptive early-detection system, and the compilation of a documented incident-response procedure that can be executed quickly and accurately.

From the Strategic Intelligence (SI) perspective, senior management has the responsibility of ensuring that medium- and long-term business planning takes into

account the dynamics of cyber-threat evolution. The foresight approach can be used as a tool for mapping various possible future digital-risk scenarios, while also serving as a basis for data-based decision-making within the context of technological, geopolitical, and regulatory uncertainty.

Meanwhile, the Cyber Resilience (CR) approach requires consistent effort from institutional leadership to build an organizational culture resilient to digital disruption. This covers strengthening technical capability, developing an incident-recovery system, and periodically testing the business-continuity-planning design. The availability of resource allocation, whether in the form of budget or expert personnel, must be adequately ensured to guarantee an institution's readiness to confront a systemic, multidimensional cyber-disruption scenario.

Finally, the role of digital leadership from senior executive ranks becomes very important to ensure that the entire process of strengthening the digital-security system is not merely symbolic or administrative. Progressive, visionary leadership is needed to cultivate an institutional awareness that protecting digital assets is part of the institution's strategic resilience. In an era of increasingly unpredictable, asymmetrically occurring cyber attacks, a financial institution's success in maintaining reputation and service continuity is highly determined by the quality of a cyber-security strategy directly led by the organization's top leadership.

Recommendations for All Stakeholders

Recommendations for OJK

As the main oversight authority in the financial-services sector, the Financial Services Authority (OJK) holds the strategic mandate of ensuring that the cyber-security aspect is thoroughly integrated into the risk-management framework of banking institutions. Digital security can no longer be positioned as a complementary technical issue, but must become an inherent part of financial-institution governance. For that, OJK needs to develop an oversight model that does not solely rest on administrative fulfillment, but also covers a comprehensive technical evaluation of the effectiveness of digital-protection-system implementation at each institution. One approach that can be applied is an in-depth audit of the cyber-security system (deep cyber audit), including testing of

early-detection capability, responsiveness to a cyber incident, and an institution's readiness to operationally run a business-continuity-planning and disaster-recovery mechanism.

Further, in order to encourage the creation of a more resilient and open ecosystem, OJK needs to formulate a policy encouraging sectoral transparency in cyber-incident reporting. This can be realized through compiling a derivative regulation of POJK No. 11/POJK.03/2022 that requires banking institutions to submit a security-incident report in real time into an integrated national reporting system. In addition, publishing part of the technical-audit results in a standardized, stakeholder-accessible format can function as a collective-learning instrument and a reference for improving security capacity between institutions.

Strengthening OJK's role in supporting the formation of cross-institutional cyber threat intelligence (CTI) is also crucial in the context of intelligence-based risk management. OJK can function as a facilitator in forming a cross-institutional-banking CTI-coordination forum, and support the compilation of a threat-intelligence-integration mechanism into the financial institution's overall risk-management process, including at the medium- and small-scale bank group that has thus far been relatively underserved. In addition, it is important for OJK to compile and apply a national banking-sector cybersecurity-readiness index as a comparative parameter and, at the same time, a driver for financial institutions to strengthen their digital-defense posture.

Within a long-term policy framework, OJK needs to adopt a foresight-based approach for compiling a regulatory strategy able to anticipate the transformation of the future digital-threat landscape. Technological development such as artificial intelligence, the misuse of deepfake for financial crime, and the increasing dependence of the banking system on third-party infrastructure vulnerable to systemic cyber attack, are indicators that regulatory policy cannot be static. Therefore, OJK is required to continuously update its oversight framework based on an escalatory-risk projection so as to remain adaptive in protecting the stability and resilience of the national financial sector.

Recommendations for Bank Indonesia

Bank Indonesia, as the monetary authority and regulator of the national payment system, holds a strategic role in ensuring that the cyber-security aspect becomes a main foundation in maintaining financial stability and preserving the level of public trust in the digital-banking-service ecosystem. This responsibility is not only limited to overseeing the payment system, but also covers strengthening resilient digital infrastructure able to adapt to the increasingly complex dynamics of cyber threat. Within this framework, an expansion of BI's oversight mandate over payment-system technology infrastructure is needed, with adjustment to an international cyber-security standard and strengthening of cross-institutional collaboration with OJK, BSSN, and the Ministry of Communication and Informatics, so as to create nationally integrated cyber-security governance.

One strategic step that can be formulated is initiating the development of a National Cybersecurity Readiness Benchmark aimed at the payment-system sector and non-bank financial institutions under BI's authority. This index is expected to be able to provide an objective parameter for assessing an institution's readiness against potential digital attack, and to become a driver for the thorough, consistent application of an international standard such as ISO/IEC 27001, ISO 22301, and ISO/IEC 27035. Further, BI needs to lead the strengthening of a cyber-threat-intelligence-based financial-intelligence system in order to monitor attack indicators against domestic and global payment-system infrastructure, while also improving an anomalous-transaction-detection system through the use of machine-learning technology to suppress digital-engineering-based financial-crime risk.

In confronting increasing risk in the digital supply chain, including at a third-party-service provider such as fintech and a payment gateway, BI is advised to form a coordinative framework in the form of a cross-entity collaborative audit. This approach is needed to anticipate the potential threat of a supply-chain compromise that could have a systemic impact on payment services. In addition, BI's support for national digital infrastructure such as the National Payment Gateway (GPN) and BI-FAST needs to be accompanied by an incident-recovery-system-strengthening policy guaranteeing service continuity and strengthening the resilience-planning framework for confronting a technology-based crisis.

Through the foresight approach, BI can also build a medium- to long-term projection of potential cyber attack directed at a real-time payment system and future financial technology. This requires strengthening strategic-analysis capability through developing a special strategic-intelligence unit focused on formulating a digital-threat scenario with implications for monetary policy, financial-system stability, and the protection of financial-service-user data.

To ensure systemic resilience in the long term, cyber-security mainstreaming needs to become an inseparable part of BI's digital-transformation agenda. This step covers forming an internal digital-security command center directly responsible to a strategic managerial structure, and strengthening internal HR competency in understanding cyber-attack tactics, operating an artificial-intelligence-based monitoring system (AI-driven monitoring platform), and improving cross-institutional coordination capacity in confronting a digital crisis situation.

Recommendations for the Ministry of Communication and Digital Affairs

The Ministry of Communication and Digital Affairs (Kemkomdigi) holds an important responsibility in building a comprehensive national information-security framework, particularly through the implementation of the latest regulation such as Ministerial Regulation No. 5 of 2025 concerning Public-Scope Electronic-System Operators (PSE). In this context, Kemkomdigi is expected not only to emphasize meeting an administrative obligation such as the registration of an electronic system by a government agency or other public-sector body, but also to actively carry out substantive oversight of the application of technical control, personal-data protection, and the effectiveness of the detection system against potential digital attack (Kemkomdigi, 2025).

One focus that needs to be strengthened is the electronic-data classification-and-protection mechanism based on the risk level set in that regulation. Kemkomdigi should encourage the creation of interoperability between national security systems to ensure that critical data, including that held by the banking sector, does not become a target of exploitation by an unauthorized party. This requires alignment between the data-management framework and a mutually integrated national security system.

Further, Kemkomdigi plays the role of key actor in facilitating the development of a shared security infrastructure usable across sectors. This infrastructure covers the existence of a national Security Operation Center (SOC), a Domain Name System (DNS) firewall, and a ransomware-attack-mitigation platform that can function as a collective early-warning system for a strategic institution, including a financial institution.

Kemkomdigi also holds the authority to halt access to an electronic system proven to violate security principles or to fail to meet the technical standard as set out in Article 16 of Ministerial Regulation No. 5 of 2025. Nevertheless, execution of this mandate needs to be accompanied by the formation of a rapid coordination mechanism together with OJK and BSSN, so that an access-termination or recovery (normalization) action does not negatively impact the operations of the financial sector, which is vital and systemic in nature.

In order to avoid cross-sector policy disharmony, Kemkomdigi is also advised to align national digital regulation with sectoral policy such as POJK No. 11/POJK.03/2022, which governs banking information-technology management, and an international standard such as ISO/IEC 27001 on the information-security-management system and ISO/IEC 27035 on information-security-incident management. This harmonization becomes a crucial element in forming effective, non-redundant digital-security governance able to drive oversight efficiency within the continuously developing national digital ecosystem.

Recommendations for BSSN

In its capacity as the main institution bearing the strategic responsibility for strengthening national cyber defense and resilience, BSSN holds a central role in formulating and coordinating digital-security policy that systemically touches every sector. The 2024 National Cyber Security Landscape document published by BSSN has provided a comprehensive policy direction related to anticipating a digital-attack pattern, and has set the financial sector as one of the priorities for strengthening vital information infrastructure. However, in order for that strategic document to be implemented effectively, a policy-actualization instrument is needed that is not only normative, but also can be concretely operationalized at the institutional level.

One priority step recommended is strengthening BSSN's role as a central intelligence integrator within the national digital-threat detection-and-response system. Expanding the National Cybersecurity Operation Center function into a strategic and tactical control center will strengthen the state's capability to monitor an attack indicator in real time and coordinate the distribution of threat data to the banking sector through a sectoral-CSIRT scheme and an early-warning system based on inter-institutional collaboration. This role also aligns with BSSN's function in orchestrating a Cyber Threat Intelligence architecture that is adaptive and responsive to the development of increasingly complex attack modus.

Further, BSSN is advised to develop a cyber-maturity-assessment framework usable for assessing digital-security readiness in the financial sector, not merely based on the existence of a formal policy or SOP, but also through field verification of implementation effectiveness, testing of the early-detection system, and an institution's readiness to run an incident-recovery procedure and operational continuity in a crisis condition. With this performance-based approach, BSSN's oversight function will become more relevant in assessing factual capability, not merely administrative.

On the other hand, to strengthen cross-institutional coordination, BSSN needs to facilitate the formation of a permanent forum involving key stakeholders such as OJK, Bank Indonesia, Kemkomdigi, and the banking sector and technology-service provider. This forum can function as a tool for harmonizing the cyber-security technical standard, exchanging best practice, and compiling a joint-response protocol for confronting a systemic incident, including one originating from a digital-supply-chain compromise involving a third party.

In developing medium- and long-term policy, BSSN is expected to involve the academic community and strategic intelligence to compile a risk projection based on a scenario-planning and horizon-scanning approach. This is important given the risk escalation marked by the increasingly widespread use of artificial intelligence for a cyber attack, the misuse of deepfake technology, and the development of a cybercrime-as-a-service ecosystem that lowers the technical entry threshold for a criminal actor.

By carrying out its role as a commanding actor in the grand design of Indonesia's digital defense, BSSN can become an important catalyst in bridging the security-

capability gap between sectors and strengthening the digital resilience of the banking sector as part of national strategic interest.

Recommendations for BIN

The State Intelligence Agency (BIN), as the main entity within the national intelligence system, is expected to play a more strategic role in mainstreaming cybersecurity issues into the state intelligence-policy framework as a whole. The continuously developing, borderless dynamics of digital threat, whether by a state-sponsored actor or a non-state entity, requires BIN's presence in a more active capacity, particularly in strategic-risk mapping and the development of an early-warning system based on multidimensional intelligence.

In the context of banking-sector protection, BIN is advised to form a special unit focused on strategic cyber-intelligence analysis. This unit's task is to monitor and analyze a national-scale, systemically impactful cyber-attack pattern, including identifying abnormal cyber activity from the early stage such as reconnaissance, and mapping the connection between digital attack and disruption of public trust in financial-system stability. The intelligence product produced by this unit will become an important component in supporting risk-mitigation policy by an institution such as OJK, BSSN, and Bank Indonesia.

Further, BIN also needs to strengthen the strategic-warning-intelligence function, namely the ability to recognize early indicators (early warning signals) of a large-scale cyber attack with the potential to create systemic disruption. Threats such as the use of deepfake for financial crime and the use of artificial intelligence to breach a bank's security system require a high level of intelligence vigilance.

As part of the national-defense-coordination system, BIN also holds an important role in building a communication bridge between the security institution and the financial sector. This particularly relates to the tactical exchange of information regarding threat indicators, infiltration technique, and mapping the relationship between cyber-crime actors and a cross-border criminal network. In this regard, a closed but verified intel-sharing system can strengthen the digital resilience of banking institutions without sacrificing the confidentiality of national strategy.

Finally, to support the effectiveness of long-term policy, BIN can become an initiator in developing a strategic-foresight strategy related to future cyber threat. Through the scenario-planning and horizon-scanning method, BIN can work together with the academic community and technology experts to compile a mitigation-policy scenario for threat with the potential to emerge five to ten years from now, such as cybercrime-as-a-service, AI-technology exploitation, and financial manipulation through digital-content engineering. With the strengthening of this role, BIN has the potential to become a main actor in building a progressive, adaptive national cyber-intelligence system relevant to the changing global technology-geopolitics landscape.

Recommendations for Future Researchers

This study provides a conceptual foundation that can be expanded by future researchers, particularly in efforts to strengthen cyber-intelligence capability and the formulation of a digital-resilience strategy in the national banking sector. One relevant research agenda to develop is the exploration of practical cyber-security-policy implementation at the institutional level. Future research can examine the extent to which cyber threat intelligence (CTI) has been functionally integrated into the managerial and operational decision-making process, and identify the structural, institutional, or resource-limitation obstacles hindering its implementation effectiveness.

Beyond that, a comparative approach across jurisdictions or sectors becomes a strategic area worth exploring further. A comparative analysis of the application of the Adaptive Security Framework and Strategic Intelligence in the financial sector in other countries has the potential to open new insight into regulatory effectiveness, technological readiness, and organizational culture in confronting the increasingly complex and layered dynamics of cyber threat. Thus, a researcher can obtain a more complete understanding of the context of success or failure of cyber policy at the sectoral or national level.

Given the limited access to cyber-incident data, which is generally closed and highly sensitive, it is recommended that further research be built through a research collaboration between academic institutions and government or financial bodies, with a formal partnership scheme enabling the acquisition of valid, reliable primary data. In addition, a study with a longitudinal design also has the potential to make a significant contribution

in monitoring the evolution of a digital-attack pattern and assessing mitigation-policy effectiveness over a broader, sustained time period.

Future researchers are also encouraged to apply a mixed-methods approach, in order to obtain triangulative validity between qualitative and quantitative data. This strategy can be used to measure the relationship between variables, such as the technical security-system readiness level, cyber-attack frequency, and the speed and quality of the operational-recovery process. Further, developing a machine-learning-based analytical model for detecting potential attack and digital anomaly is also a prospective research topic in line with the demand of technological progress and the needs of the financial industry.

In closing, it is important for future researchers to explore in depth the integration of the strategic-foresight approach within the national digital-security-policy framework. A method such as scenario planning, horizon scanning, and roadmapping can become a crucial instrument for formulating an anticipatory response to the spectrum of future cyber threat, which is disruptive and crosses sectoral and geographic boundaries. This approach also opens opportunity for strengthening collaboration between the academic community and policymakers in forming a national digital-defense system that is adaptive, structured, and intelligence-based.

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Author Biography



The author, Eggan Nachson Silueta, was born in Jakarta on 21 June 2000. The author is the first of two children, son of Mr. Bartholomeus Laurenz Silueta, S.H., and Mrs. Adolfina Kakalang, S.H. The author currently resides at Jl. Gang Fajar IX No. 30, Central Jakarta.

The author's educational history began at the elementary level at SD Strada Van Lith 1, then continued to the junior-high level at Gandhi International School, and completed senior-high education at SMA Negeri 20 Jakarta. The author pursued higher education in the Vocational Program of Universitas Indonesia with a concentration in Administration, and obtained a Bachelor's degree in Informatics Engineering from ITB Swadharma.

In the professional field, the author has four years of work experience as a Relationship Manager Micro in the banking sector starting in 2020. Since November 2024, the author has begun building and developing a digital-education startup called MatchupSkills.id, which focuses on developing technology competency and future-ready skills.

The author can be reached by email at: nachson21@gmail.com

Appendices

REVISION MATRIX
NATIONAL RESILIENCE STUDIES PROGRAM
SCHOOL OF STRATEGIC AND GLOBAL STUDIES – UNIVERSITAS
INDONESIA

Name : Eggan Nachson
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Track : Coursework
Concentration : Intelligence Strategic Studies
Research Title : Intelligence Analysis of Cyberattacks on the Security of
Banking Institutions in Indonesia

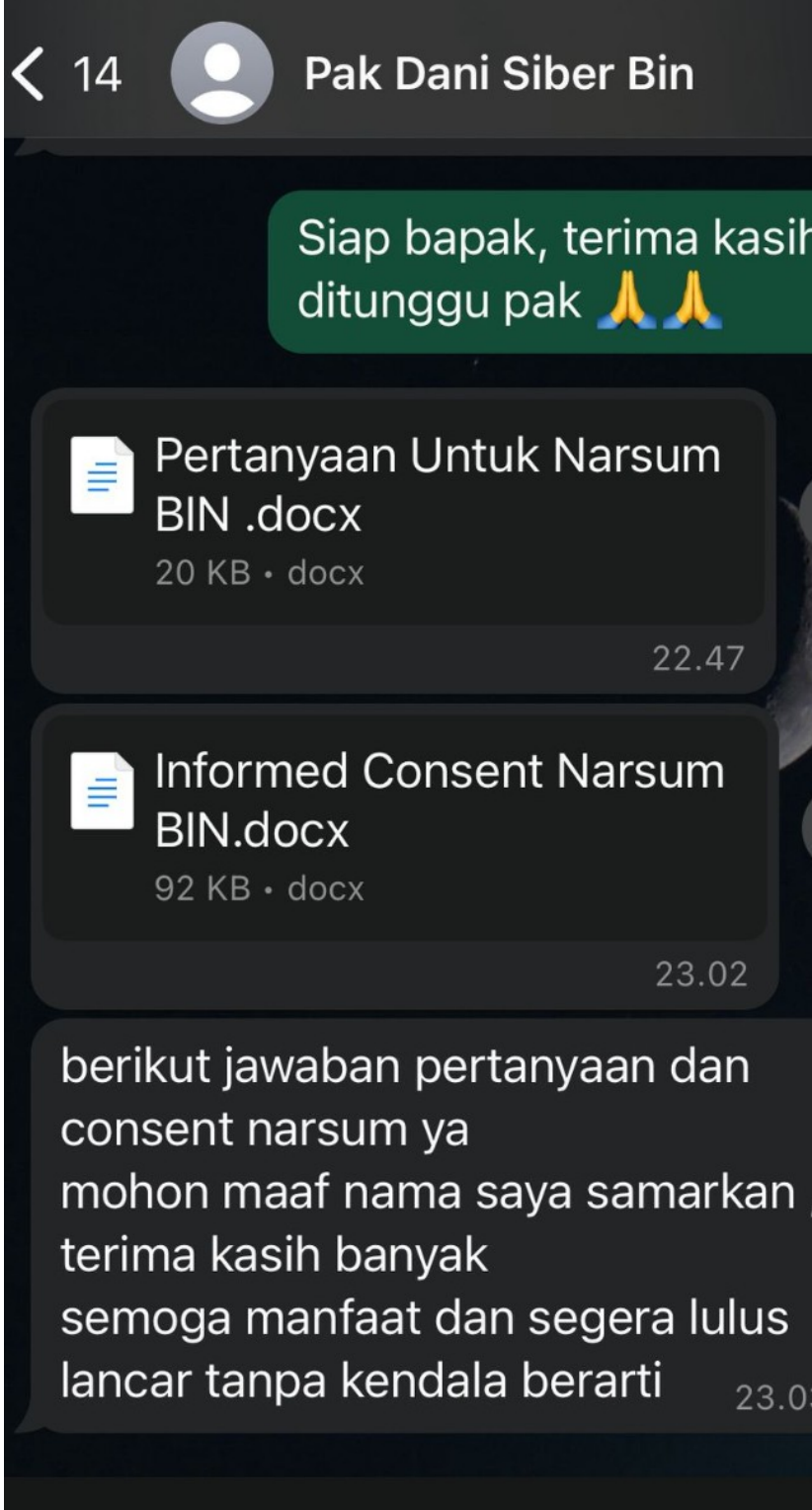
Exam Type : Thesis Proposal Seminar Exam/

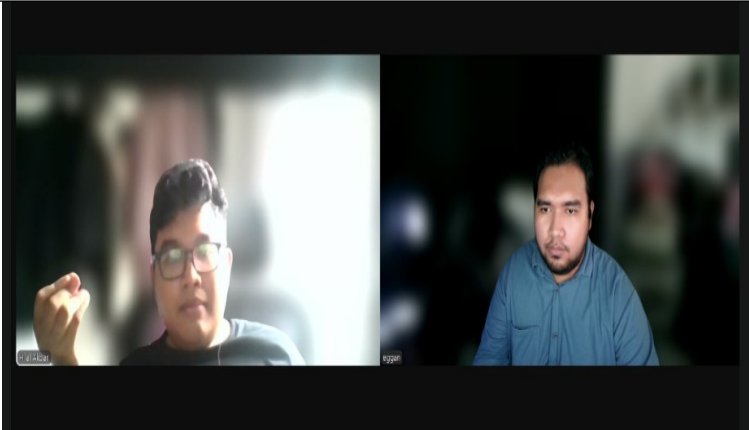
No	Comment/ Suggestion	Revision Made	Page
Examiner 1, Name: Bondan Widiawan, S.Kom., M.Si			
1	Revision related to the DDoS threat, which is not relevant to the threat in banking	Removal of the DDoS threat as one of the threats studied	Removed from the Abstract through Chapter 3
2	Update the 2023 cyber-security-landscape reference to 2024	Added the 2024 Cyber Security Landscape reference in Chapter 1	P. 1
3	Change of Research Question and Research Objective, which originally only analyzed the pattern — how to prevent it early	Changed the Research Question and Research Objective to: 1. Analyze a proactive cyber-intelligence-system approach in detecting, identifying, and responding to a cyber-attack pattern targeting banking institutions 2. Analyze the application of an adaptive security approach in confronting the escalation of cyber threat to the banking sector in Indonesia 3. Analyze the integration of a progressive, strategic information-	P. 9

No	Comment/ Suggestion	Revision Made	Page
		protection policy within the banking cyber-security system	
Advisor 1, Name: Dr. Puspitasari, S.Sos., M.Si.			
1	Addition of the Indonesian national banking locus to the title	Changed the research title to Intelligence Analysis of Cyberattacks on the Security of Banking Institutions in Indonesia	Cover — Abstract, p. i
2	Removal of the citation within the abstract	Removed the citation within the abstract	Abstract, p. i
3	Addition of an Informant Profile at point 3.3 related to the Data Collection Method	Added an Informant Profile, 4 informant profiles	P. 87 P. 88
4	Requested removal of the informant-count/unit column in the List of Informants table	Removed the Number column in Table 3.1, List of Informants	P. 89
Advisor 2, Name: Dr. Aloysius Mado, B.Sc, S.Sos, M.Han			
1	In Chapter 3 regarding the Foresight analysis method, a revision of the x and y axes in the research was requested	Revised the x and y axes in determining the foresight quadrant — the x axis, previously "resilience-capability level," changed to "banking-institution resilience capability"; the Y axis, from "hacker attack-capability level," changed to "hacker attack capability"	P. 100
2	In Table 3.3, Foresight Stages, the column containing the correlation between the foresight stage and the four main concepts was removed	In that table, the column correlating the foresight stage with the four main concepts was removed and replaced directly with a description of that foresight stage	P. 98

No	Comment/ Suggestion	Revision Made	Page
Committee Chair, Name: Dr. Palupi Lindiasari Samputra, S.Pi., M.M.			
1	Addition of the Indonesian national banking locus to the title	Changed the research title to Intelligence Analysis of Cyberattacks on the Security of Banking Institutions in Indonesia	Cover — Abstract, p. i
2	The explanation regarding the Scope of Strategic Intelligence, page 1.6.1, was removed	Removed section 1.6.1, Description of the Scope of Strategic Intelligence	P. 10
3	Addition of research-gap narrative in sub-section 1.2, Strategic Intelligence Focus	Added research-gap narrative in sub-section 1.2, Strategic Intelligence Focus	P. 6
4	Addition of an opening narrative in sub-section 2.1, Literature Study	Added narrative describing how many books were used and from what year range, and the correlation of those books to the theory to be used in the research	P. 13
5	Revision of the conceptual-framework flow	Redrew the conceptual-framework flow in sub-section 2.4, Researcher's Conceptual Framework	P. 80

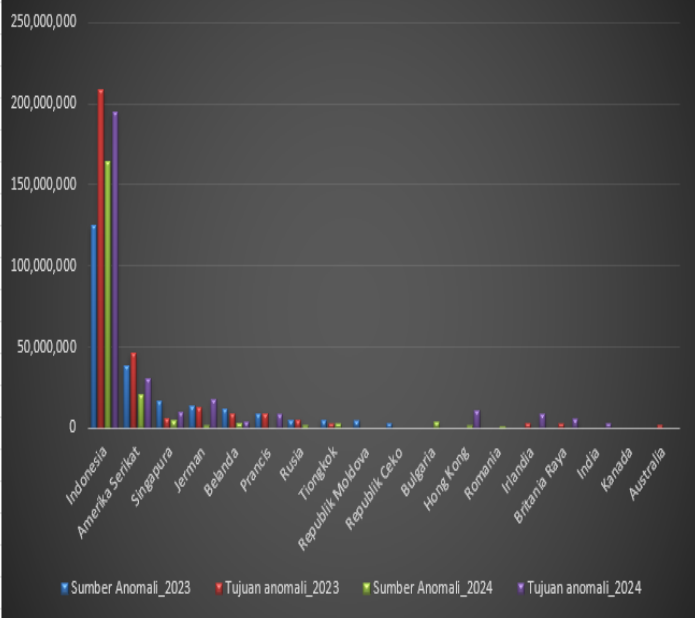
Informant	Documentation	Time & Execution
Dr. Pratama Dahlian Persadha, S.Sos., M.M. Chairman , CISSReC		Via Zoom Meeting , on 3-06-2025

Informant	Documentation	Time & Execution
Dani (alias), Deputy for Cyber Intelligence, BIN	 The screenshot shows a WhatsApp chat interface. At the top, there is a back arrow, the number '14', a profile picture, and the name 'Pak Dani Siber Bin'. The chat history includes a green bubble with the text 'Siap bapak, terima kasih ditunggu pak 🙏🙏'. Below this are two document attachments: 'Pertanyaan Untuk Narsum BIN .docx' (20 KB) and 'Informed Consent Narsum BIN.docx' (92 KB). At the bottom, a text bubble contains the message: 'berikut jawaban pertanyaan dan consent narsum ya mohon maaf nama saya samarkan terima kasih banyak semoga manfaat dan segera lulus lancar tanpa kendala berarti'.	Via WhatsApp chat, on Friday 30-05-2025

Informant	Documentation	Time & Execution
Hilal Akbar, S.Kom, CyberSecurity Division Head, PT Mitra Integrasi Informatika		Via Zoom Meeting, on Friday 30-05-2025
Julio (alias), White Hat Hacker (JadiHacker community)		Via Zoom Meeting, on Friday 30-05-2025

Data Tabulation and Data Visualization Process

Negara	Sumber Anomali_2023	Tujuan anomali_2023	Sumber Anomali_2024	Tujuan anomali_2024
Indonesia	124,646,656	208,061,734	164,432,795	194,562,390
Amerika Serikat	38,773,173	46,221,919	20,606,740	30,670,349
Singapura	16,774,825	6,277,960	4,821,163	10,221,545
Jerman	13,499,932	12,617,940	1,592,177	17,644,837
Belanda	12,177,139	9,299,270	3,214,422	4,321,678
Prancis	8,908,443	9,160,106		8,416,562
Rusia	5,173,922	5,289,342	1,810,428	
Tiongkok	5,161,362	2,791,992	2,615,326	
Republik Moldova	4,507,275			
Republik Ceko	3,205,356			
Bulgaria			3,776,822	
Hong Kong			1,462,164	11,232,698
Romania			1,182,215	
Irlandia		3,134,935		8,592,002
Britania Raya		2,613,607		5,567,376
India				3,120,616
Kanada				
Australia		2,325,622		



Kode Awal	Kategori	Tema Utama
Technical Level, Threat Indicator, Operational Level, Strategic Level, Tactical Level, Threat Intel Sharing, Intel Lifecycle Process, Malware Exploit Profiling, Threat Actor Attribution, Threat Intel Tools.	Cyber Threat Intelligence	Deteksi Dini dan Proyeksi Pola Ancaman Siber
Detect, Predict, Prevent, Respond.	Adaptive Security Framework	Respon Adaptif terhadap Eskalasi Ancaman
Event, Trend, Driver, Institutional Readiness, Leadership Intel, Risk Based Policy, Scenario Planning, Strategic Planning, Resilience		Integrasi

Tahap Cyber Kill Chain	BIN	PT MII	White Hat Hacker	Pakar Siber	Validasi
Reconnaissance	—	✓	✓	✓	Tiga Sumber
Weaponization	—	✓	✓	—	Dua Sumber
Delivery	—	—	✓	—	Satu Sumber
Exploitation	—	✓	✓	—	Dua Sumber
Installation	—	✓	✓	✓	Tiga Sumber
Command and Control (C2)	—	✓	✓	✓	Tiga Sumber
Action on Objectives	✓	✓	✓	—	Tiga Sumber

Narasumber	Probabilistic Attack (2021–2022)	Targeted Attack (2023)	AI-powered Attack (Prediksi 2024–2025)
Dr. Pratama Persadha	Insiden awal didominasi phishing klasik dan malware tidak terpersonalisasi.	Serangan ransomware pada BSI menunjukkan pemetaan target dan eksploitasi sistem kritis.	Perlu waspada serangan AI-driven seperti deepfake phishing dan malware yang memodifikasi dirinya saat jalan.
Hilal Akbar	Serangan masih berbasis eksploitasi sederhana; fokus pada volume bukan presisi.	Serangan mulai fokus ke sektor finansial dengan rekayasa sosial spesifik.	Ancaman berikutnya adalah ransomware yang digerakkan oleh AI (polymorphic & evasive).
	Awal era dimana phishing masal dan	Mulai muncul cyber phishing dengan profiling	Sudah ada malware dengan fitur adaptif dan ekspansi

PERNYATAAN KESEDIAAN MENJADI NARASUMBER PENELITIAN
(INFORMED CONSENT)

Yang bertanda tangan di bawah ini, saya:

Nama : **DR. Pratama Dahlian Persadha, S.Sos. MM.**
Jabatan : **Chairman**
Instansi : **CISSReC**

Setelah mendapat penjelasan tentang maksud dan tujuan serta memahami penelitian yang dilakukan dengan judul “Analisis Intelijen terhadap Serangan Siber pada Keamanan Institusi Perbankan Indonesia” yang dilakukan oleh :

Nama : **Eggan Nachson**
NPM : **2306186431**
Program Studi : **Kajian Ketahanan Nasional, SKSG Universitas Indonesia**

PERNYATAAN KESEDIAAN MENJADI NARASUMBER PENELITIAN

(INFORMED CONSENT)

Yang bertanda tangan di bawah ini, saya:

Nama :- Dani
Jabatan :- Sub Bid Siber BIN
Instansi :-BIN

Setelah mendapat penjelasan tentang maksud dan tujuan serta memahami penelitian yang dilakukan dengan judul “Analisis Intelijen terhadap Serangan Siber pada Keamanan Institusi Perbankan Indonesia” yang dilakukan oleh :

Nama :- Eggan Nachson
NPM :- 2306186431
Program Studi :- Kajian Ketahanan Nasional, SKSG Universitas Indonesia

Dengan ini saya menyatakan kesediaan untuk berperan serta menjadi narasumber penelitian, bersedia mengikuti rangkaian wawancara penelitian, serta memberikan informasi dan pernyataan sesuai dengan data yang diperlukan. Demikian pernyataan ini

PERNYATAAN KESEDIAAN MENJADI NARASUMBER PENELITIAN
(INFORMED CONSENT)

Yang bertanda tangan di bawah ini, saya:

Nama : Julio
Jabatan : Whitehat Hacker

Setelah mendapat penjelasan tentang maksud dan tujuan serta memahami penelitian yang dilakukan dengan judul “Analisis Intelijen terhadap Serangan Siber pada Keamanan Institusi Perbankan Indonesia” yang dilakukan oleh :

Nama : **Eggan Nachson**
NPM : **2306186431**
Program Studi : **Kajian Ketahanan Nasional, SKSG Universitas Indonesia**

Dengan ini saya menyatakan kesediaan untuk berperan serta menjadi narasumber penelitian, bersedia mengikuti rangkaian wawancara penelitian, serta memberikan informasi dan pernyataan sesuai dengan data yang diperlukan. Demikian pernyataan ini dibuat dengan penuh kesadaran, tanpa ada paksaan dari pihak manapun.

Jakarta,

PERNYATAAN KESEDIAAN MENJADI NARASUMBER PENELITIAN

(INFORMED CONSENT)

Yang bertanda tangan di bawah ini, saya:

Nama : Hilal Akbar
Jabatan : Cyber Security Division Head
Instansi : PT MII

Setelah mendapat penjelasan tentang maksud dan tujuan serta memahami penelitian yang dilakukan dengan judul “Analisis Intelijen terhadap Serangan Siber pada Keamanan Institusi Perbankan Indonesia” yang dilakukan oleh :